IUBAT – INTERNATIONAL UNIVERSITY OF BUSINESS AGRICULTURE AND TECHNOLOGY

Founded 1991 by Md. Alimullah Miyan

CSC 470 – Software Engineering Lab

Lab Report

Title of the Project: Telemedicine & Consultation Booking System

Name of the Team: The Anonymous

Submitted By

ID	Name
22303364	Abir Hasan
22303235	Sadia Yasmin
22303195	Samiul Basher Shaan
22303297	Md Shamiul Kabir Alvi
22303258	Umana Tahrim

Submitted To

Engr. A.S.M. Shakil Ahamed

Senior Lecturer

Department of Computer Science and Engineering, IUBAT

Table of Contents

Chapter 1: Introduction 5
1.1 Introduction 5
1.2 Objectives of the Project 6
1.3 System Benefits 7
1.4 Software Process Model 8
1.5 Reason for Choosing the Process Model 9
Chapter 2: Requirement Engineering 10
2.1 User Requirements 10
2.2 System Requirements 11
2.3 Functional Requirements 15
2.4 Non-Functional Requirements 15
Use Case Diagram: 17
Chapter 3: Analysis Modelling 18
Activity Diagrams 18
Swim lane diagrams 27
Sequence Diagram 35
CRC 44
Class diagrams 51
Chapter 4: Project Management 52
Risk identification Table 52
Risk Analysis Table 53
Risk Planning Table 54
Chapter 5: Project Planning and Scheduling 55
Function Point Estimation 55
Project Schedule Chart 58
Chapter 6: Project Estimation 59
1. Introduction 59

2. Identification of Software Processes 59
3. Effort Estimation (Person-Month) 61
4. Effort Distribution (Process Based) 61
5. Pie Chart – Effort Distribution 61
6. Project Schedule Estimation 62
7. Project Schedule Chart 63
8. Personnel Requirement Chart 63
9. Cost Estimation 64
10. Personnel Cost Estimation 64
11. Hardware Cost Estimation (Depreciation Calculation) 65
12. Accounts Table 66
13. Accounts Chart 66
14. Quality Management & Software Development 68
15. Process Based Quality 69
16. Process Based Quality 71
17. Conclusion 72
Chapter 7: Designing 73
7.1 Data Flow Design 73
7.2 Database Design 81
7.3 ERD 83
7.4 Interface Design 85
Chapter 8: Testing 92
Functional Testing: 92
Module 1: User Registration & Authentication (UR#1) 92
Module 2: Doctor Registration & Approval (UR#2) 93
Module 3: Doctor Search & Filtering (UR#3, UR#7) 94
Module 4: Appointment Booking & Management (UR#3, UR#4, UR#5) 94
Module 5: Online Consultation (UR#4, UR#6) 95
Module 6: Prescription & Medical Records (UR#5, UR#7) 95
Module 7: Payment & Receipt (UR#6, UR#8) 96
Module 8: Admin Monitoring & Reports (UR#8, UR#9) 97

Non-Functional Testing 97
Performance Testing 97
Security Testing 98
Usability Testing 98
Reliability & Availability 99
Chapter 9: Conclusion 100
9.1 Brief Overview of the Project 100
9.2 Limitations of the Project 101

Chapter 1: Introduction

1.1 Introduction

Software engineering plays a fundamental role in the development of modern systems that are reliable, efficient, and capable of solving real-world problems in a structured manner. As technology continues to evolve, software systems are increasingly expected to handle complex operations while maintaining high standards of usability, security, and performance. In the healthcare sector, these expectations are even higher due to the direct impact such systems have on human well-being.

In recent years, the demand for digital healthcare solutions has grown rapidly as patients and healthcare providers seek more accessible, time-efficient, and organized ways to deliver medical services. Factors such as long waiting times, overcrowded hospitals, and geographical limitations have highlighted the need for alternative healthcare delivery models. Telemedicine has emerged as a practical and effective response to these challenges by enabling remote consultations, reducing unnecessary hospital visits, and facilitating digital communication between patients and medical professionals.

This project focuses on the development and analysis of a Telemedicine & Consultation Booking System, a web-based platform designed to support online medical consultations, appointment scheduling, and secure management of medical records. The system brings together essential healthcare services within a single digital environment, allowing patients, doctors, and administrators to interact through clearly defined roles and controlled access mechanisms. Beyond implementing core functionalities, the project places strong emphasis on the application of software engineering principles such as systematic estimation, quality management, software testing, and process planning. By doing so, the project demonstrates how disciplined engineering practices contribute to the development of dependable and maintainable healthcare software systems.

1.2 Objectives of the Project

The primary objective of this project is to design, analyze, and document a telemedicine system by applying established software engineering methodologies rather than focusing solely on software implementation. The project is intended to bridge the gap between theoretical knowledge gained in the course and its practical application in a controlled lab environment. Through this project, emphasis is placed on understanding the complete software development lifecycle, including planning, estimation, quality assurance, and testing.

The specific objectives of the project are as follows:

- To analyze and document both functional and non-functional requirements of a healthcare-oriented software system in order to clearly define system behavior, performance expectations, and security needs.
- To design a well-structured system that supports core healthcare functionalities such as patient registration, doctor consultation management, appointment scheduling, and digital prescription handling while maintaining clarity and modularity.
- To select and apply an appropriate software process model that supports systematic development, iterative refinement, and continuous quality improvement throughout the project lifecycle.
- To perform process-based estimation and cost analysis suitable for an academic lab project, demonstrating an understanding of effort distribution, resource allocation, and financial planning in software engineering.
- To implement quality management practices and software testing strategies at different stages of development to ensure correctness, reliability, and maintainability of the system.
- To gain practical experience in project planning, technical documentation, and disciplined software development practices, preparing for real-world software engineering challenges.

1.3 System Benefits

The Telemedicine & Consultation Booking System provides significant benefits to users as well as to the overall healthcare process by introducing digital efficiency, structured workflows, and improved accessibility. The system is designed to address common challenges faced in traditional healthcare environments, such as long waiting times, limited access to specialists, and inefficient record management.

From a patient perspective, the system offers convenient access to medical services without the need for frequent physical hospital visits. Patients can register on the platform, search for doctors based on specialization, book appointments, and attend online consultations from their location. This reduces travel time, minimizes waiting periods, and makes healthcare more accessible, especially for individuals living in remote areas or those with limited mobility. Digital access to prescriptions and consultation records also helps patients manage their health information more effectively.

For doctors, the system simplifies daily operations by providing a structured way to manage appointments, consultation schedules, and patient interactions. Doctors can review appointment requests, communicate with patients, and issue digital prescriptions through the platform. This organized workflow improves time management, reduces administrative burden, and ensures that medical records are properly maintained.

From an administrative perspective, the system enables centralized control and monitoring of platform activities. Administrators can verify doctor registrations, manage user accounts, monitor system usage, and ensure that security and privacy policies are enforced. This helps maintain trust in the system and prevents misuse or unauthorized access.

From a software engineering viewpoint, the system demonstrates the benefits of applying structured design, role-based access control, and modular architecture. These design choices improve maintainability, scalability, and system reliability. Additionally, the project showcases how proper estimation, testing, and quality management contribute to the development of dependable software systems.

Overall, the Telemedicine & Consultation Booking System enhances healthcare service delivery while serving as a practical example of how software engineering principles can be applied to develop efficient, secure, and user-centered systems in a real-world context.

1.4 Software Process Model

The development of the Telemedicine & Consultation Booking System follows the Iterative Software Process Model. In this model, the system is developed through a sequence of repeated cycles, known as iterations. Each iteration consists of requirement analysis, system design, implementation, testing, and review activities. Instead of delivering the complete system in a single phase, the iterative model allows the system to evolve gradually through incremental improvements.

At the beginning of each iteration, requirements are reviewed and refined based on previous feedback and evaluation results. New features or enhancements are then designed, implemented, and tested within that iteration. This approach ensures that errors and design issues are identified early, reducing the risk of major failures at later stages. Testing is not postponed until the end of development; rather, it is integrated into every iteration, supporting continuous quality improvement.

The iterative model encourages flexibility and adaptability, making it suitable for systems that involve multiple user roles and complex workflows, such as telemedicine platforms. By delivering functionality in increments, the development team can evaluate progress more effectively and ensure that the system remains aligned with its objectives throughout the development lifecycle.

1.5 Reason for Choosing the Process Model

The Iterative Process Model was chosen for this project due to its ability to support gradual development, early feedback, and continuous improvement. Telemedicine systems involve diverse user requirements, including patients, doctors, and administrators, which can be difficult to fully define at the initial stage. The iterative approach allows these requirements to be refined over time without disrupting the entire development process.

Another important reason for selecting this model is its strong emphasis on testing and quality assurance. Since testing is conducted at the end of each iteration, defects can be detected and corrected early, reducing rework and improving system reliability. This is particularly important for healthcare-related systems, where data accuracy and system stability are critical.

From an academic perspective, the iterative model is well suited for lab projects because it aligns with incremental learning and development schedules. It allows students to apply theoretical concepts step by step while observing how planning, estimation, quality management, and testing interact across multiple development cycles. Overall, the iterative process model provides a balanced, flexible, and quality-focused approach that supports the successful development of the Telemedicine & Consultation Booking System.

Chapter 2: Requirement Engineering

2.1 User Requirements

User requirements describe what patients, doctors, and administrators expect from the Telemedicine System. These requirements reflect real-world needs and outline the interactions, services, and experiences the system must provide. They ensure that the platform supports accessible healthcare, smooth appointment processes, secure information handling, and reliable communication.

The following user requirements apply to the TeleMed System:

i. Patients can create an account using email or phone and manage a personal profile containing basic demographic and medical information.

ii. Doctors can register on the platform by submitting personal information, qualifications, professional documents, consultation preferences, and availability.

iii. Patients can search for doctors based on specialization, availability, consultation type, fees, and other relevant filters.

iv. Patients can book in-person or online consultations by selecting preferred doctors, dates, and time slots.

v. Both patients and doctors can manage appointments, including viewing upcoming sessions, rescheduling, and cancellations (according to system rules).

vi. Patients and doctors can join secure online consultation sessions with support for chat, file sharing, and video communication.

vii. Doctors can issue prescriptions, medical notes, and consultation summaries, while patients can view or download them anytime.

viii. Patients can complete payments for consultations through integrated payment methods and receive digital receipts.

ix. Admins can monitor system activities, approve doctor registrations, manage users, and generate reports.

2.2 System Requirements

System requirements define the technical, functional, and operational capabilities the TeleMed System must implement to satisfy user requirements. These requirements guide system design, development, and evaluation.

Below is the corrected and structured set of system requirements.

UR#1: Patients can create an account using email or phone and manage a personal profile containing basic demographic and medical information.

The system must allow users to register, authenticate, secure their accounts, and manage profiles.

1.1 The system shall provide registration via email or phone, collecting identity information and login credentials.

1.2 The system shall allow uploading of profile photographs and relevant medical or professional documents.

1.3 The system shall validate uploaded document types and enforce secure storage.

1.4 Users may update or delete non-sensitive profile data.

1.5 The system shall maintain audit logs for profile changes and protect sensitive fields using encryption.

1.6 The system shall authenticate users with valid credentials before granting access.

1.7 Incorrect login attempts beyond a limit shall trigger account lockout.

1.8 Users may reset passwords through verified OTP or email-based recovery.

1.9 The system shall prevent unauthorized access to protected resources.

UR#2: Doctors can register on the platform by submitting personal information, qualifications, professional documents, consultation preferences, and availability.

The system must allow secure registration and validation of doctor credentials.

2.1 The system shall allow doctors to upload credentials, qualification documents, and certifications.

2.2 Admins shall review and approve doctor registrations before activation.

2.3 The system shall verify document authenticity where possible.

2.4 Doctors may manage consultation preferences, fees, and availability settings.

UR#3: Patients can search for doctors based on specialization, availability, consultation type, fees, and other relevant filters.

The system must support searching, booking, and managing consultations.

3.1 Patients can search for doctors based on specialty, availability, language, fee, and location.

3.2 Patients can select doctor, date, time, and consultation type during booking.

3.3 The system shall check slot availability and prevent overlapping appointments.

3.4 Users can reschedule or cancel appointments according to system rules.

3.5 Both patient and doctor can view appointment status and details.

UR#4: Patients can book in-person or online consultations by selecting preferred doctors, dates, and time slots.

The system must support secure online and offline consultations.

4.1 The system shall generate secure, time-limited video session links for online appointments.

4.2 Consultation screens shall support chat, attachments, and session controls.

4.3 Doctors may mark sessions as started, paused, completed, or cancelled.

4.4 The system shall log timestamps for billing and auditing.

UR#5: Both patients and doctors can manage appointments, including viewing upcoming sessions, rescheduling, and cancellations.

The system must support secure creation and sharing of medical records.

5.1 Doctors can create digital prescriptions and consultation notes.

5.2 Patients may view and download prescriptions.

5.3 The system shall maintain version history and immutable audit logs.

5.4 Access to patient data shall require proper consent and role-based permissions.

UR#6: Patients and doctors can join secure online consultation sessions with support for chat, file sharing, and video communication.

The system must support secure and transparent payment handling.

6.1 Patients can pay consultation fees using integrated payment gateways.

6.2 The system shall update appointment payment status automatically.

6.3 Paid sessions generate downloadable digital receipts.

6.4 Admins and doctors can view transaction histories and earnings summaries.

UR#7: Doctors can issue prescriptions, medical notes, and consultation summaries, while patients can view or download them anytime.

The system must make it easy to find relevant doctors.

7.1 Search filters include specialization, location, availability, rating, and consultation fees.

7.2 Results should show next available slot and quick booking options.

7.3 Search queries must be optimized for speed through indexed and cached mechanisms.

UR#8: Patients can complete payments for consultations through integrated payment methods and receive digital receipts.

The system must provide dashboards and exportable reports.

8.1 Admins can view KPIs such as total appointments, revenue, top doctors, and cancellations.

8.2 Reports can be generated on daily, weekly, and monthly bases.

8.3 Doctors can view personal performance and earning statistics.

UR#9: Admins can monitor system activities, approve doctor registrations, manage users, and generate reports.

The system must ensure data protection and continuity.

9.1 The system shall automatically back up the database every 24 hours.

9.2 Admins can trigger manual backups and restore operations.

9.3 All critical actions must be logged with timestamps and user identifiers.

2.3 Functional Requirements

Functional requirements define what the TeleMed System must do:

- Allow users to register, authenticate, and access the system.
- Enable patients to manage profiles, book appointments, and view records.
- Support doctors in managing schedules, consultations, and medical documents.
- Handle payments, generate receipts, and maintain transaction integrity.
- Provide secure online consultation capabilities.
- Ensure admin oversight through user management, reports, and audits.

2.4 Non-Functional Requirements

Performance

- Pages must load within two seconds under normal conditions.
- Searches should return results within one second.
- The system should support multiple simultaneous consultations without degraded performance.

Reliability & Availability

- The system must maintain at least 99.5% uptime.
- Database backups must ensure a maximum of 24 hours of data loss.

Security

- All communication must use HTTPS encryption.
- Passwords must be stored securely using hashing and salting.
- Unauthorized access attempts must be logged.

Usability

- Users should be able to reach key features within three clicks.
- UI elements must remain consistent across the platform.

Scalability

- The platform should support horizontal scaling as user count grows.

Maintainability

- Code should follow modular architecture and standard naming conventions.

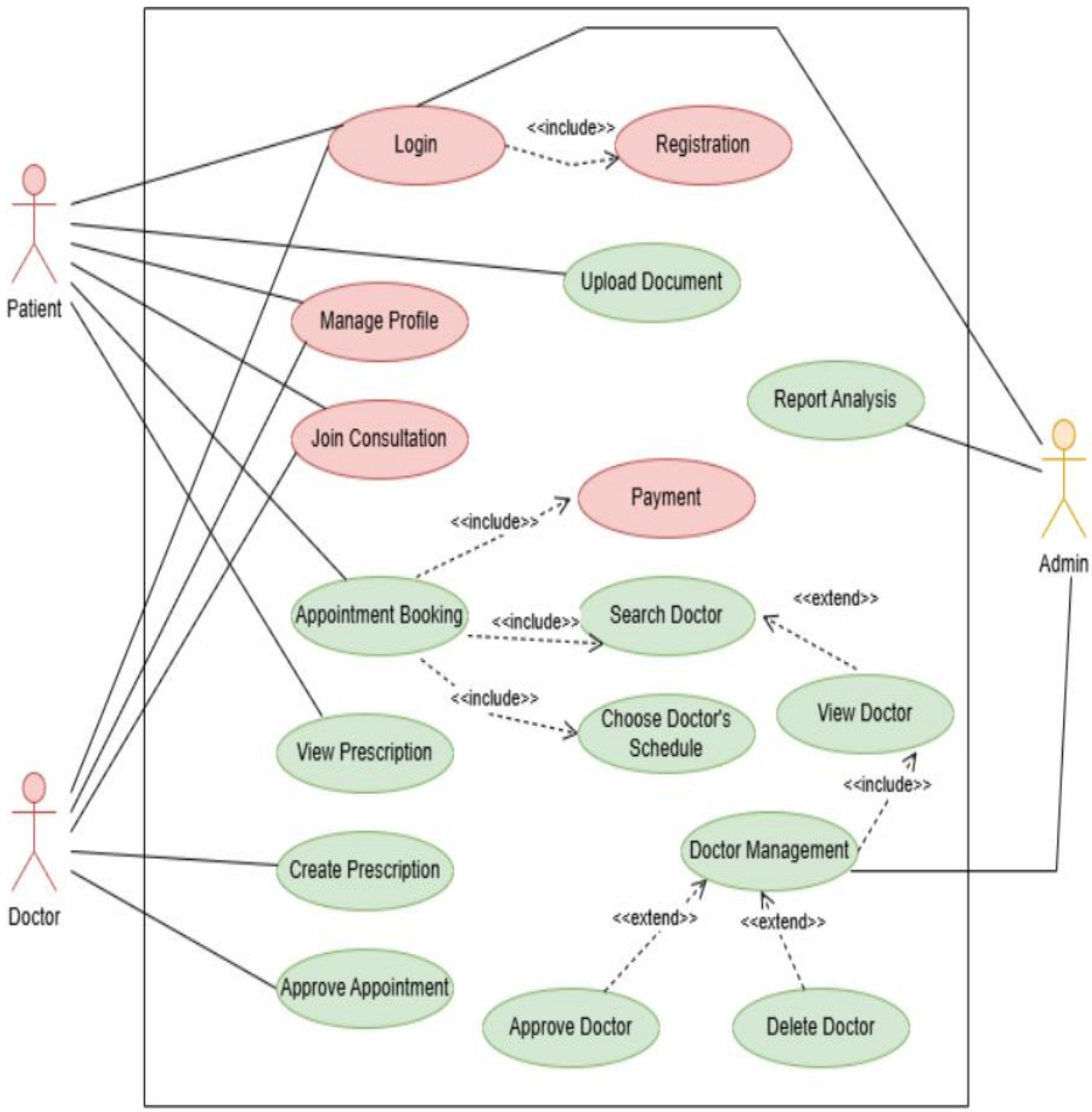
Portability

- The system must run on Windows or Linux servers and support container deployment.

Compliance

- Medical data must comply with regional privacy regulations.

Use Case Diagram:



Chapter 3: Analysis Modelling

Activity Diagrams

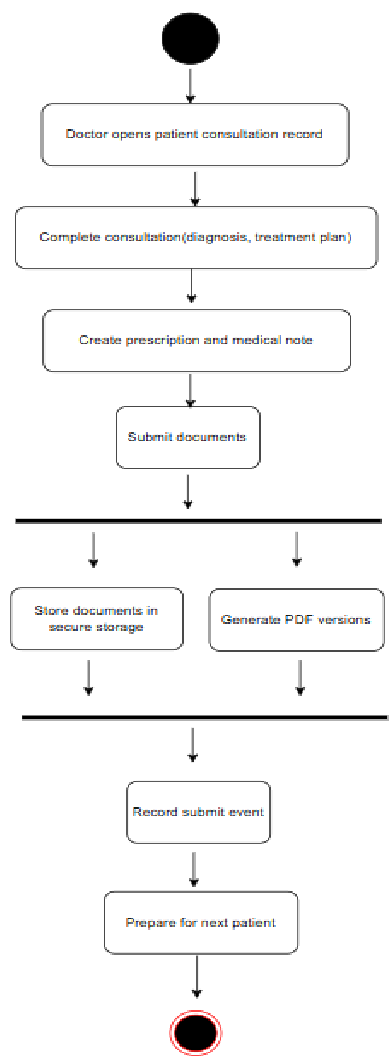
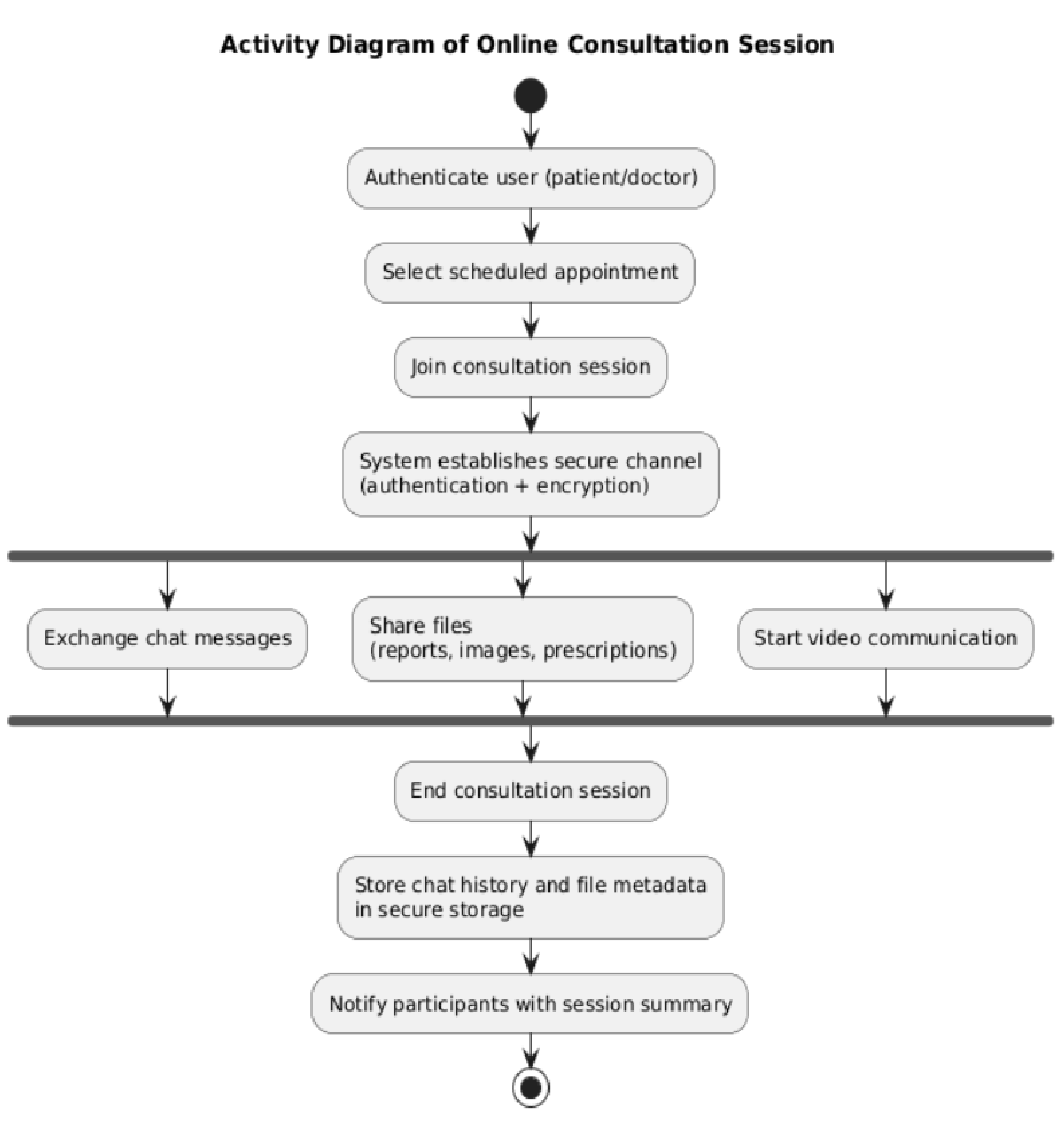


Figure : Activity Diagram for "Issue Prescription"


Activity Diagram of Online Consultation Session
Authenticate user (patient/doctor)
Select scheduled appointment
Join consultation session
System establishes secure channel
(authentication + encryption)
Exchange chat messages
Share files
(reports, images, prescriptions)
Start video communication
End consultation session
Store chat history and file metadata
in secure storage
Notify participants with session summary

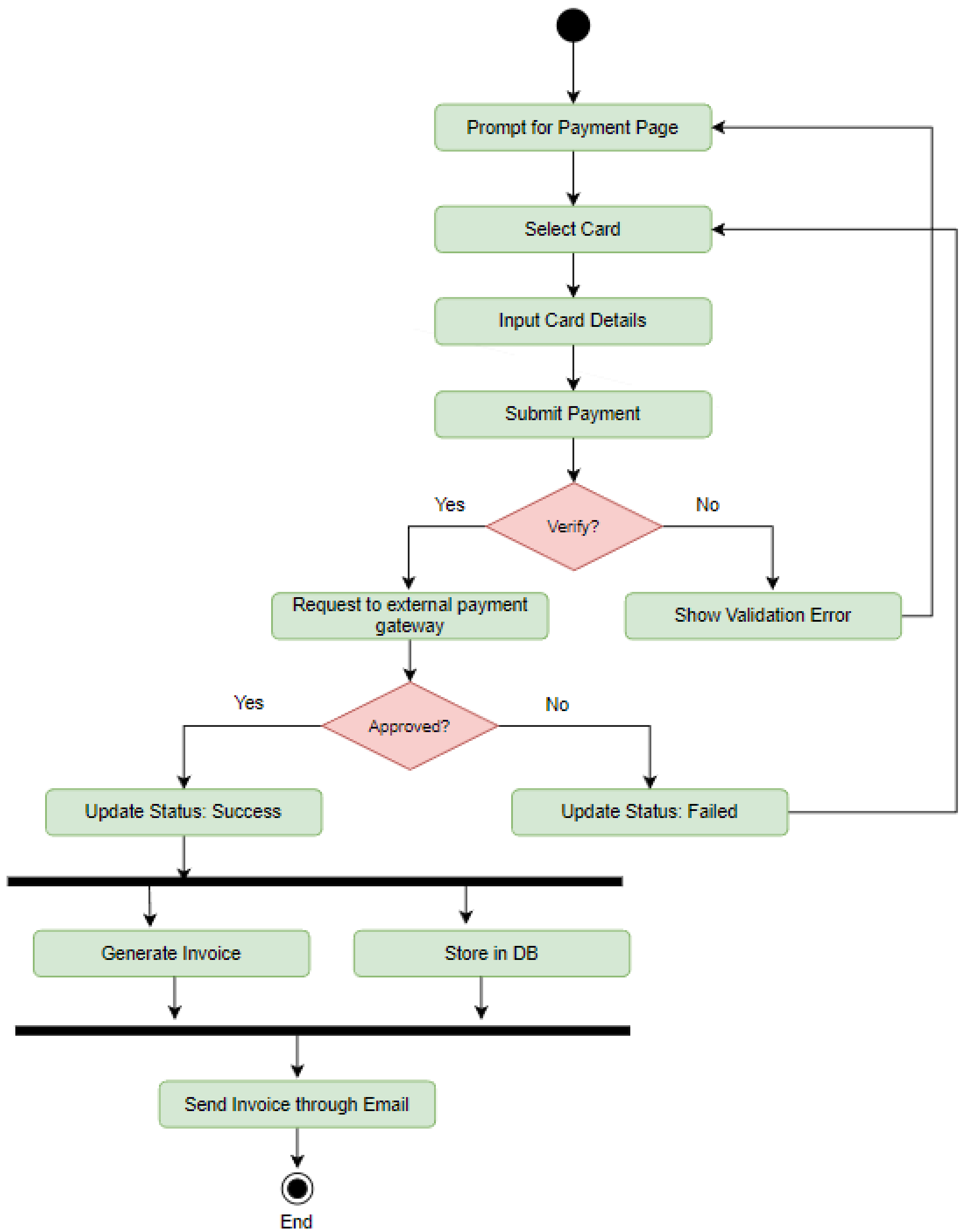


Fig: Activity Diagram for Payment

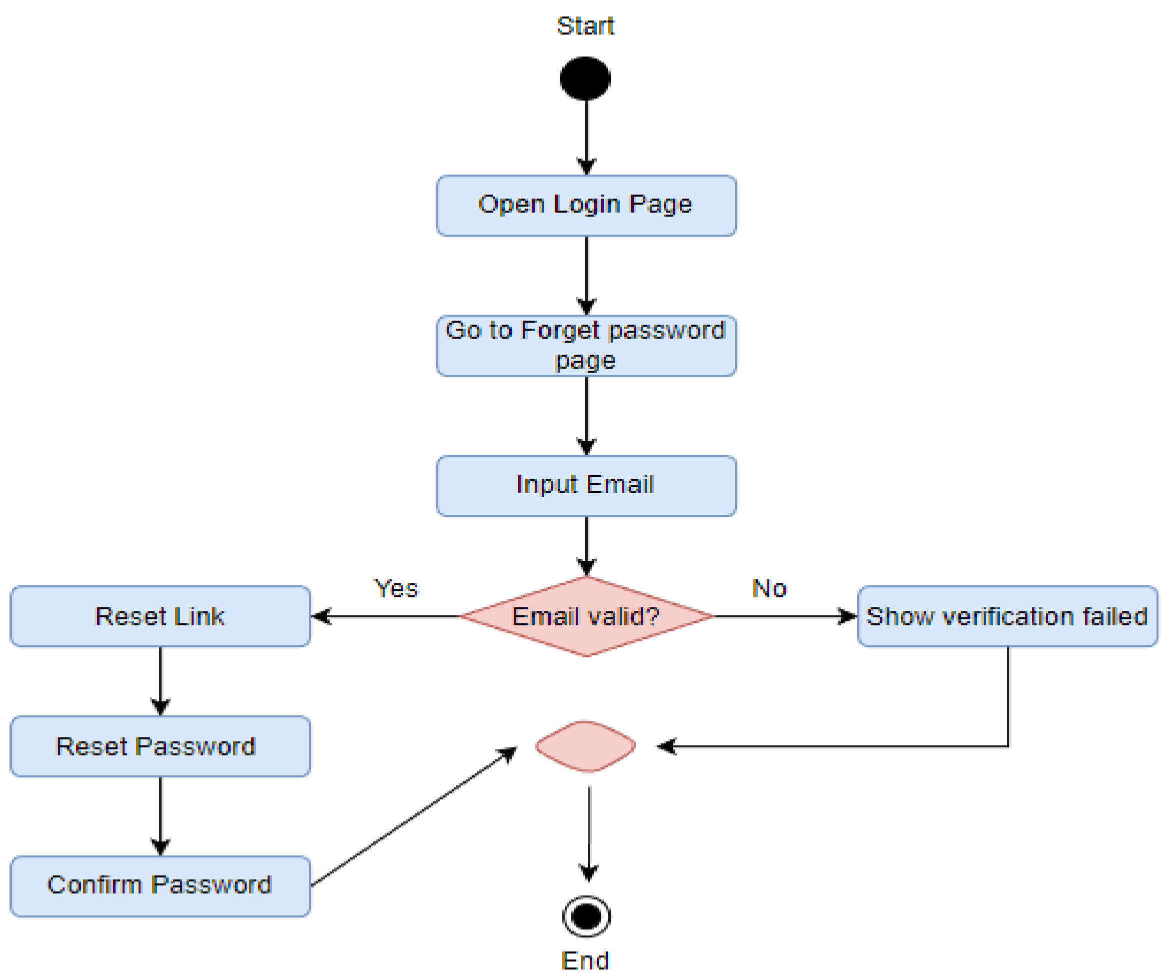


Fig: Reset Password

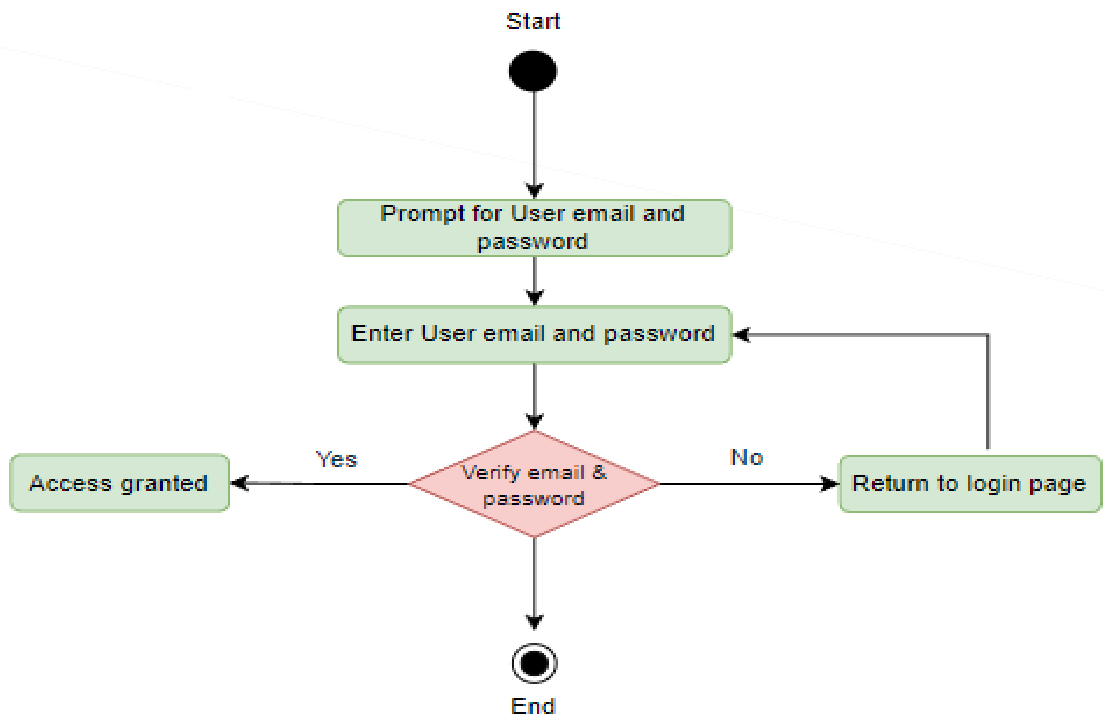


Fig: User Login

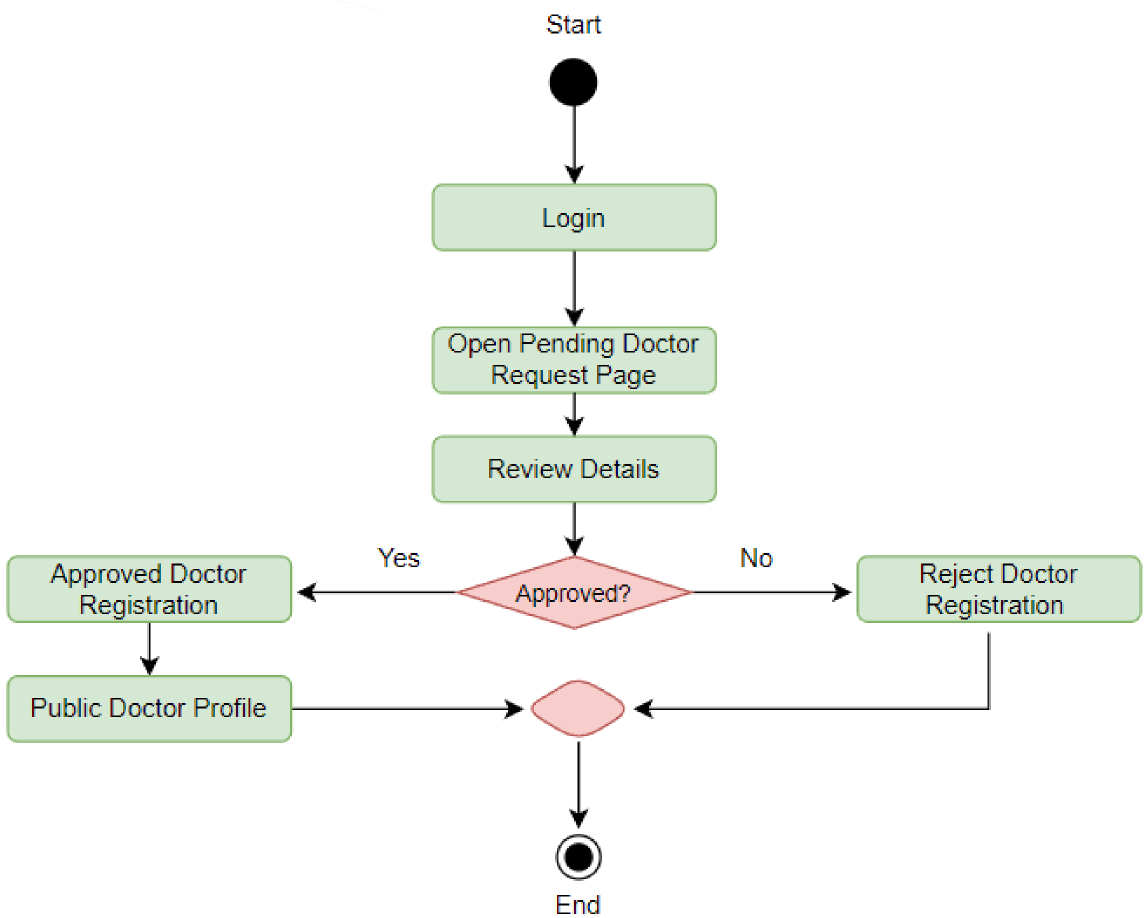


Fig: Admin Approve Doctor

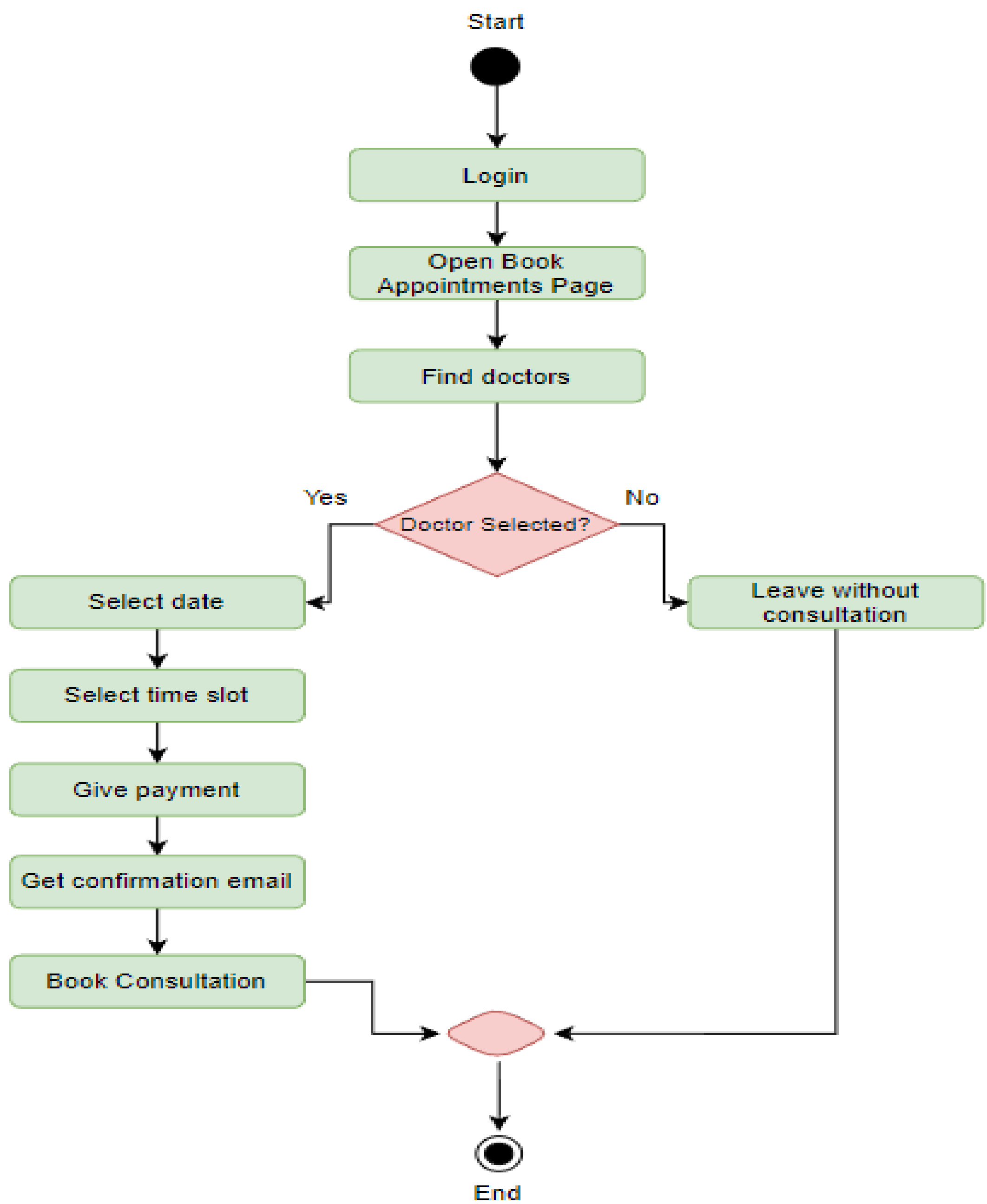


Fig: Book Appointments

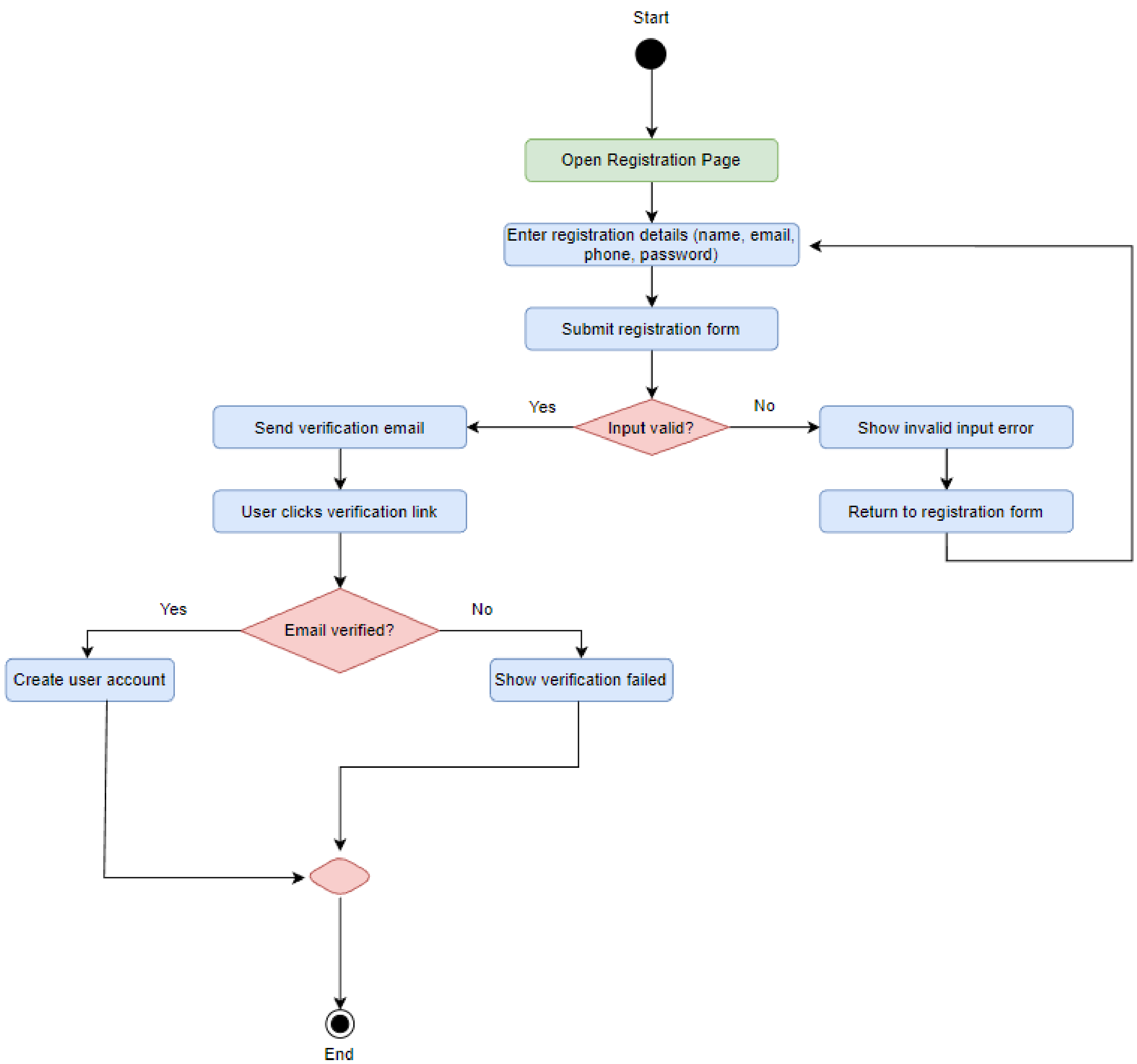


Fig: Create Account

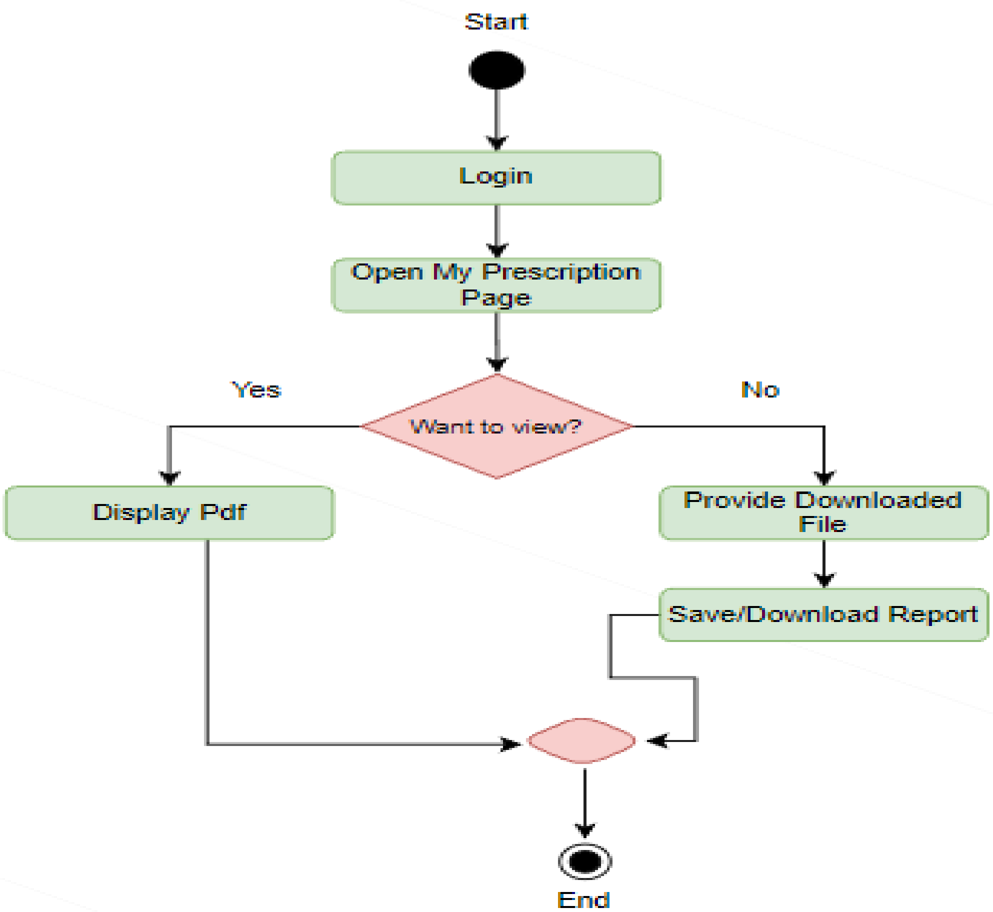


Fig: Download Prescription

Swim lane diagrams

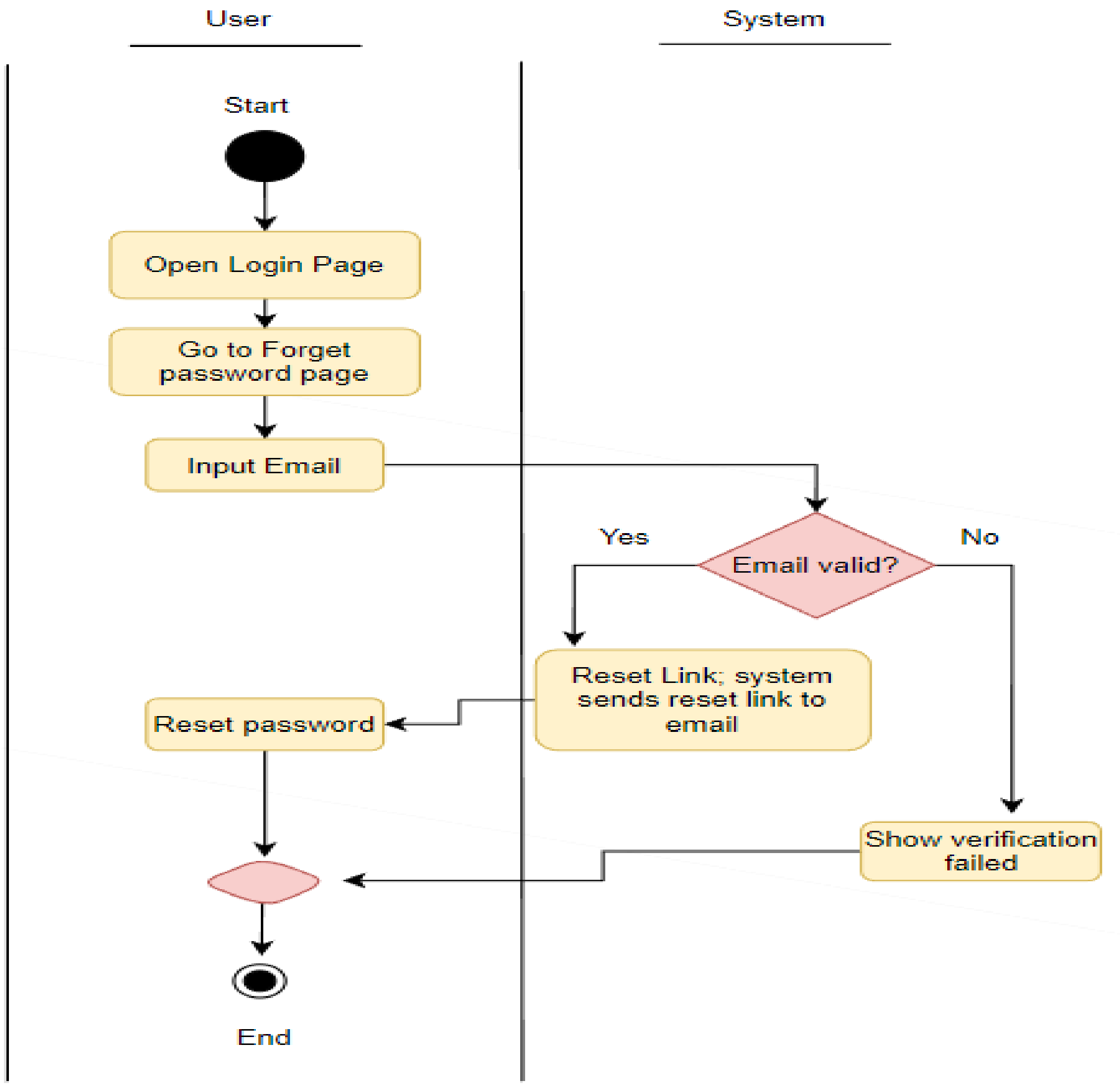


Fig: Reset Password

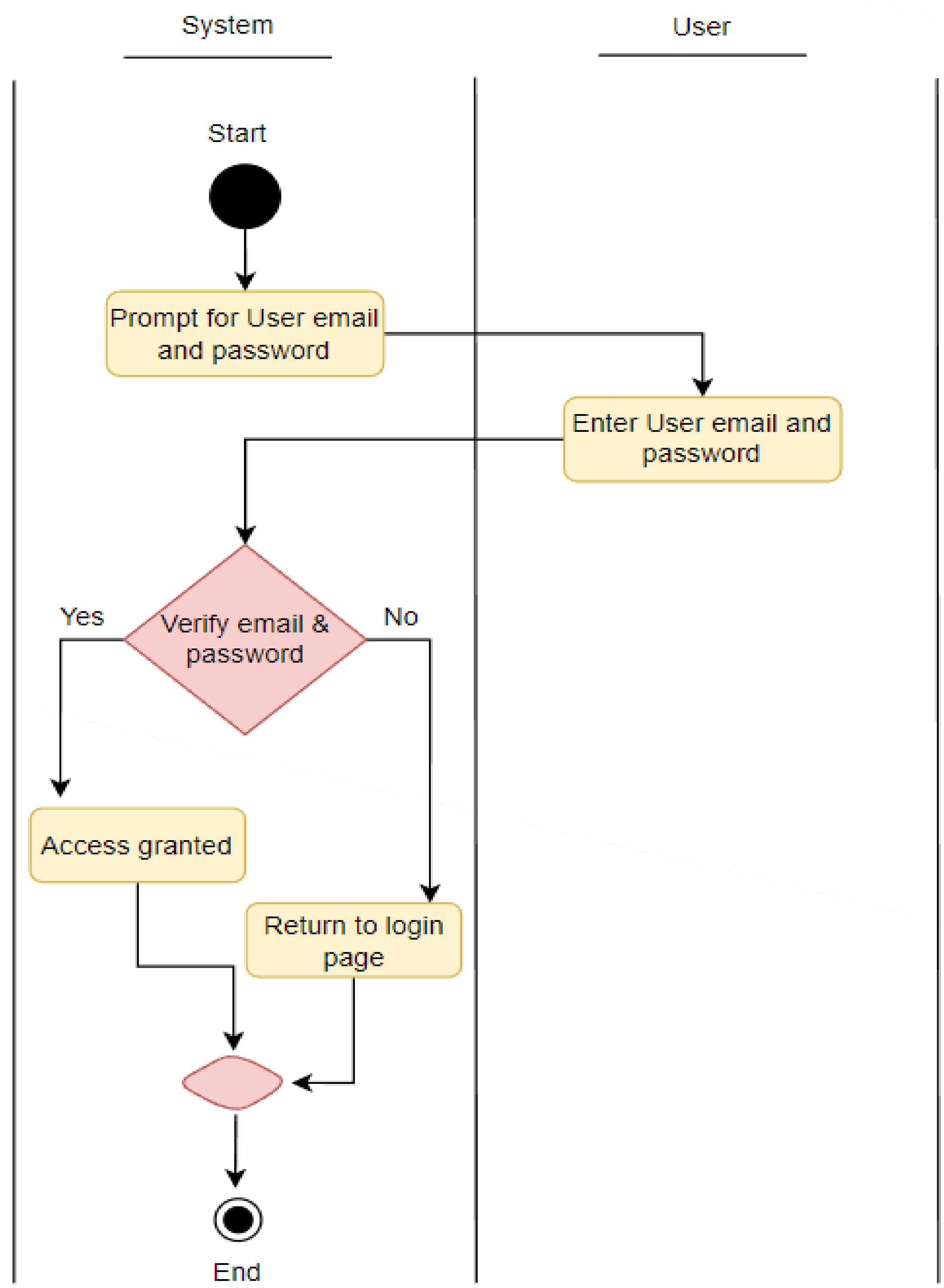


Fig: User Login

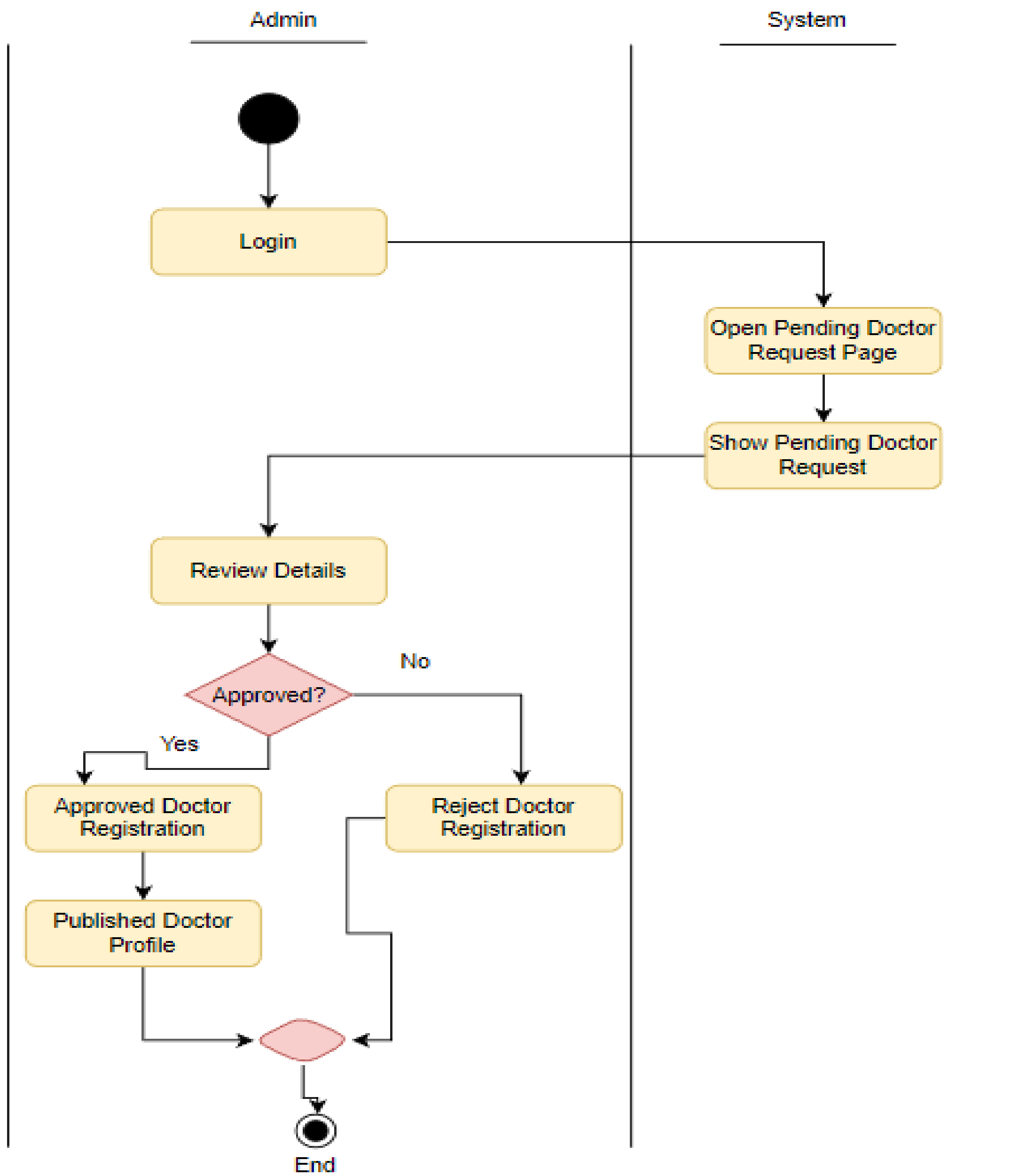


Fig: Approve Doctor

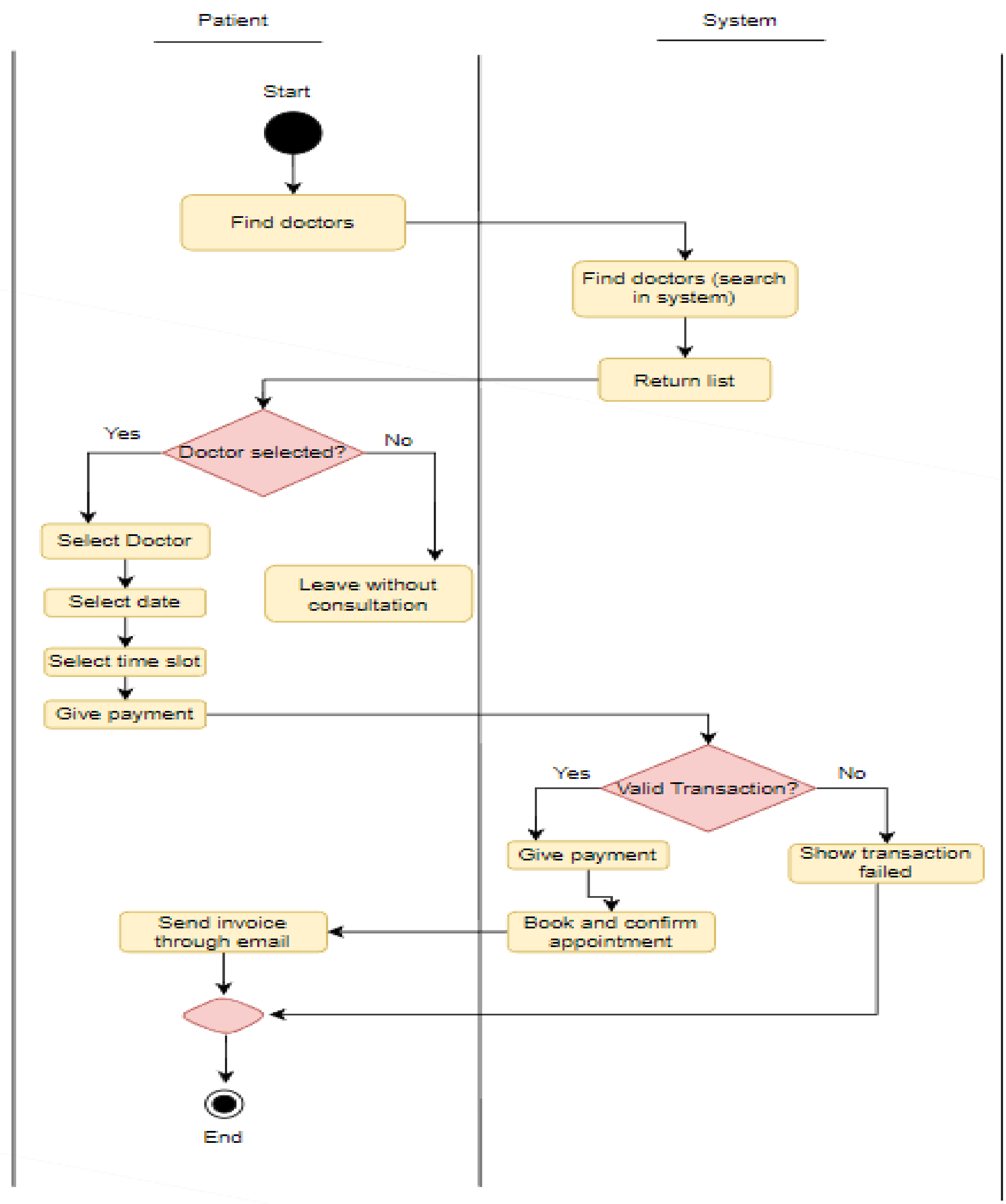


Fig: Booking Appointment

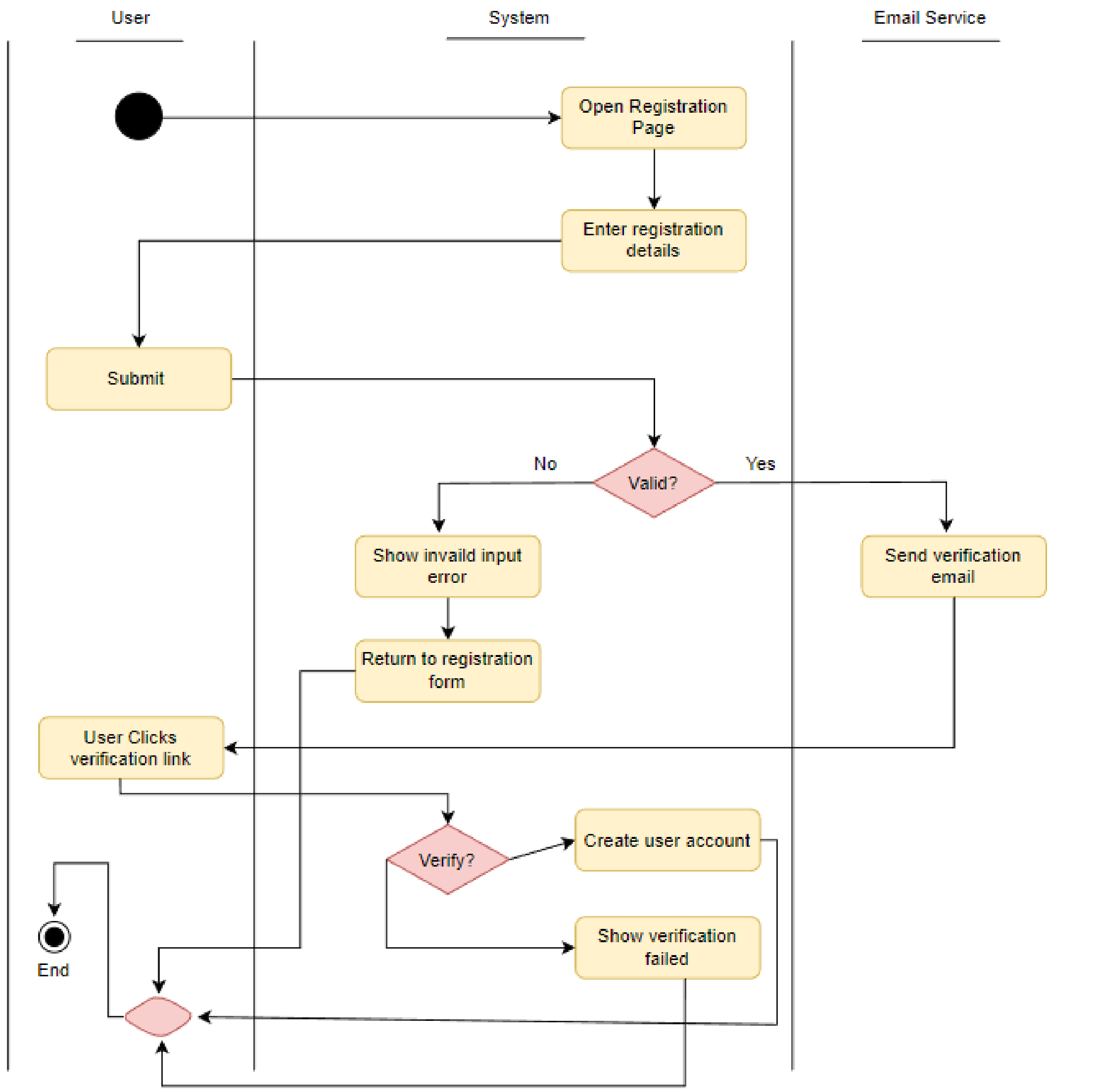


Fig: Swim lane diagram for create account

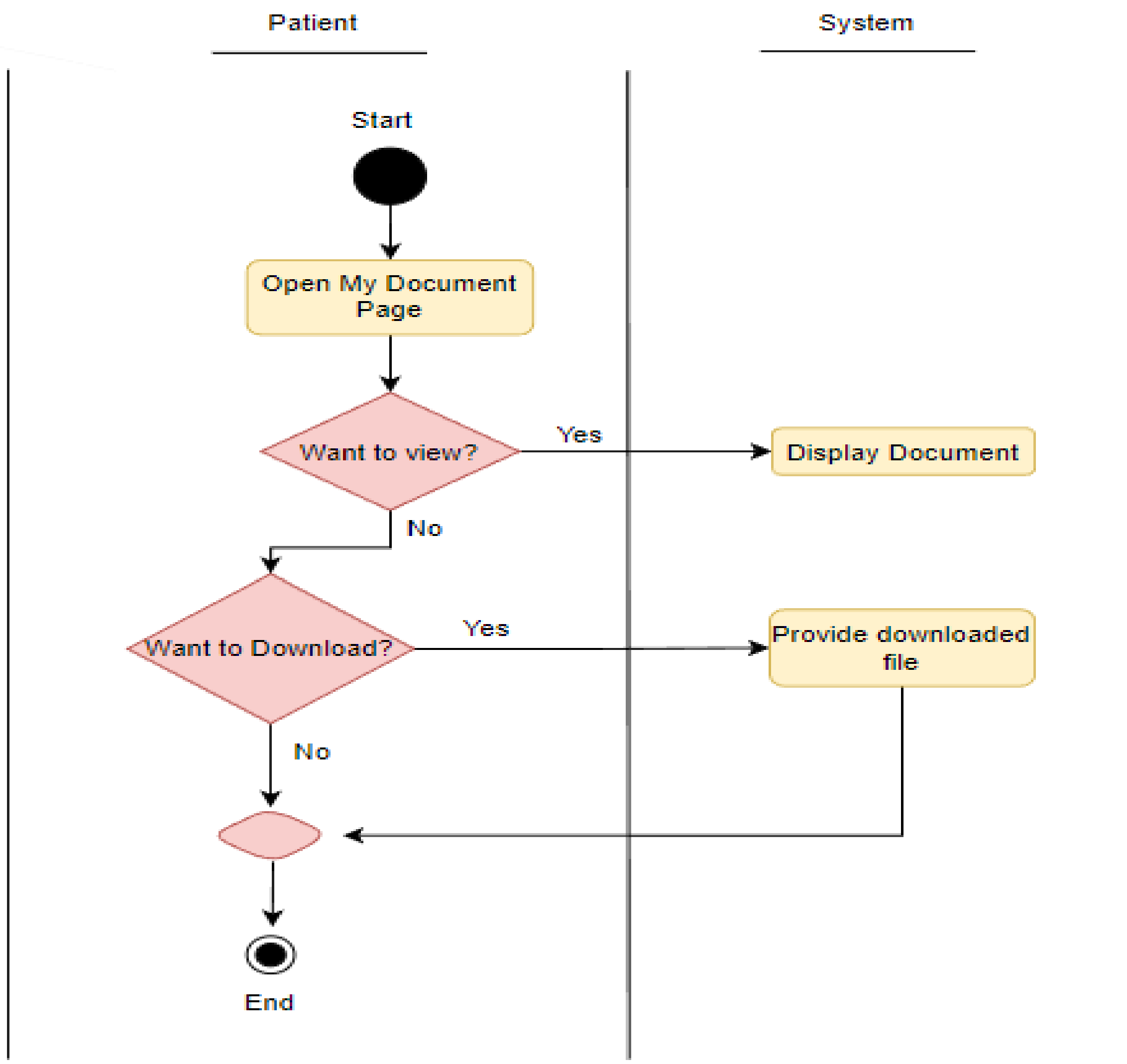
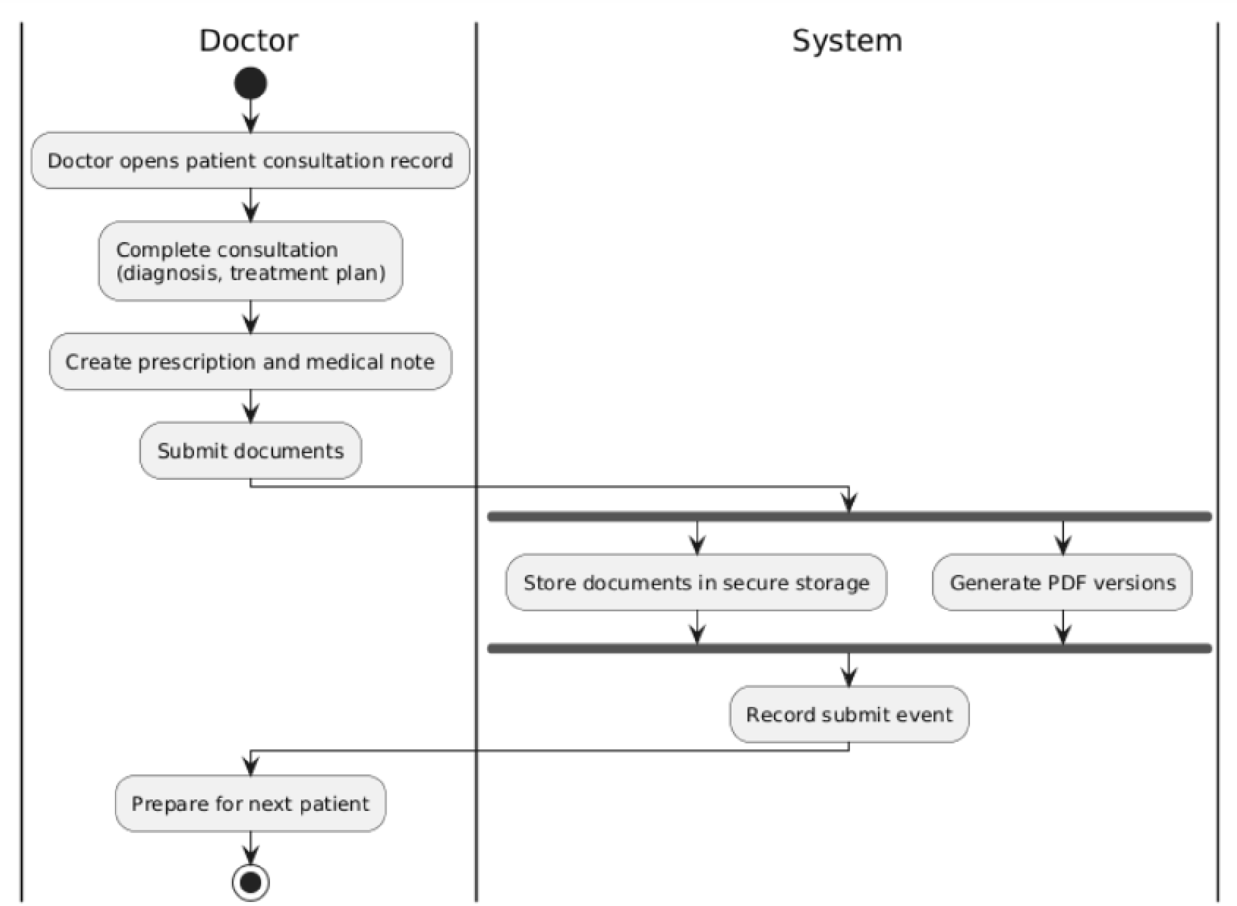


Fig: Download Prescription


Doctor
System
Doctor opens patient consultation record
Complete consultation
(diagnosis, treatment plan)
Create prescription and medical note
Submit documents
Store documents in secure storage
Generate PDF versions
Record submit event
Prepare for next patient

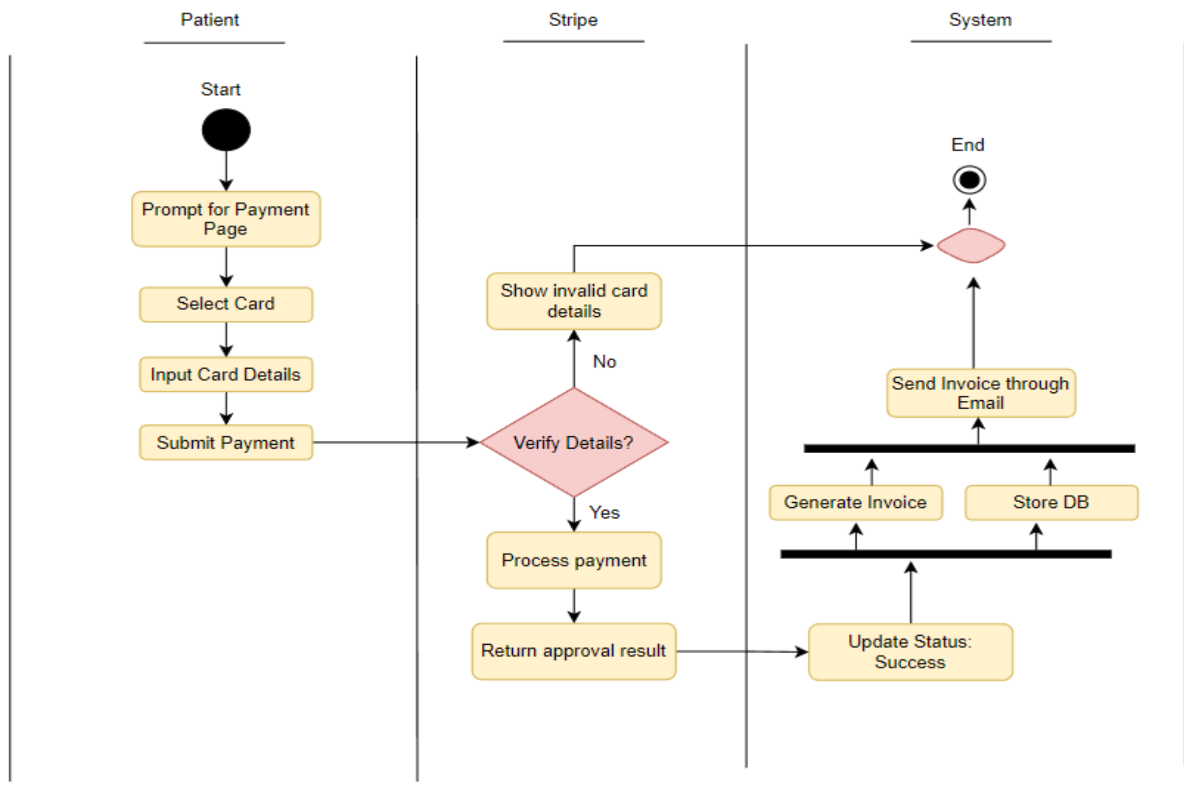
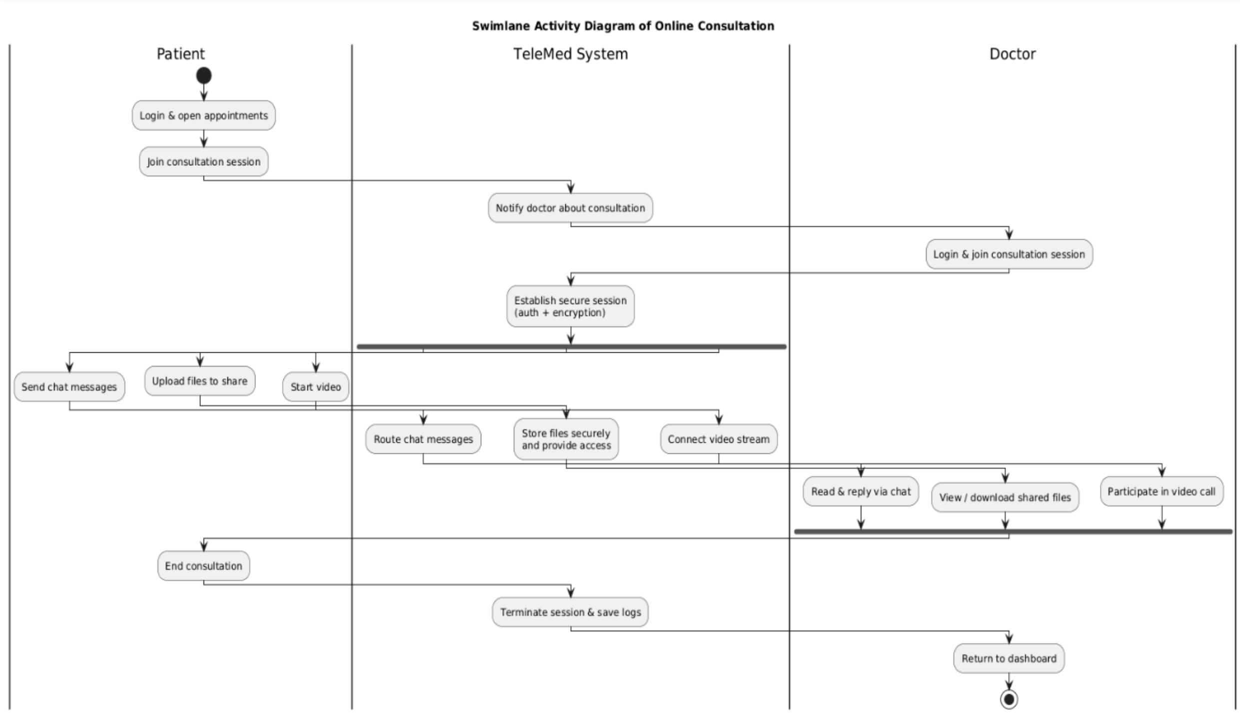
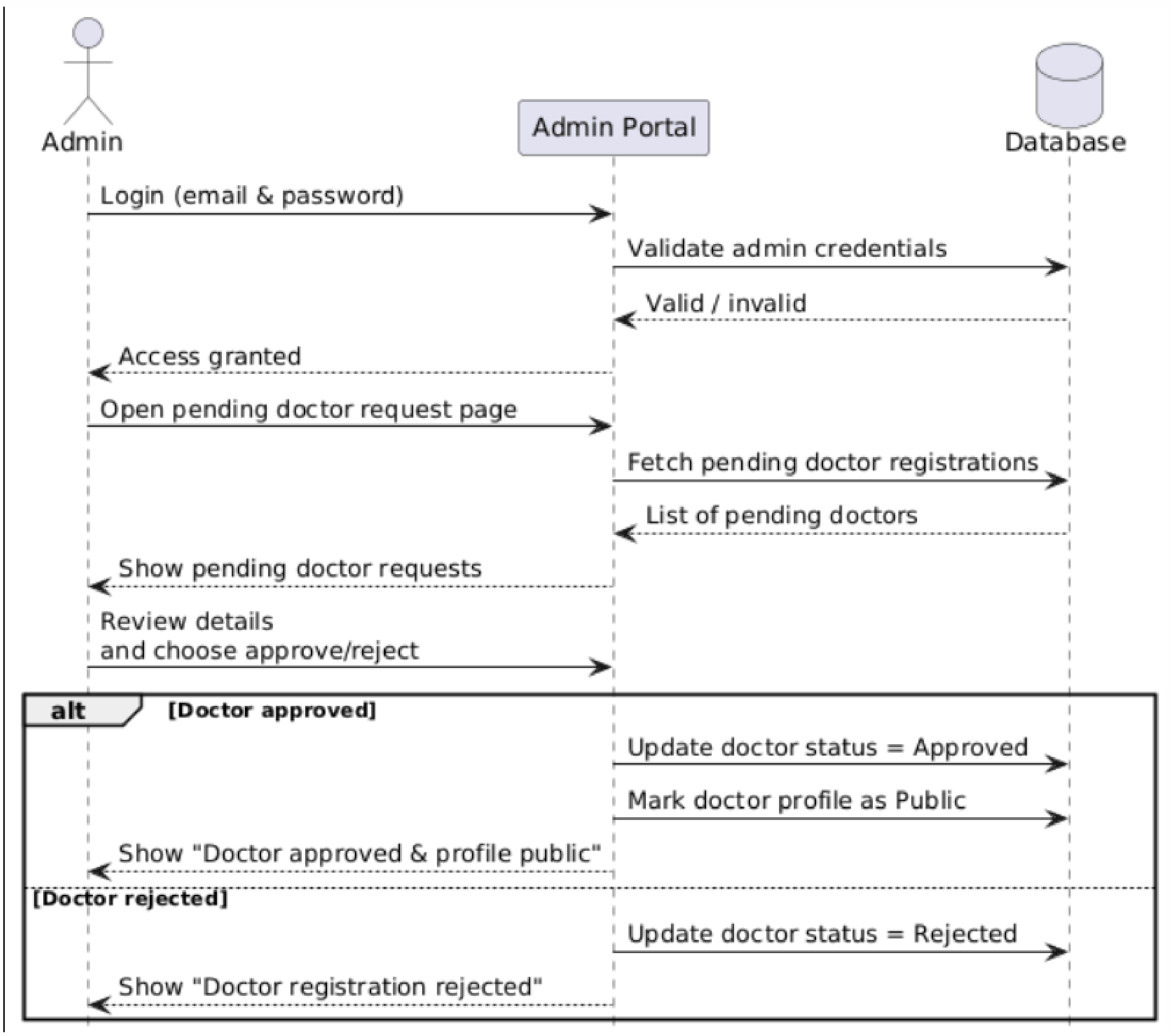
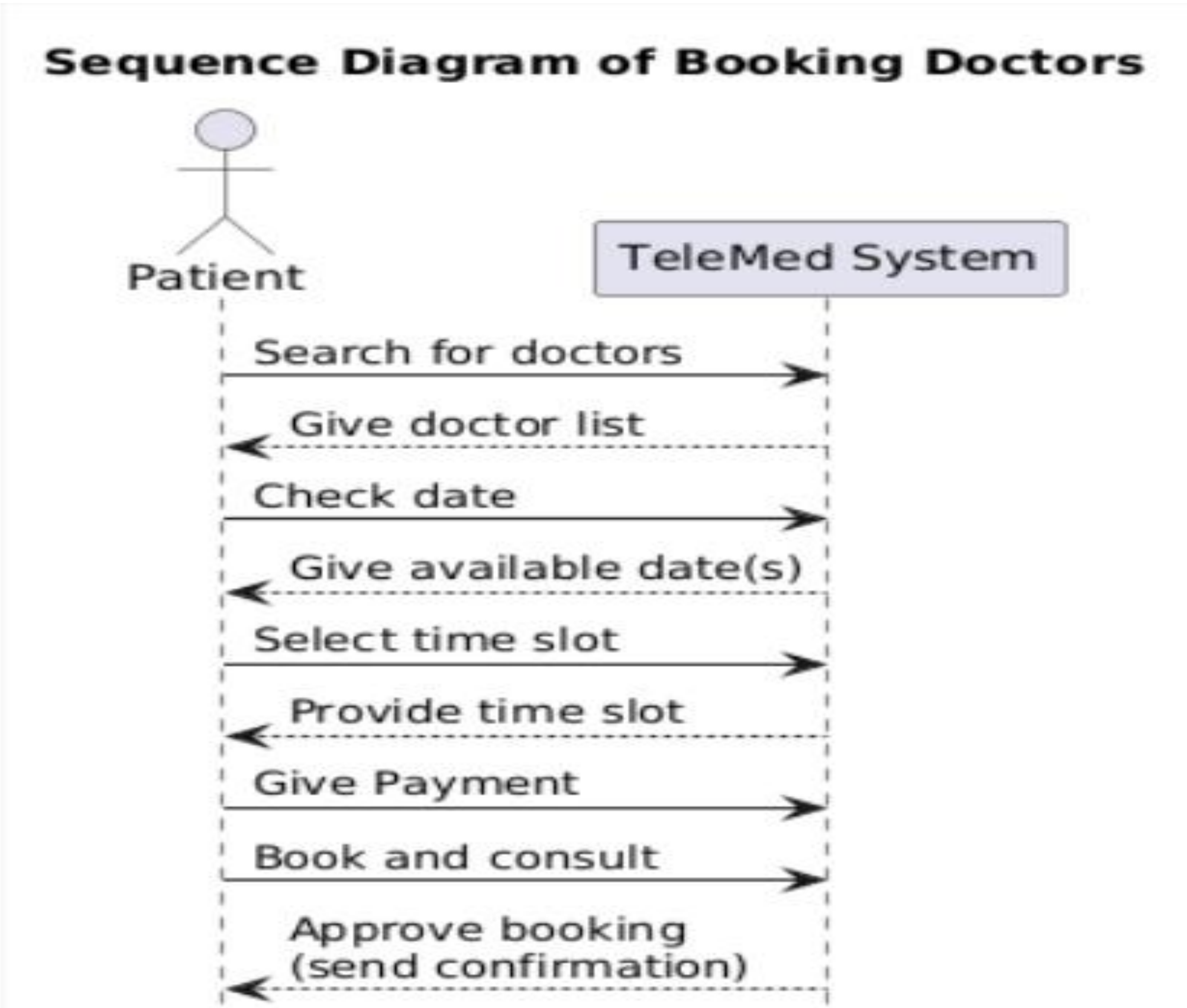

Patient
Stripe
System
Start
Prompt for Payment Page
Select Card
Input Card Details
Submit Payment
Verify Details?
No
Yes
Show invalid card details
Process payment
Return approval result
End
Send Invoice through Email
Generate Invoice
Store DB
Update Status: Success

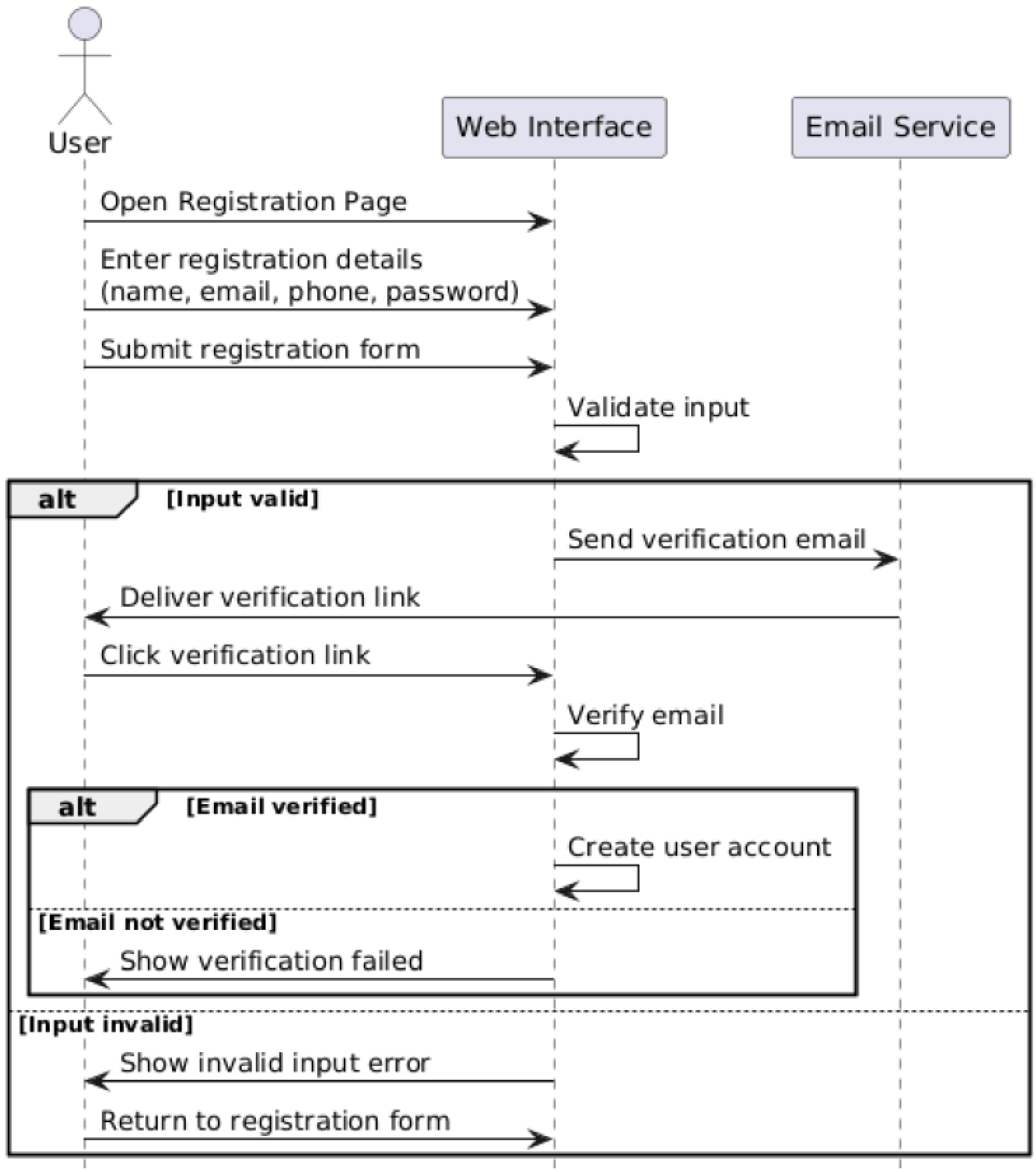
Fig: Payment


Swimlane Activity Diagram of Online Consultation
Patient
TeleMed System
Doctor
Login & open appointments
Join consultation session
Notify doctor about consultation
Login & join consultation session
Establish secure session (auth + encryption)
Send chat messages
Upload files to share
Start video
Route chat messages
Store files securely and provide access
Connect video stream
Read & reply via chat
View / download shared files
Participate in video call
End consultation
Terminate session & save logs
Return to dashboard

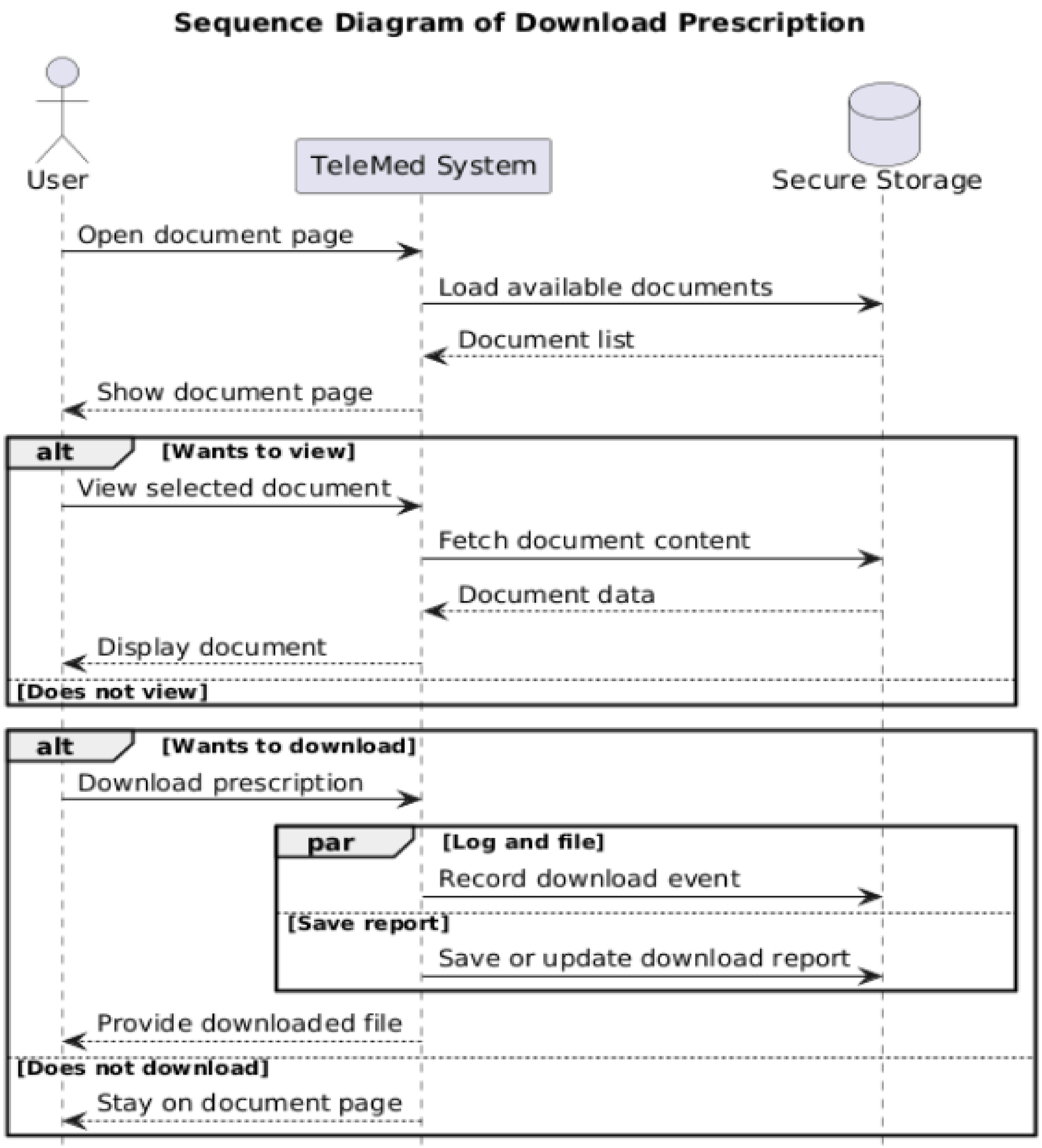
Sequence Diagram

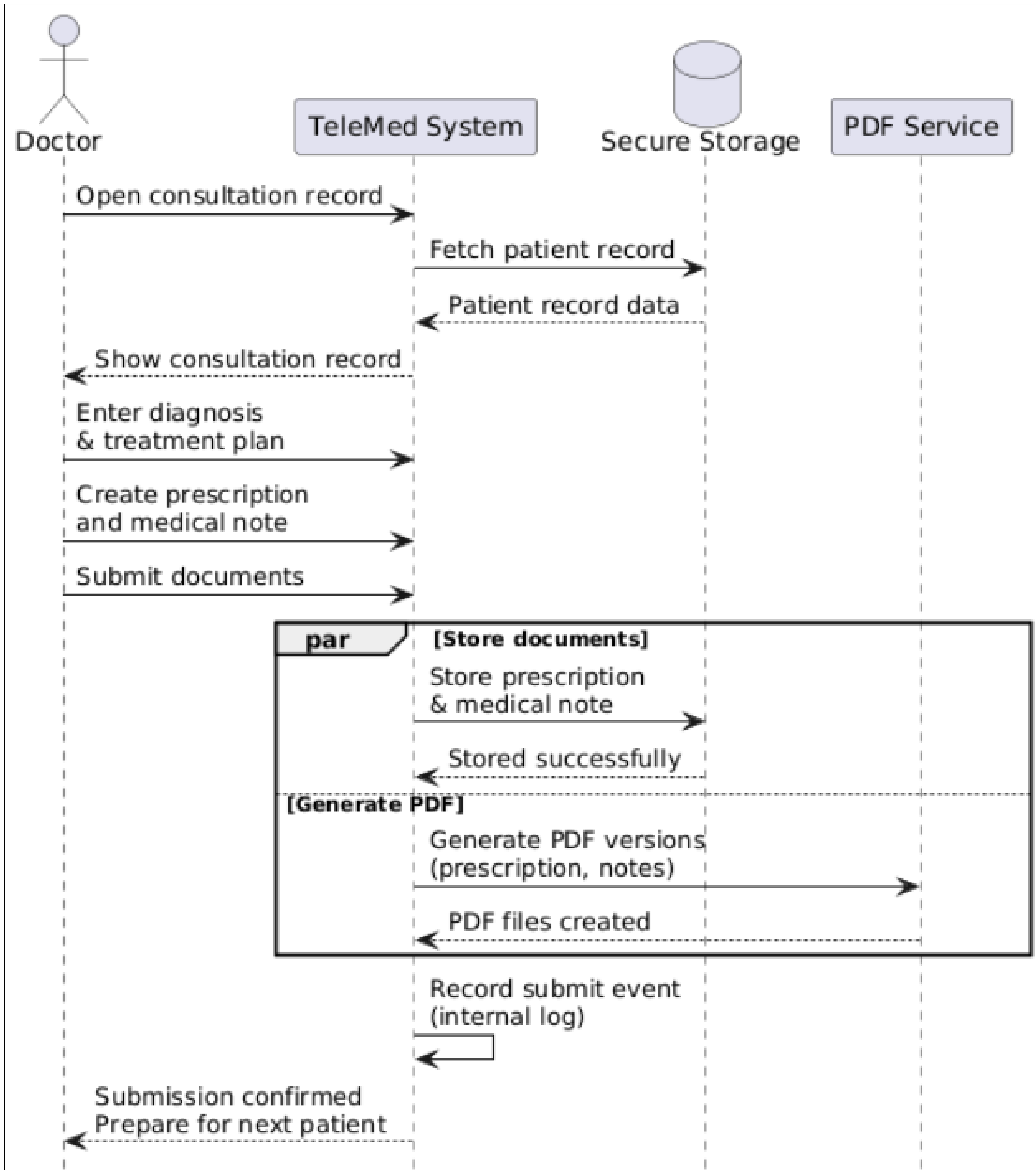


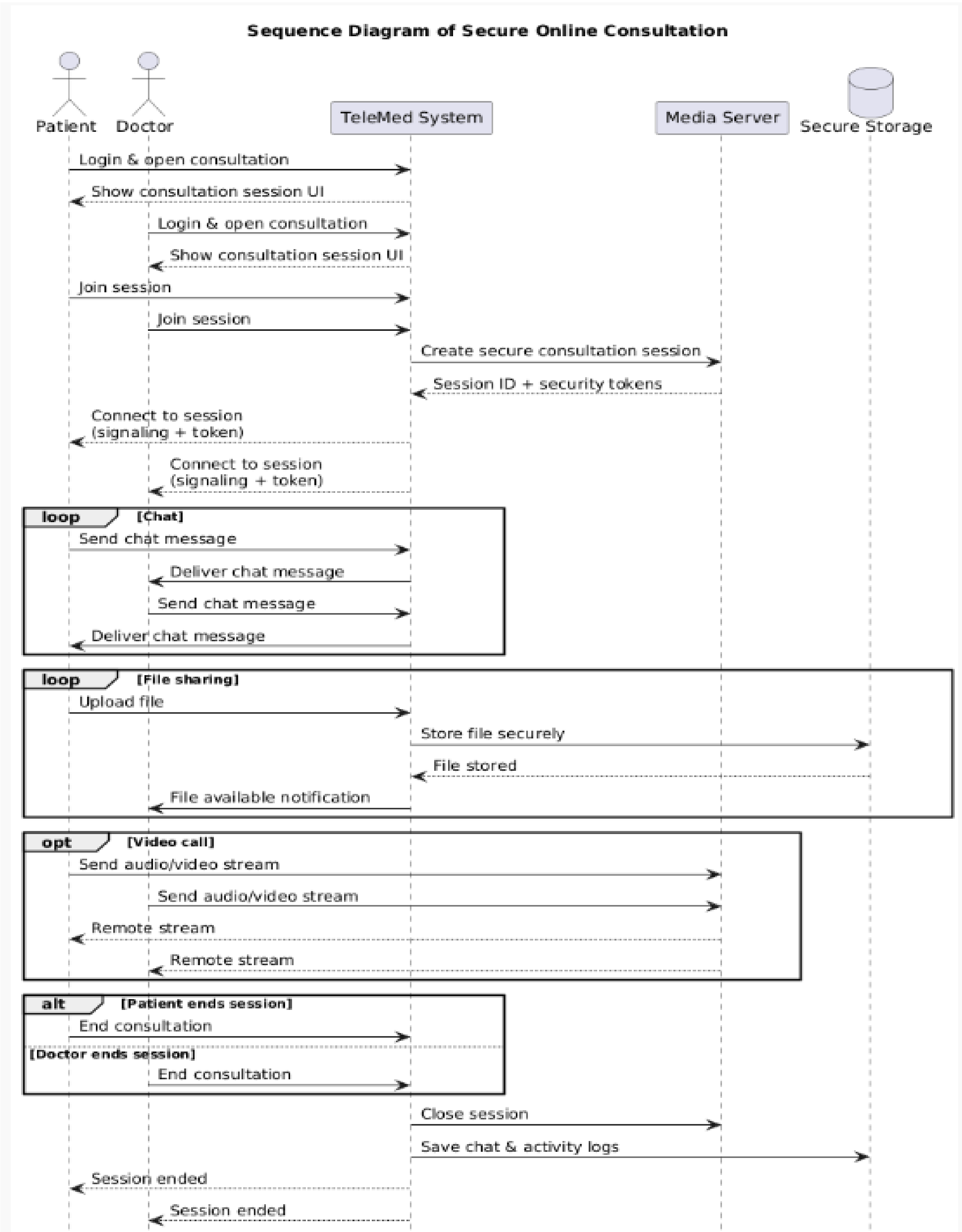

Sequence Diagram of Booking Doctors
Patient
TeleMed System
Search for doctors
Give doctor list
Check date
Give available date(s)
Select time slot
Provide time slot
Give Payment
Book and consult
Approve booking
(send confirmation)

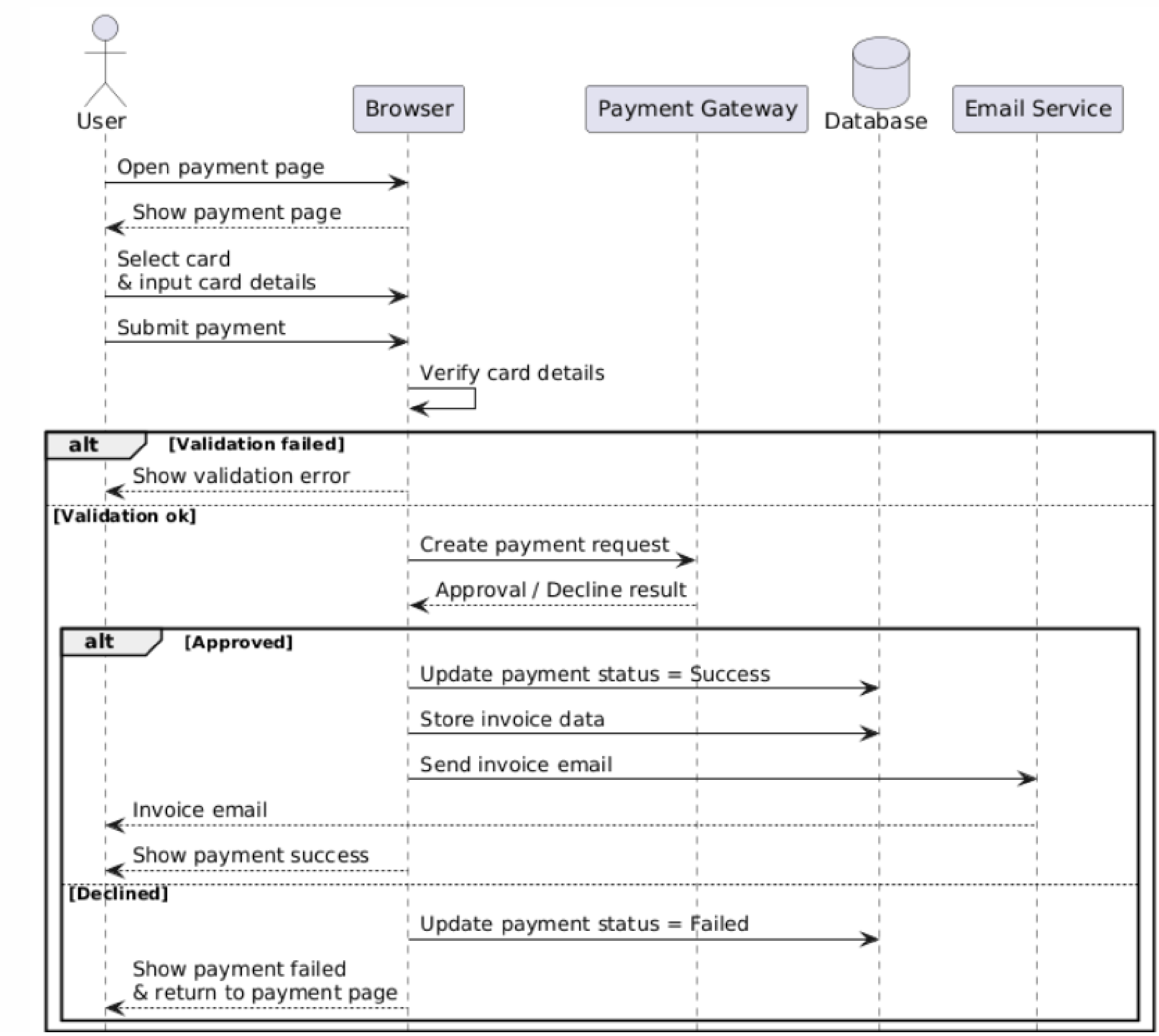


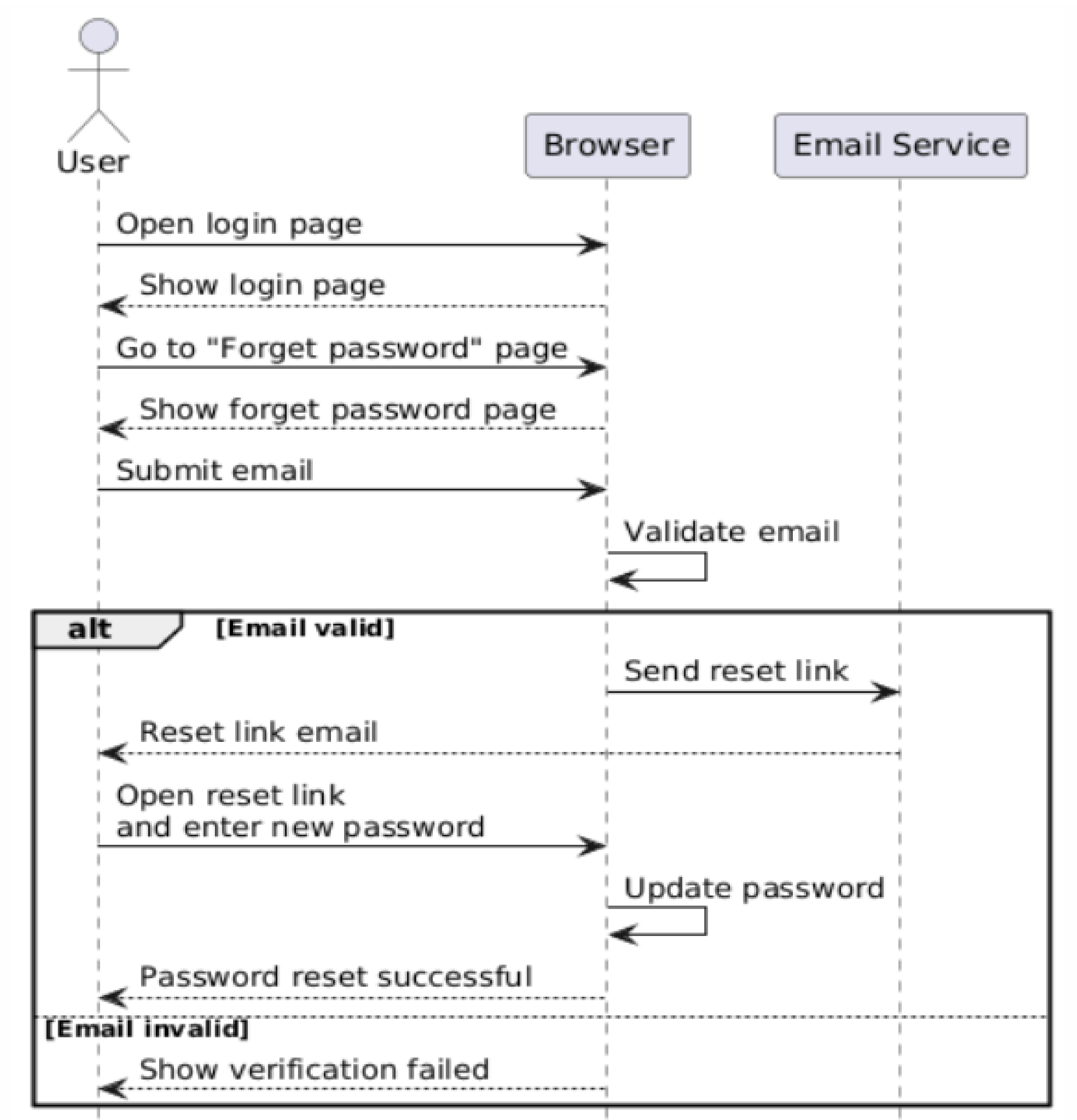
User
Web Interface
Email Service
Open Registration Page
Enter registration details
(name, email, phone, password)
Submit registration form
Validate input
alt
[Input valid]
Send verification email
Deliver verification link
Click verification link
Verify email
alt
[Email verified]
Create user account
[Email not verified]
Show verification failed
[Input invalid]
Show invalid input error
Return to registration form

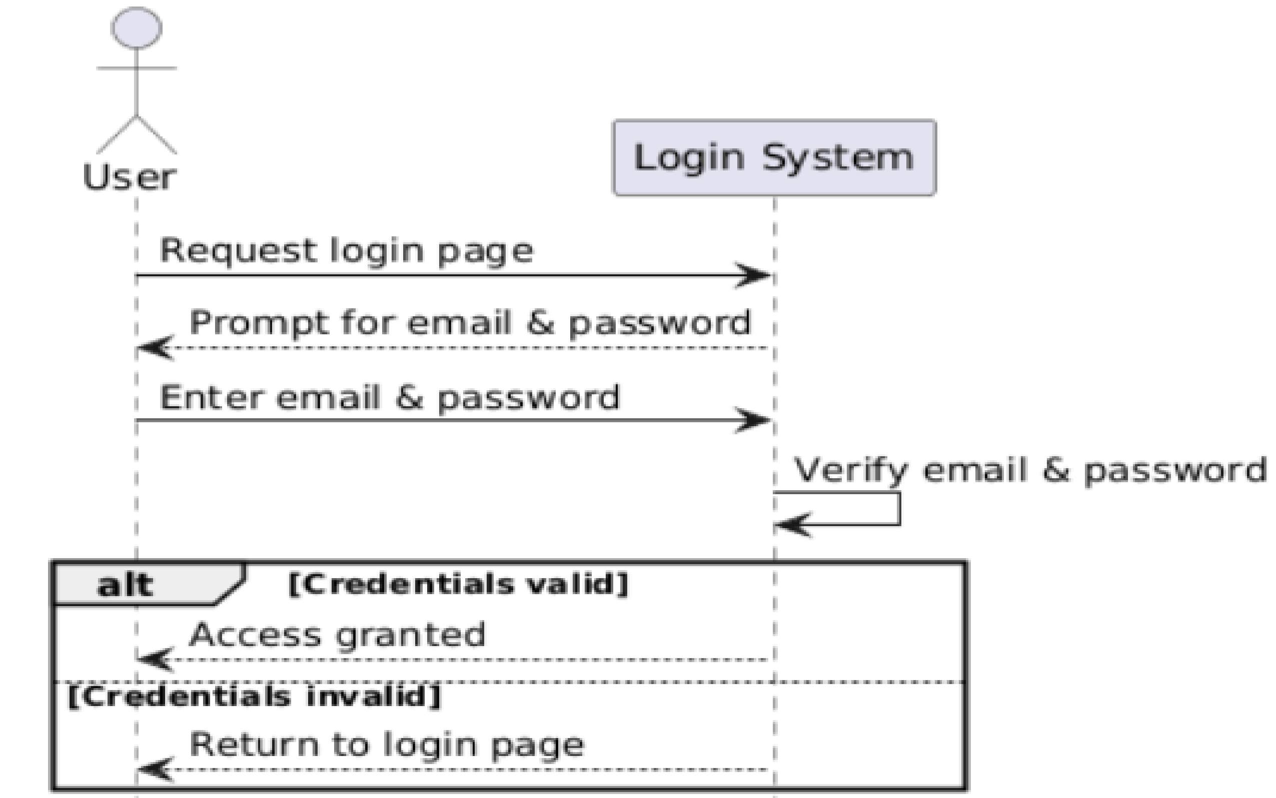

Sequence Diagram of Download Prescription
User
TeleMed System
Secure Storage
Open document page
Load available documents
Document list
Show document page
alt
[Wants to view]
View selected document
Fetch document content
Document data
Display document
[Does not view]
alt
[Wants to download]
Download prescription
par
[Log and file]
Record download event
[Save report]
Save or update download report
Provide downloaded file
[Does not download]
Stay on document page


Doctor
TeleMed System
Secure Storage
PDF Service
Open consultation record
Fetch patient record
Patient record data
Show consultation record
Enter diagnosis
& treatment plan
Create prescription
and medical note
Submit documents
par
[Store documents]
Store prescription
& medical note
Stored successfully
[Generate PDF]
Generate PDF versions
(prescription, notes)
PDF files created
Record submit event
(internal log)
Submission confirmed
Prepare for next patient


Sequence Diagram of Secure Online Consultation
Patient
Doctor
TeleMed System
Media Server
Secure Storage
Login & open consultation
Show consultation session UI
Login & open consultation
Show consultation session UI
Join session
Join session
Create secure consultation session
Session ID + security tokens
Connect to session
(signaling + token)
Connect to session
(signaling + token)
loop
[Chat]
Send chat message
Deliver chat message
Send chat message
Deliver chat message
loop
[File sharing]
Upload file
Store file securely
File stored
File available notification
opt
[Video call]
Send audio/video stream
Send audio/video stream
Remote stream
Remote stream
alt
[Patient ends session]
End consultation
[Doctor ends session]
End consultation
Close session
Save chat & activity logs
Session ended
Session ended


User
Browser
Payment Gateway
Database
Email Service
Open payment page
Show payment page
Select card
& input card details
Submit payment
Verify card details
alt
[Validation failed]
Show validation error
[Validation ok]
Create payment request
Approval / Decline result
alt
[Approved]
Update payment status = Success
Store invoice data
Send invoice email
Invoice email
Show payment success
[Declined]
Update payment status = Failed
Show payment failed
& return to payment page


User
Browser
Email Service
Open login page
Show login page
Go to "Forget password" page
Show forget password page
Submit email
Validate email
alt
[Email valid]
Send reset link
Reset link email
Open reset link
and enter new password
Update password
Password reset successful
[Email invalid]
Show verification failed


User
Login System
Request login page
Prompt for email & password
Enter email & password
Verify email & password
alt
[Credentials valid]
Access granted
[Credentials invalid]
Return to login page

CRC

Class name: Payment	
Class type: Occurrence / Event	
Characteristics: Abstract, atomic, sequential, persistent, guarded	
Responsibilities:	**Collaborators:**
Store payment amount.	Appointment
Update status.	Invoice
Trigger invoice creation	Invoice

Class name: Invoice	
Class type: Thing	
Characteristics: Abstract, aggregate, sequential, persistent, guarded	
Responsibilities:	**Collaborators:**
Store Invoice metadata	
Hold PDF path	
Provide proof of payment	Payment

<table>
<tr><td colspan="2">Class name: ConsultationSession</td></tr>
<tr><td colspan="2">Class type: Occurrence / Event</td></tr>
<tr><td colspan="2">Characteristics: Abstract, aggregate, concurrent, persistent, guarded</td></tr>
<tr><td>Responsibilities:

Start and end the session
Hold chat and file sharing data
Support video sharing metadata</td><td>Collaborators:

Appointment, Doctor, Patient
ChatMessage, FileAttachment</td></tr>
</table>

<table>
<tr><td colspan="2">Class name: ChatMessage</td></tr>
<tr><td colspan="2">Class type: Occurrence / Event</td></tr>
<tr><td colspan="2">Characteristics: Abstract, atomic, sequential, persistent, guarded</td></tr>
<tr><td>Responsibilities:

Show sender and content</td><td>Collaborators:

ConsultationSession, Doctor, Patient</td></tr>
</table>

<table>
<tr><td colspan="2">Class name: FileAttachment</td></tr>
<tr><td colspan="2">Class type: Thing</td></tr>
<tr><td colspan="2">Characteristics: Abstract, atomic, sequential, persistent, guarded</td></tr>
<tr><td>Responsibilities:

Store file metadata and path
Provide downloadable content</td><td>Collaborators:

ConsultationSession</td></tr>
</table>

<table>
<tr><td colspan="2">Class name: PrescriptionDocument</td></tr>
<tr><td colspan="2">Class type: Thing</td></tr>
<tr><td colspan="2">Characteristics: Abstract, aggregate, sequential, persistent, guarded</td></tr>
<tr><td>Responsibilities:

Store Prescription Data
Provide downloadable file path</td><td>Collaborators:

Appointment</td></tr>
</table>

Class Name: ApplicationUser	
Class type: External entity	
Characteristics: Abstract, atomic, sequential, persistent, guarded	
Responsibilities: Store user login information Maintain role association Provide contact details for notifications	**Collaborators:** Doctor Patient Admin

Class Name: Doctor	
Class type: Role	
Characteristics: Tangible, aggregate, sequential, persistent, guarded	
Responsibilities: Manage professional profile Control schedule Accept or reject appointments Conduct consultations Create prescriptions View Feedback	**Collaborators:** ApplicationUser DoctorSchedule Appointment ConsultationSession PrescriptionDocument Feedback

Class Name: Patient	
Class type: Role	
Characteristics: Tangible, aggregate, sequential, persistent, guarded	
Responsibilities:	**Collaborators:**
Browse doctors	ApplicationUser
Book and cancel appointments	Appointment
Join consultation sessions	ConsultationSession
Download prescriptions and invoices	PrescriptionDocument
Give Payment	Payment, Invoice
Give feedback	Feedback

Class Name: Admin	
Class type: Role	
Characteristics: Tangible, atomic, sequential, persistent, guarded	
Responsibilities:	**Collaborators:**
Approve or reject doctor applications	ApplicationUser, Doctor
Manage system data	
View reports	Patient, Doctor

Class Name: DoctorSchedule	
Class type: Event	
Characteristics: Abstract, aggregate, sequential, persistent, guarded	
Responsibilities: Store available time slots Validate booking availability	**Collaborators:** Doctor Appointment

Class Name: Appointment	
Class type: Occurrence / Event	
Characteristics: Abstract, aggregate, sequential, persistent, guarded	
Responsibilities:	**Collaborators:**
Track date, time, and status	DoctorSchedule
Link doctor and patient	Doctor, Patient
Maintain payment status	Payment
Link consultations, prescriptions, and feedback	ConsultationSession, PrescriptionDocument, Feedback

Class name: Feedback	
Class type: Occurrence / Event	
Characteristics: Abstract, Atomic, Concurrent, Persistent, Guarded	
Responsibilities: Show and store rating and comment Associate to appointment and doctor	**Collaborators:** Appointment, Patient Appointment, Doctor

Class diagrams

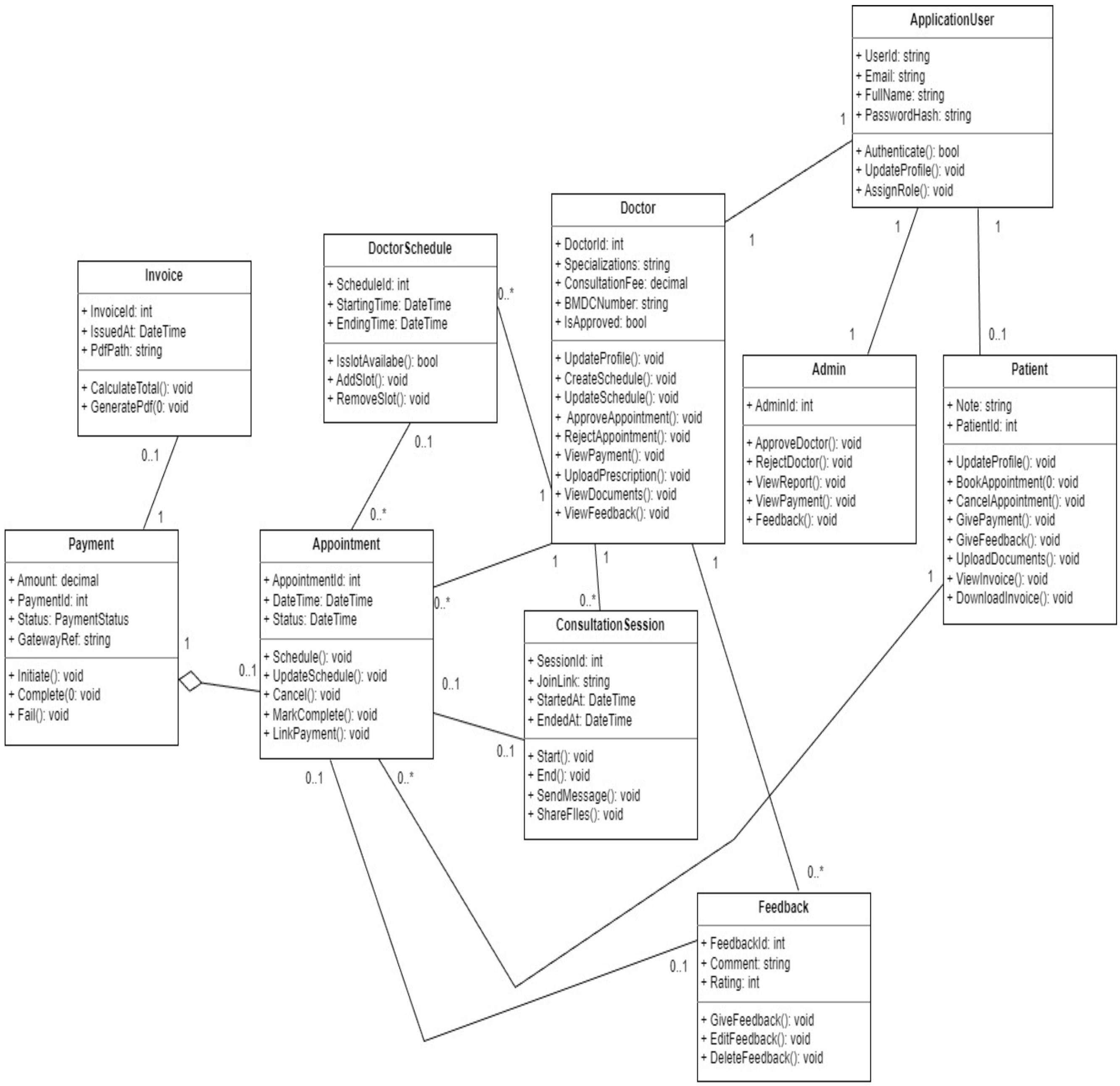


Fig:
Class Diagram

Chapter 4: Project Management

Risk identification Table

Risk Type	Possible Risks
Technology	The system database cannot process as many appointment and payment transactions as expected. (1) Video consultation and real-time chat components contain defects and fail under load. (2)
People	It is difficult to recruit team members with required technical skills. (3) Key team members may be unavailable at critical times. (4)
Organizational	Academic or organizational restructuring affects project responsibility. (5) Financial constraints force reductions in project scope or resources. (6)
Tools	Development tools generate inefficient or incompatible code. (7) Software tools cannot work together in an integrated manner. (8)
Requirements	Major requirement changes require extensive redesign. (9) Stakeholders fail to understand the impact of requirement changes. (10)
Estimation	Development time is underestimated. (11) Testing and defect repair effort is underestimated. (12)

Risk Analysis Table

Risk	Probability	Effects
The system database cannot process expected transactions. (1)	Moderate	Serious
Defects in real-time consultation components. (2)	Moderate	Serious
Inability to recruit skilled team members. (3)	High	Catastrophic
Key team members unavailable at critical times. (4)	Moderate	Serious
Organizational or academic restructuring. (5)	High	Serious
Financial constraints reduce project resources. (6)	Low	Catastrophic
Inefficient or incompatible development tools. (7)	Low	Tolerable
Tool integration failures. (8)	Moderate	Tolerable
Major requirement changes require redesign. (9)	Moderate	Serious
Stakeholders misunderstand requirement impact. (10)	Moderate	Serious
Development time underestimated. (11)	High	Serious
Testing and defect fixing underestimated. (12)	High	Serious

Risk Planning Table

Risk	Strategy
Database performance limitations	Investigate database optimization or higher-performance database solutions.
Defective real-time components	Replace unreliable components with stable and well-tested libraries.
Recruitment problems	Notify stakeholders of potential delays and investigate third-party solutions.
Staff unavailability	Reorganize the team so responsibilities overlap and knowledge is shared.
Organizational restructuring	Clearly communicate project value to new academic or organizational management.
Financial constraints	Prepare justification documents showing why budget cuts are not cost-effective.
Inefficient development tools	Evaluate alternative tools or stabilize tool versions.
Tool integration failures	Simplify architecture and limit tool dependencies.
Requirement changes	Maintain requirement traceability and minimize design dependencies.
Misunderstanding requirement impact	Formally document and review all requirement changes with stakeholders.
Underestimated development time	Add schedule buffer and reuse existing components.
Underestimated testing effort	Allocate additional time and prioritize critical test cases.

Chapter 5: Project Planning and Scheduling

Function Point Estimation

Sl	Functionality	Input	Output
1	Patient creates account	Name, Email, Phone, Age, IdType, Nid/Passport, RegisterAs, DOB, Gender, Password, ConfirmPassword	Account created message or error
2	User logs in	Email, Password	My Profile or invalid login message
3	Reset password	Email, ConfirmEmail, New Password, ConfirmPassword	Password reset success or failure
4	Patient updates profile	Personal info, Photo upload, Change Password	Profile updated confirmation
5	Doctor registers	Consultation Fee, Short bio/clinic address, BM&DC Registration Number, IdType, Nid/Passport, RegisterAs, Specialization, Qualification, Password, Con	Submitted message, status pending approval
6	Admin approves doctor	Approve/Reject button	Approval success or rejection message
7	Search doctor	Search text, Specialization, Name	Doctor list results
8	Book appointment	Doctor, Date, Available Time Slots, consultation fee, Additional Notes	Appointment confirmation or slot unavailable
9	Cancel appointment	Cancel button, Cancellation reason	Cancellation confirmation
10	Join online consultation	Join button or session link	Video/chat consultation screen opens
11	Doctor gives prescription	Medicines, Notes	Prescription saved and downloadable
12	Patient views/download prescription	Prescription view/download button	Prescription displayed or downloaded
13	Make payment	Card/wallet details, amount	Payment success or error message and receipt
14	Generate admin report	View report button, Date range	Report displayed
15	Backup database	Automatically backup in 24hours	Backup completed confirmation
16	Logout	Logout button	Redirected to login screen

Sl	Transection Functions	TF	Fields/File involvement	FTR	DET
1	Patient creates account	EI	Fields: Name, Email, Phone, Age, IdType, NID/Passport, RegisterAs, DOB, Gender, Password, ConfirmPassword. Files: UserTable, PatientTable	2	11
2	User logs in	EI	Fields: Email, Password. File: UserTable	1	2
3	Reset password	EI	Fields: Email, ConfirmEmail, New Password, ConfirmPassword. File: UserTable	1	4
4	Patient updates profile	EI	Fields: Personal info, photo upload, change password. File: PatientTable	1	3
5	Doctor registers	EI	ields: Name, Email, Consultation Fee, Bio/Address, BM&DC Reg No, IdType, NID/Passport, RegisterAs, Specialization, Qualification, Password, ConfirmPassword. Files: DoctorTable, DocumentTab	2	12
6	Admin approves doctor	EI	Fields: Approve/Reject button. File: DoctorTable	1	1
7	Search doctor	EQ	Fields: Search text, Specialization, Name. File: DoctorTable	1	3
8	Book appointment	EI	Fields: Doctor, Date, Available Time Slots, Consultation fee, Notes. File: AppointmentTable	1	5
9	Cancel appointment	EI	Fields: Cancel button, Cancellation reason. File: AppointmentTable	1	2
10	Join online consultation	EO	Fields: Join button or session link. File: SessionTable	1	2
11	Doctor gives prescription	EI	Fields: Medicines, Notes. Files: PrescriptionTable, PatientTable	2	2
12	Patient views/download prescription	2xEQ	Fields: View/download button. File: PrescriptionTable	1	1
13	Make payment	EI	Fields: Card/Wallet details, amount. File: PaymentTable	1	2
14	Generate admin report	EO	Fields: View report button, Date range. Files: ReportTable, AppointmentTable, PaymentTable	3	2
15	Backup database	EO	Fields: Automatically backup trigger. File: BackupTable	1	1
16	Logout	EI	Fields: Logout button. File: Session/UserSession	1	1

Sl	Data Functions	Fields/File involvement	RET	DET
1	UserTable (ILF)	UserID, Name, Email, Phone, Role, PasswordHash, Status	1	7
2	PatientTable (ILF)	PatientID, UserID, Age, Gender, Address, Medical info	1	6
3	DoctorTable (ILF)	DoctorID, UserID, Specialization, Fee, Status, Rating	1	6
4	DocumentTable (ILF)	DocumentID, DoctorID, FileName, Path, Type, Verified	1	6
5	AppointmentTable (ILF)	AppointmentID, DoctorID, PatientID, Date, Slot, Type, Status	3	7
6	PrescriptionTable (ILF)	PrescriptionID, PatientID, DoctorID, Medicines, Notes	3	5
7	PaymentTable (ILF)	PaymentID, AppointmentID, Amount, Method, Status	2	5
8	SessionTable (ILF)	SessionID, AppointmentID, Link, Status, StartTime, EndTime	2	6
9	ReportTable (ILF)	ReportID, Type, Filters, GeneratedDate	1	4
10	PaymentGatewayData (EIF)	ExternalTransactionRef, GatewayStatus, Amount	1	3
11	VideoServiceData (EIF)	ExternalSessionID, Provider, ConnectionStatus	1	3

Unadjusted Funtion Point Contribution

	Transection Functions	TF	FTR	DET	Complexity	UFP	Total UFP
1	Patient creates account	EI	2	11	Average	4	
2	User logs in	EI	1	2	Low	3	
3	Reset password	EI	1	4	Low	3	
4	Patient updates profile	EI	1	3	Low	3	
5	Doctor registers	EI	2	12	Average	4	
6	Admin approves doctor	EI	1	1	Low	3	
7	Search doctor	EQ	1	3	Low	3	
8	Book appointment	EI	1	5	Low	3	
9	Cancel appointment	EI	1	2	Low	3	
10	Join online consultation	EO	1	2	Low	4	
11	Doctor gives prescription	EI	2	2	Low	3	
12	Patient views/download prescription	2xEQ	1	1	Low	6	
13	Make payment	EI	1	2	Low	3	
14	Generate admin report	EO	3	2	Low	4	
15	Backup database	EO	1	1	Low	4	
16	Logout	EI	1	1	Low	3	
							56

EI Table

FTR's	DATA ELEMENTS		
	1-4	5-15	> 15
0-1	Low	Low	Ave
2	Low	Ave	High
3 or more	Ave	High	High

Shared EO and EQ Table

FTR's	DATA ELEMENTS		
	1-5	6-19	> 19
0-1	Low	Low	Ave
2-3	Low	Ave	High
> 3	Ave	High	High

Values for transactions

Rating	VALUES		
	EO	EQ	EI
Low	4	3	3
Average	5	4	4
High	7	6	6

	Unadjusted Funtion Point Contribution						
	DataFunctions		RET	DET	Complexity	UFP	Total UFP
1	UserTable (ILF)	ILF	1	7	Low	7	
2	PatientTable (ILF)	ILF	1	6	Low	7	
3	DoctorTable (ILF)	ILF	1	6	Low	7	
4	DocumentTable (ILF)	ILF	1	6	Low	7	
5	AppointmentTable (ILF)	ILF	3	7	Low	7	
6	PrescriptionTable (ILF)	ILF	3	5	Low	7	
7	PaymentTable (ILF)	ILF	2	5	Low	7	
8	SessionTable (ILF)	ILF	2	6	Low	7	
9	ReportTable (ILF)	ILF	1	4	Low	7	
10	PaymentGatewayData (EIF)	EIF	1	3	Low	5	
11	VideoServiceData (EIF)	EIF	1	3	Low	5	
							73

RET's	DATA ELEMENTS		
	1-19	20 - 50	> 50
1	Low	Low	Ave
2-5	Low	Ave	High
> 5	Ave	High	High

Rating	Values	
	ILF	EIF
Low	7	5
Average	10	7
High	15	10

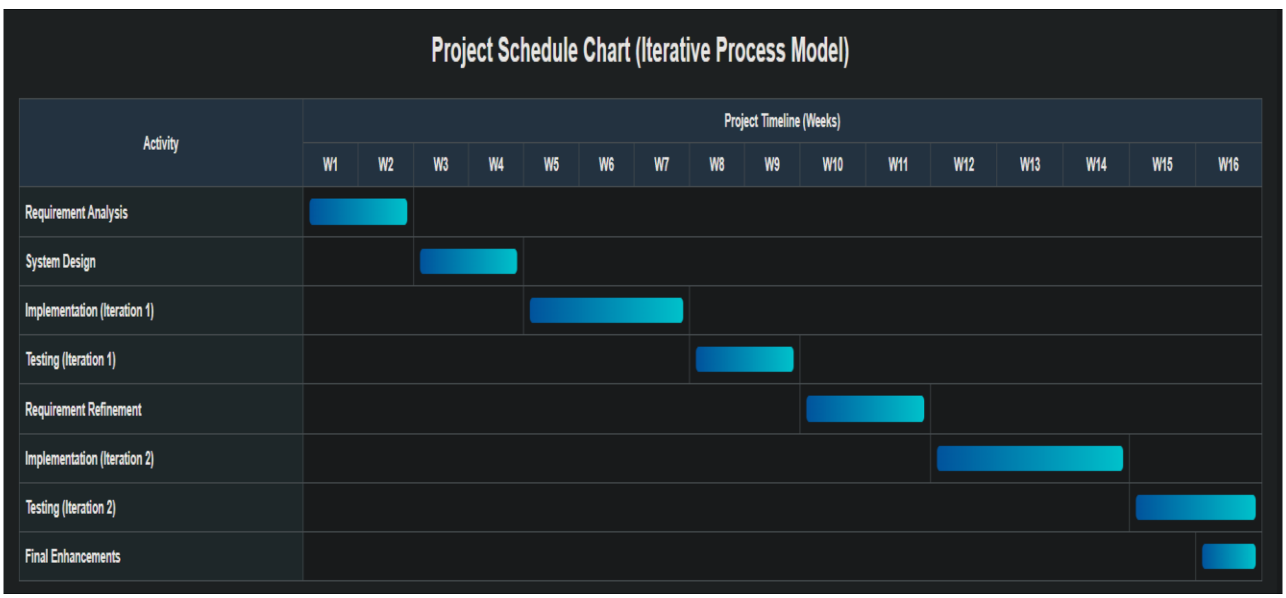
General System Characteristic		Brief Description	DI	
1	Data communications	How many communication facilities are there to aid in the transfer or exchange of information with the application or system?	5	
2	Distributed data processing	How are distributed data and processing functions handled?	0	
3	Performance	Was response time or throughput required by the user?	3	
4	Heavily used configuration	How heavily used is the current hardware platform where the application will be executed?	5	
5	Transaction rate	How frequently are transactions executed daily, weekly, monthly, etc.?		
6	On-Line data entry	What percentage of the information is entered On-Line?	5	
7	End-user efficiency	Was the application designed for end-user efficiency?	0	
8	On-Line update	How many ILF's are updated by On-Line transaction?	0	
9	Complex processing	Does the application have extensive logical or mathematical processing?	5	
10	Reusability	Was the application developed to meet one or many user's needs?	1	
11	Installation ease	How difficult is conversion and installation?	3	
12	Operational ease	How effective and/or automated are start-up, back-up, and recovery procedures?	3	
13	Multiple sites	Was the application specifically designed, developed, and supported to be installed at multiple sites for multiple organizations?	3	
14	Facilitate change	Was the application specifically designed, developed, and supported to facilitate change?	5	
		Total Degree of Influence (TDI)	38	30 to 40
		Value Adjusrment Factor (VAF) =	.65+(.01xTDI)	
			1.03	

UFP =	UFP (TF) + UFP (DF)
UFP(Total) =	129

Adjusted Function Point Estimation (AFP) =	UFP (Total) * TDI
Adjusted Function Point Estimation (AFP) =	132.87

Effort for ASP =	ASP * Productivit	ASP	
Effort for ASP =	132.87	* 06.1	
Effort for ASP =	810.507		Person Hours
Effort for ASP =	101.313375		Person Days (8 Hours Per Day)
Effort for ASP =	4.404929348		Man Months (23 Days Per Month)
Effort for ASP =	1.101232337		Months (4 person for developing the project)

Project Schedule Chart



Chapter 6: Project Estimation

1. Introduction

Process based estimation is a structured software estimation technique in which the overall project effort is divided into clearly defined software development processes. Instead of estimating the project as a single unit, each process is analyzed individually, and its required effort, time, and cost are estimated separately. These individual estimates are then combined to calculate the total project cost and schedule. This approach improves estimation accuracy and provides better insight into how resources are distributed across different development activities.

In this project, the Iterative Process Model is followed. Development is carried out through multiple cycles, where each iteration includes analysis, design, implementation, and testing activities. This model supports incremental delivery of features, early validation of requirements, and continuous feedback from previous iterations. By applying process-based estimation within an iterative framework, the project benefits from better planning, improved risk management, and more realistic estimation of effort and cost.

2. Identification of Software Processes

To apply process-based estimation effectively, the Telemedicine Consultation Booking System was divided into a set of major software development processes. Each process represents a distinct phase of work that contributes to the overall system development. Breaking the project into these processes makes it easier to estimate effort, monitor progress, and control quality throughout the development lifecycle.

The major software processes identified for this project are:

1. **Requirement Analysis and Planning**

This process involves gathering functional and non-functional requirements, understanding user needs, defining system scope, and planning project activities. Clear requirement analysis reduces ambiguity and prevents costly changes later in development.

2. **System Design**

   System design focuses on defining the overall architecture, database structure, and module interactions. Design decisions made in this stage influence system performance, security, and maintainability.

3. **Implementation (Coding)**

   During implementation, system features are developed according to the design specifications. This includes backend logic, frontend interfaces, database integration, and security mechanisms.

4. **Testing and Debugging**

   Testing is performed to verify that the system functions correctly and meets specified requirements. Bugs and defects identified during testing are corrected to improve system reliability and stability.

5. **Deployment and Documentation**

   This process covers preparing the system for final use and creating technical and user documentation. Proper documentation ensures that the system can be understood, maintained, and extended in the future.

6. **Project Management and Review**

   Project management activities include progress tracking, iteration planning, review meetings, and risk monitoring. Regular reviews help ensure that the project stays on schedule and aligned with its objectives.

By identifying and estimating these processes separately, the project achieves a clearer understanding of effort distribution and supports more accurate and manageable cost estimation.

3. Effort Estimation (Person-Month)

- Team members = 4
- Project duration = 4 months

Total Effort = 4 × 4 = 16 Person-Months (PM)

4. Effort Distribution (Process Based)

Process	**Percentage**	**Effort (PM)**
Requirement Analysis & Planning	15%	2.4
System Design	20%	3.2
Implementation (Coding)	35%	5.6
Testing & debugging	20%	3.2
Deployment & Documentation	5%	0.8
Project Management & Review	5%	0.8
Total	**100%**	**16 PM**

5. Pie Chart – Effort Distribution

- 35% Implementation
- 20% System Design
- 20% Testing
- 15% Requirement Analysis
- 5% Deployment & Documentation
- 5% Project Management

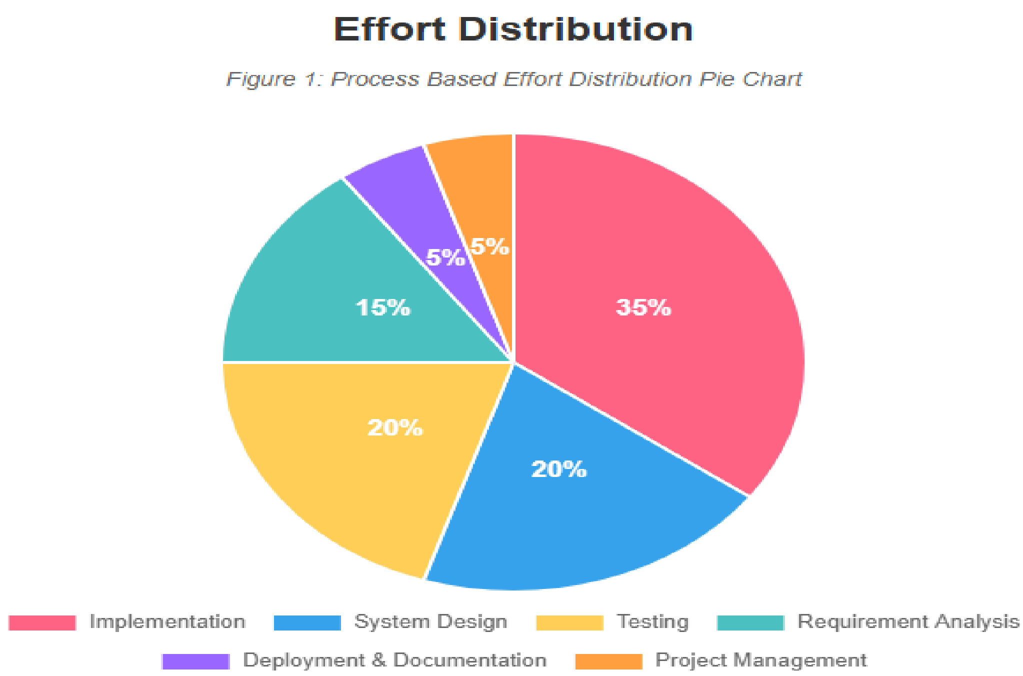


Figure: Process Based Effort Distribution Pie Chart

6. Project Schedule Estimation

Each iteration lasts 1 month.

Month	**Activities**
Month 1	Requirement analysis, initial design, planning
Month 2	Core implementation (booking, login, dashboards)
Month 3	Advanced features, testing, integration
Month 4	System testing, deployment, documentation

7. Project Schedule Chart

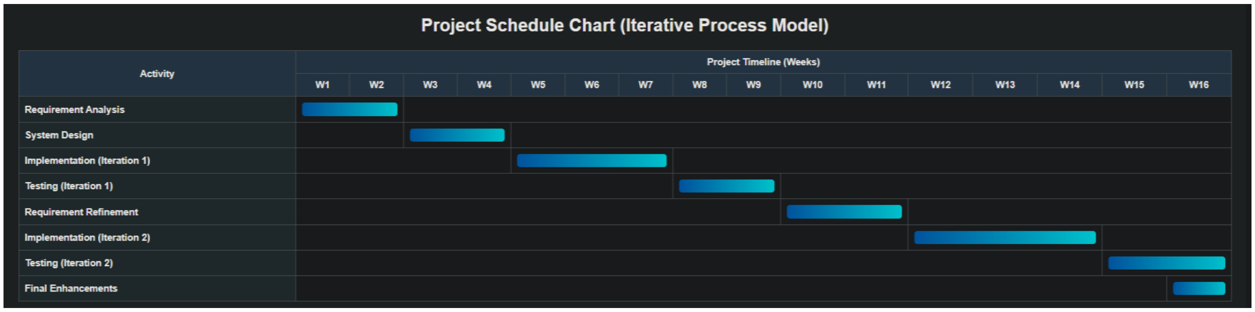


Figure: Project Schedule Chart

8. Personnel Requirement Chart

Role	No. of Persons
System Analyst	1
Backend Developer	1
Frontend Developer	1
Tester & Documentation	1

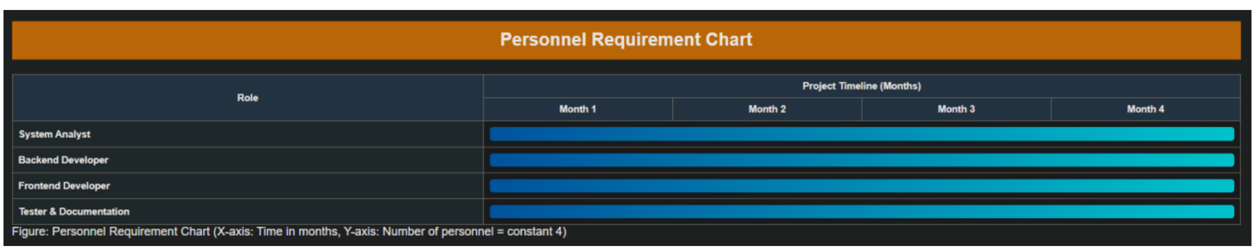


Figure: Personnel Requirement Chart (X-axis: Time in months, Y-axis: Number of personnel = constant 4)

Figure: Personnel Requirement Chart

9. Cost Estimation

Cost estimation for the Telemedicine & Consultation Booking System was performed to understand how project resources are distributed.

The total project cost is divided into four main categories:

- Personnel cost
- Hardware cost
- Software & utilities cost
- Miscellaneous cost

Personnel cost is considered the highest because human effort is the most significant contributor to software development.

10. Personnel Cost Estimation

The project was developed by a team of four members, each contributing throughout the project lifecycle.

Team Composition

Role	**No. of Persons**
System Analyst	1
Backend Developer	1
Frontend Developer	1
Tester & Documentation	1
Total	**4 Members**

Salary Calculation (Academic Estimation)

Assumptions:

- Project duration = 4 months
- Estimated effort value per member = Tk 2,500 per month

Cost per Member

2,500 × 4 months = Tk 10,000

Total Personnel Cost (4 Members)

10,000 × 4 = Tk 40,000

11. Hardware Cost Estimation (Depreciation Calculation)

The project used existing hardware resources such as personal laptops and internet devices. Therefore, only **depreciation cost** is considered instead of full purchase cost.

Assumptions

- Average laptop cost = Tk 60,000
- Useful life = 4 years
- Project duration = 4 months

Monthly Depreciation

60,000 ÷ (4 × 12) = Tk 1,250 per month

Hardware Cost for Project

$1,250 \times 4 \text{ months} \approx \text{Tk } 5,000$

Additional peripherals, power, and maintenance are estimated conservatively.
Total Hardware Cost = Tk 15,000

12. Accounts Table

Cost Category	**Estimated Cost (Tk)**
Personnel Cost	40,000
Hardware Cost	15,000
Software & Utilities	5,000
Miscellaneous	3,000
Total Estimated Cost	**63,000**

- Personnel cost is highest
- Hardware cost is medium
- Software & utilities cost is low

13. Accounts Chart

The Accounts Chart provides a visual representation of how the total project cost is distributed across different cost categories in the Telemedicine & Consultation Booking System. It helps in understanding which resources contribute most significantly to the overall project expenditure and supports better financial planning and analysis.

In the chart, the X-axis represents the cost categories, including Personnel Cost, Hardware Cost, Software & Utilities, and Miscellaneous expenses, while the Y-axis represents the estimated cost in Bangladeshi Taka (Tk). The height of each bar reflects the relative contribution of each category to the total project cost.

Personnel cost appears as the highest bar in the chart, emphasizing that human effort is the primary factor in software development. This includes activities such as requirement analysis, system design, implementation, testing, and documentation carried out by the project team members. Hardware cost is shown as a medium-level expense, calculated using depreciation of existing devices rather than full purchase cost. Software and utility costs remain comparatively low due to the use of open-source tools and existing infrastructure.

Overall, the Accounts Chart complements the Accounts Table by offering a clear and intuitive view of cost distribution. It confirms that the project follows a realistic and balanced cost structure suitable for an academic lab project while demonstrating proper application of software engineering cost estimation principles.

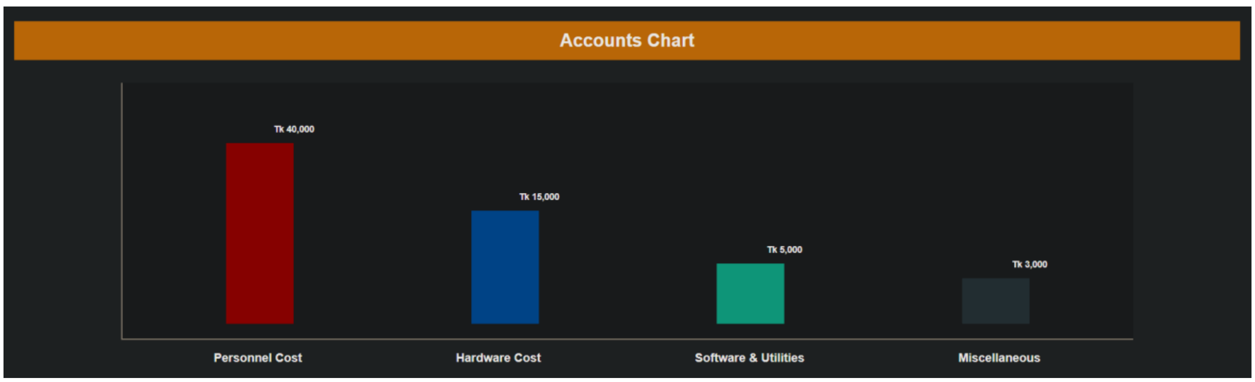


Figure: Accounts Chart (Bar Chart)

14. Quality Management & Software Development

Quality management plays a vital role in ensuring that software not only functions correctly but also meet user expectations, reliability standards, and long-term usability requirements. In the Telemedicine & Consultation Booking System, quality management was treated as a continuous activity integrated into every stage of the software development lifecycle rather than a one-time verification step at the end.

A clear quality plan was defined at the beginning of the project to guide development activities and ensure consistency across iterations. This plan helped align technical implementation with functional requirements, ethical considerations, and usability expectations.

Quality Plan Components

- **Product Introduction**

  The quality plan began with a clear understanding of the system's purpose, target users, and operational environment. Defining the scope and objectives early ensured that development remained focused on delivering a reliable telemedicine platform rather than unnecessary features.

- **Development Process**

  The project followed an iterative development model, allowing features to be designed, implemented, tested, and refined incrementally. This approach supported early detection of issues and enabled improvements based on feedback received after each iteration.

- **Quality Goals**

  Specific quality goals were identified, including correctness, security, usability, maintainability, and performance. These goals guided design decisions and testing activities throughout the project and helped maintain a consistent standard across all modules.

- **Risk Management**

  Potential risks such as data security breaches, appointment conflicts, system misuse, and performance limitations were identified early. Preventive measures, including role-based access control, validation checks, and controlled workflows, were applied to minimize these risks and protect system integrity.

- **Review and Testing Strategy**

  Regular reviews were conducted at different stages, including requirement reviews, design reviews, and code reviews. Testing was performed at the end of each iteration using functional and non-functional test cases to ensure that new features worked correctly and existing functionality remained stable.

  The iterative development model strongly supported quality management by enabling continuous testing and feedback in every cycle. By validating features incrementally, the system evolved in a controlled and reliable manner, ensuring that quality was consistently maintained throughout development.

Overall, effective quality management ensured that the Telemedicine & Consultation Booking System was developed in a disciplined, reliable, and user-focused manner, reflecting professional software engineering practices even within an academic lab environment.

15. Process Based Quality

Process based quality emphasizes the importance of *how* software is developed rather than focusing only on the final output. The main idea behind process-based quality is that if each stage of the software development process is carefully planned, monitored, and reviewed, the final product is more likely to be reliable, maintainable, and error-free. In this project, quality was treated as a continuous activity that was applied throughout all phases of the iterative development process.

To ensure process quality, several quality assurance activities were performed at different stages of development. Each activity helped identify issues early and prevented defects from propagating to later stages.

Quality Activities

- **Requirement Validation**

  At the beginning of each iteration, requirements were reviewed to ensure they were clear, complete, and feasible. This step helped confirm that the system requirements accurately reflected user needs and reduced the risk of misunderstandings during development.

- **Design Review**

  System design components such as architecture, database structure, and module interactions were reviewed before implementation. Design reviews helped identify potential issues related to scalability, security, and maintainability, allowing improvements to be made before coding began.

- **Code Review**

  During implementation, source code was reviewed to ensure proper coding standards, logical correctness, and consistency across modules. This practice helped reduce bugs, improve readability, and make the code easier to maintain and extend.

- **Iteration-wise Testing**

  Testing was conducted at the end of each iteration rather than waiting until the final stage. Functional and non-functional test cases were executed to verify newly implemented features and ensure that existing functionality was not broken. This incremental testing approach helped detect defects early and improve overall system stability.

- **Final System Validation**

After completing all iterations, the entire system was validated against the original requirements. This ensured that all functional and quality requirements were satisfied and that the system was ready for final submission.

Overall, applying process-based quality practices ensured that quality was built into the system from the beginning rather than added at the end. This approach improved reliability, reduced rework, and provided valuable experience in following disciplined software engineering processes.

16. Process Based Quality

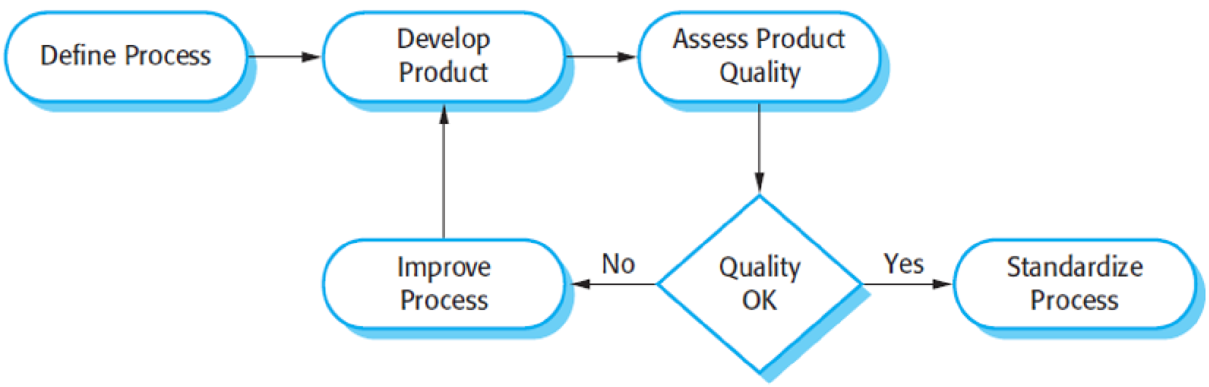


Figure: Process Based Quality Diagram

17. Conclusion

The project estimation process provided a structured understanding of how resources are planned and allocated during the development of the Telemedicine & Consultation Booking System. Although this is an academic lab project, the estimation was carried out using realistic assumptions to demonstrate proper software engineering cost analysis and planning techniques.

Personnel effort was identified as the most significant contributor to the overall project cost. This reflects the fact that software development is primarily driven by human expertise, analysis, design, implementation, testing, and documentation activities. The estimated personnel cost represents effort-based valuation rather than actual salary payment, which is appropriate for a student project. Hardware cost was calculated using depreciation principles, acknowledging the use of existing devices without inflating costs. Software and utility costs remained minimal due to the use of open-source tools and available infrastructure.

By separating costs into personnel, hardware, software & utilities, and miscellaneous categories, the estimation clearly illustrates how different resources contribute to the project. The Accounts Table and Accounts Chart further help visualize this distribution, ensuring transparency and consistency across all estimation components.

Overall, this estimation exercise highlights the importance of planning and resource management in software engineering. Even in a lab environment, systematic cost estimation helps in understanding project feasibility, managing effort effectively, and preparing for real-world software development scenarios.

Chapter 7: Designing

7.1 Data Flow Design

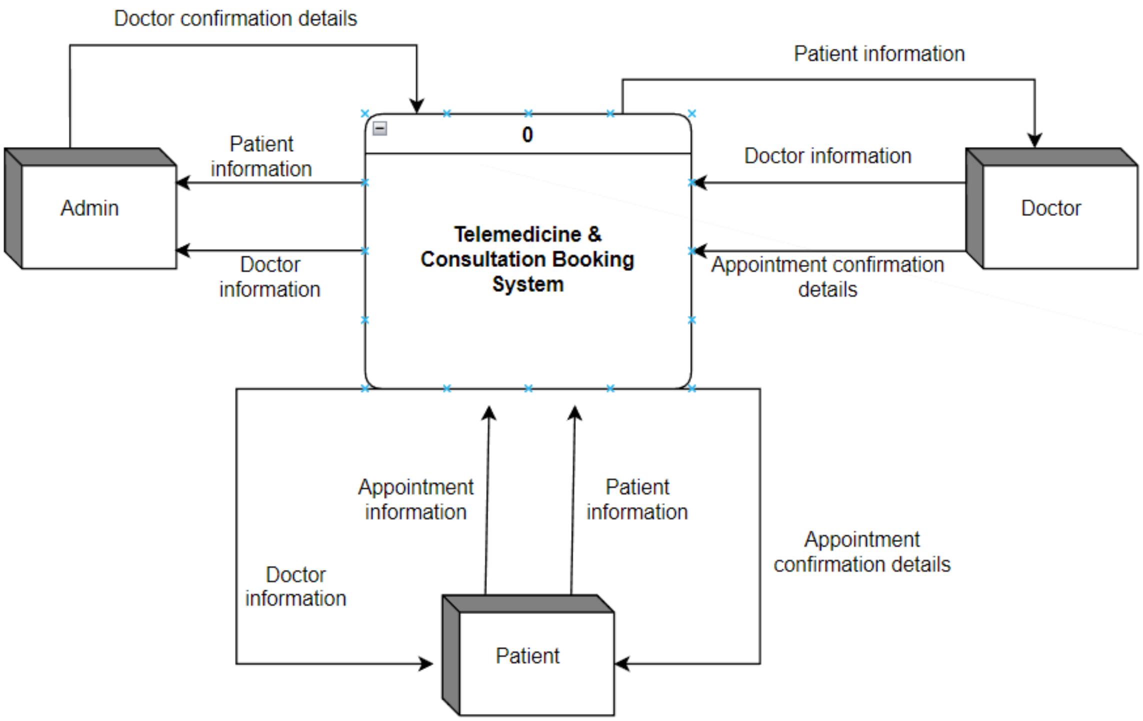


Fig: Context level DFD for TMCBS

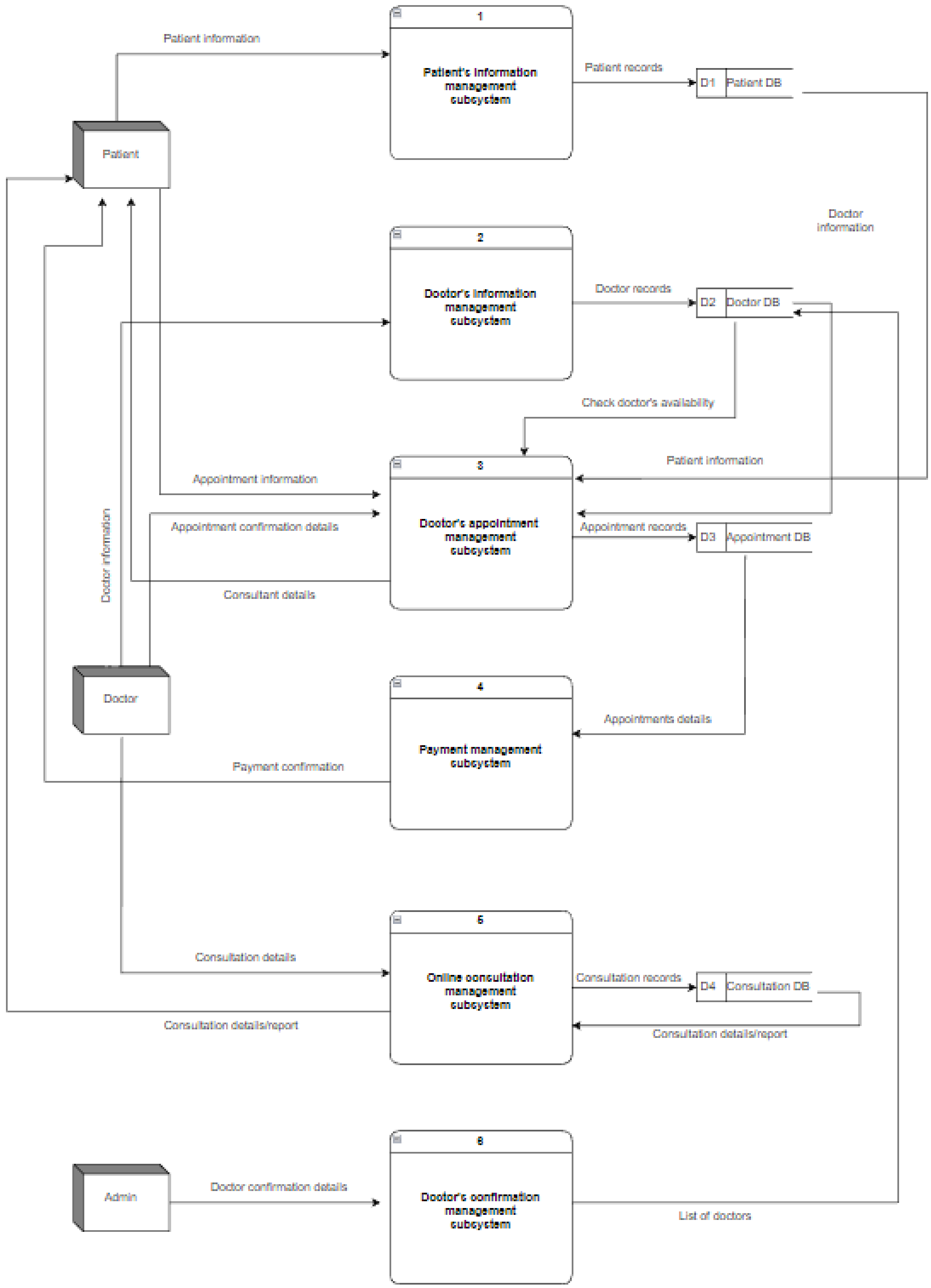


Fig: Level 1 DFD

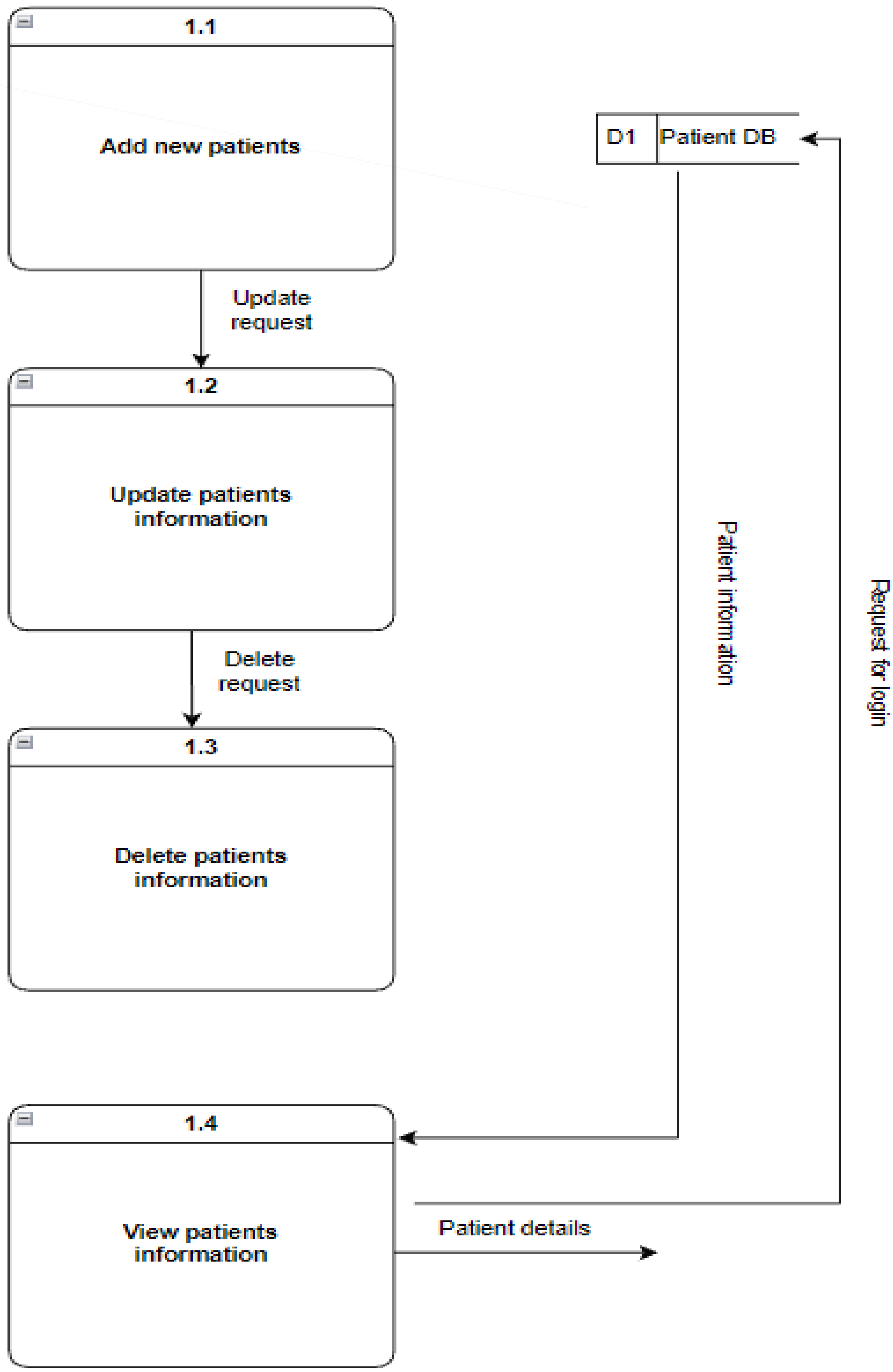
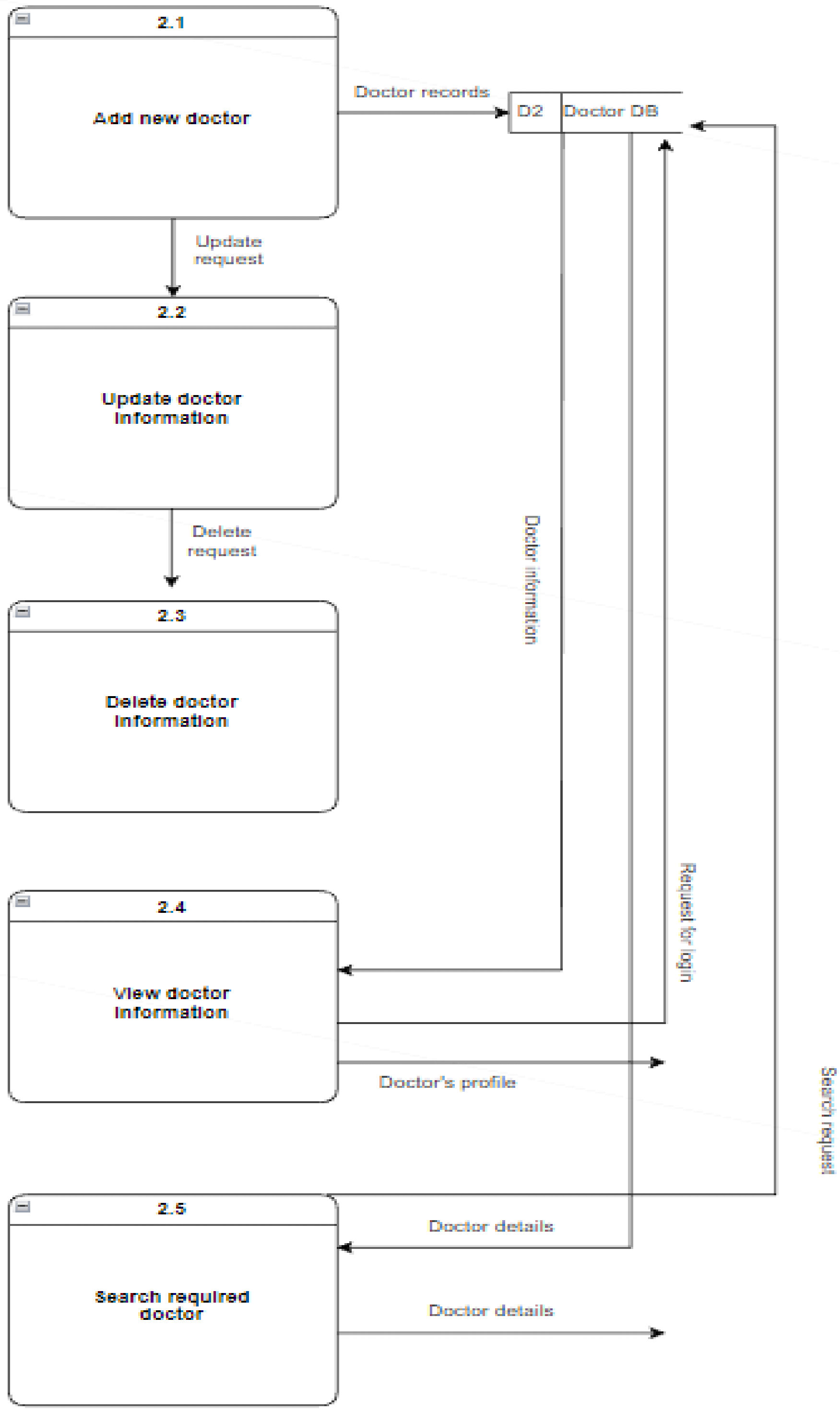


Fig: Level 2 process 1



i

Fig: Level 2 process 2

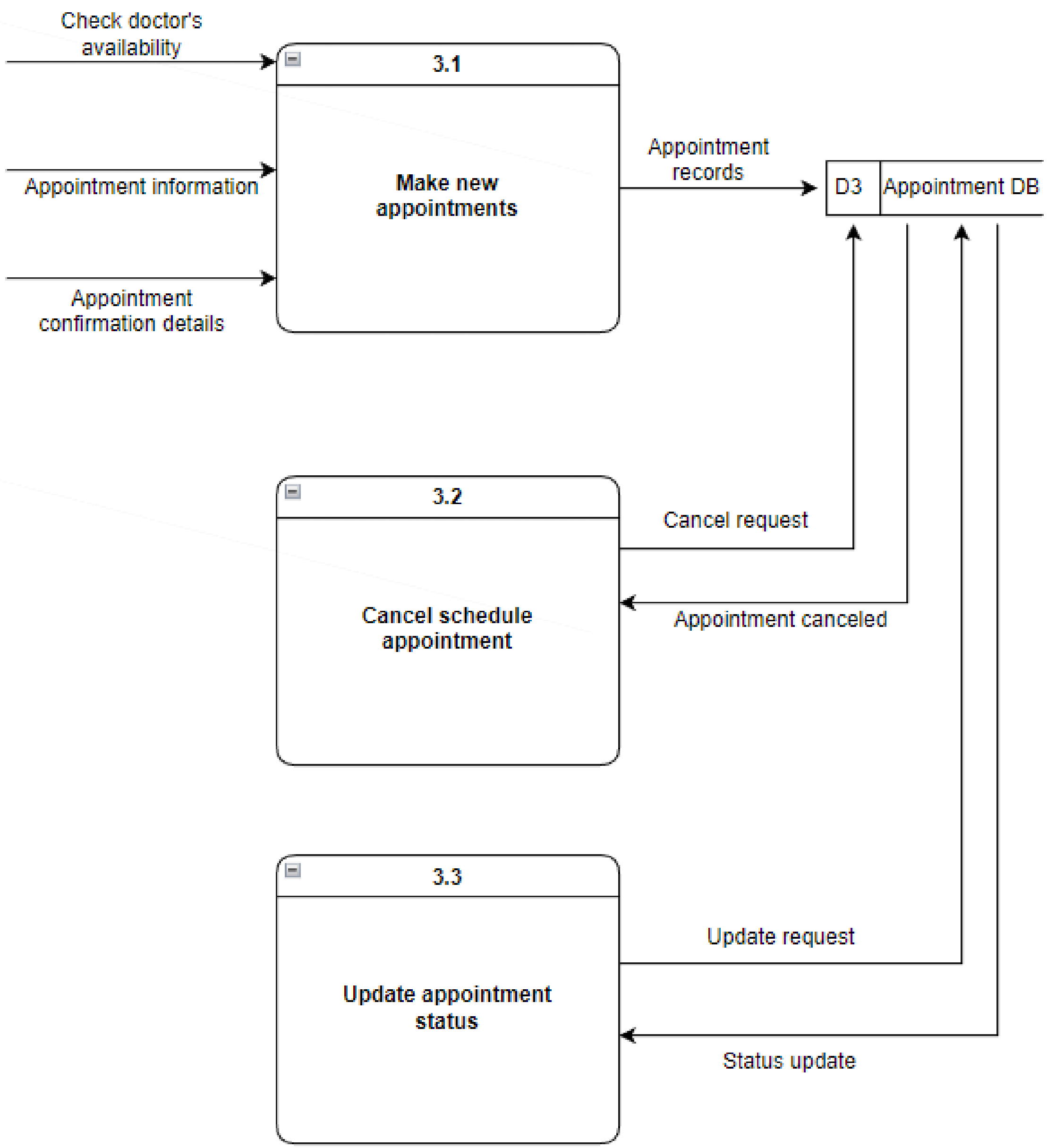


Fig: Level 2 process 3

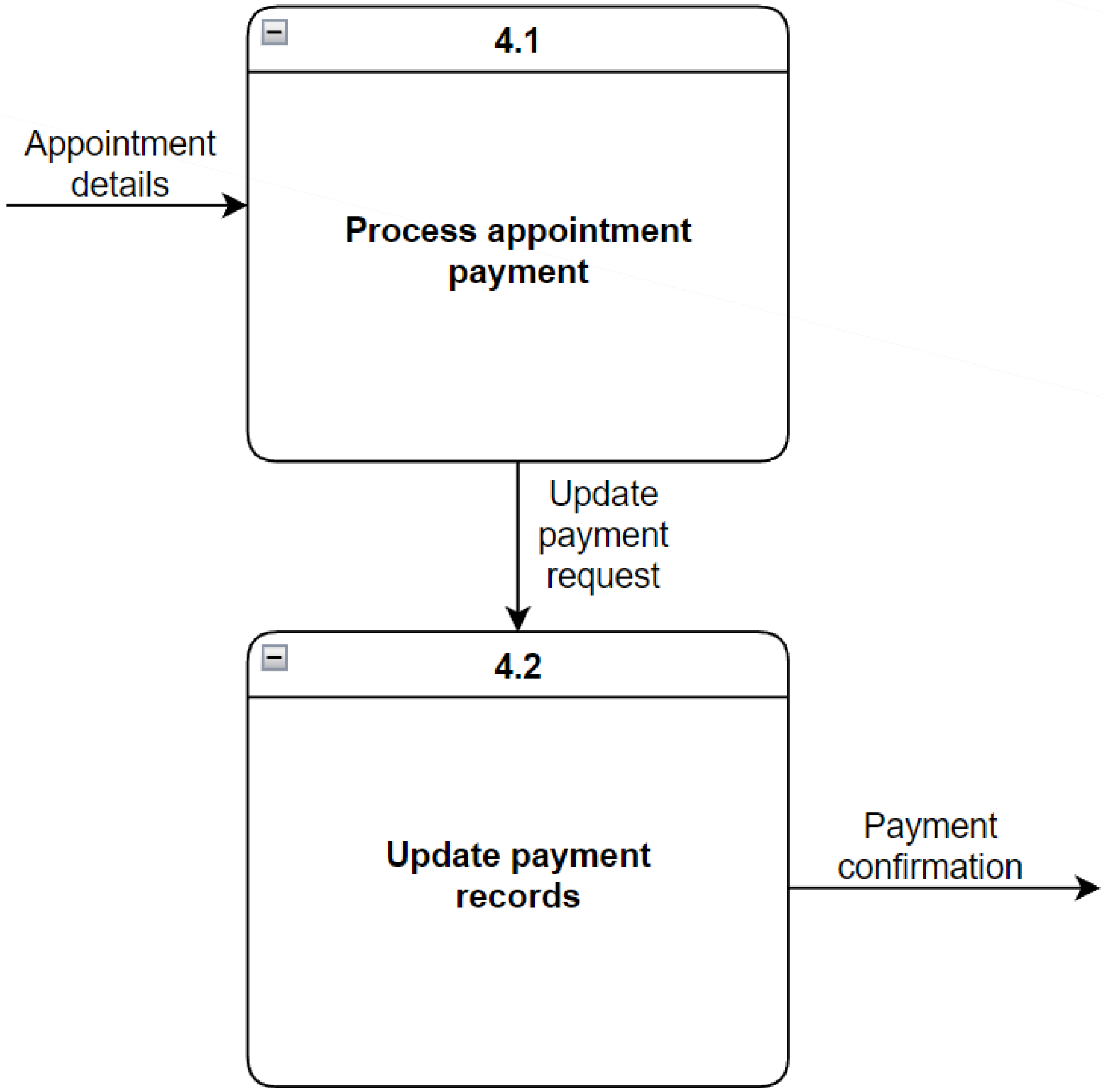


Fig: Level 2 process 4

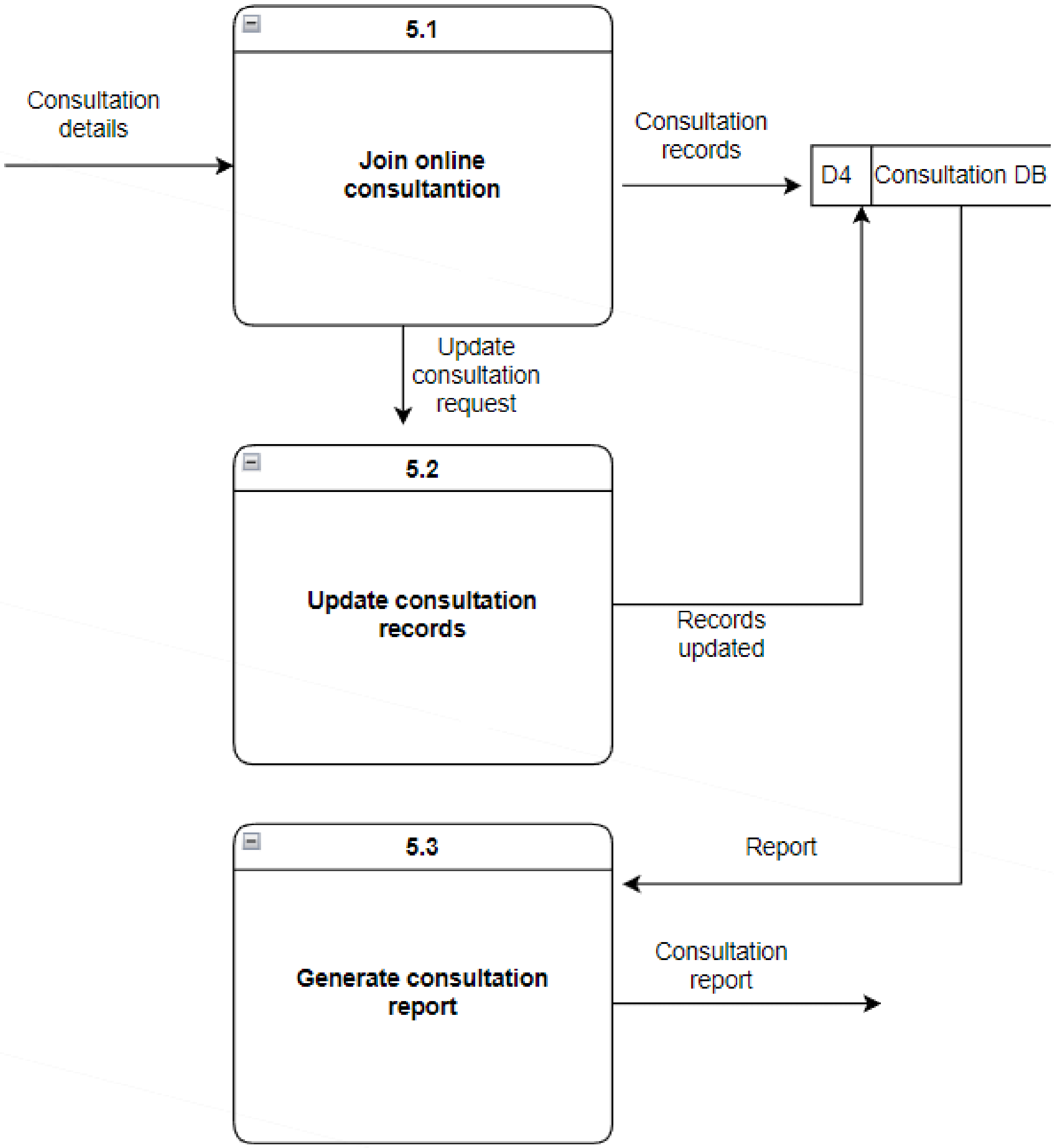


Fig: Level 2 process 5

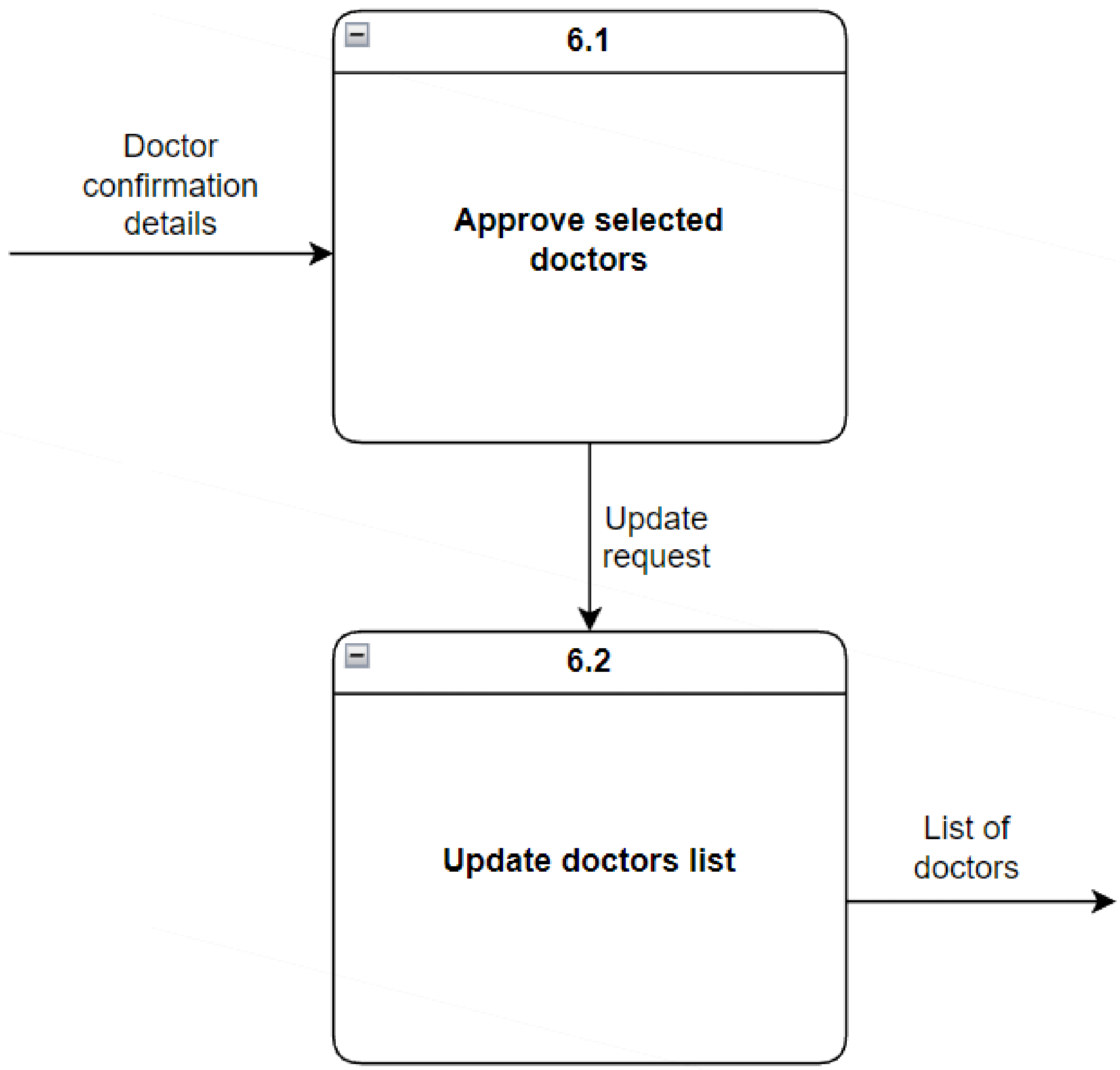


Fig: Level 2 process 6

7.2 Database Design

The Telemedicine & Consultation Booking System uses a relational database that closely follows the structure of the ASP.NET MVC models used in the application. The database design is intentionally kept simple and practical to match the academic scope of the project while ensuring data integrity and ease of maintenance.

Instead of creating separate tables for each role, the system follows a **role-based user model**, where a single Users table stores common user information. Additional domain-specific tables reference the user table using foreign keys.

Users Table

Field Name	Data Type	Null	Default	Extra
UserId	int	No	—	Primary Key, Auto Increment
FullName	string(100)	No	—	—
Email	string (100)	No	—	Unique
Password	string (255)	No	—	—
Role	string (20)	No	—	(Patient / Doctor / Admin)
IsActive	bool	No	1	—

Doctors Table

Field Name	Data Type	Null	Default	Extra
DoctorId	int	No	—	Primary Key, Auto Increment
UserId	int	No	—	Foreign Key (Users)
Specialization	string (100)	No	—	—
Qualification	string (150)	Yes	NULL	—
IsApproved	bool	No	0	Admin controlled

Appointments Table

Field Name	Data Type	Null	Default	Extra
AppointmentId	int	No	—	Primary Key, Auto Increment
PatientId	int	No	—	FK → Users
DoctorId	int	No	—	FK → Doctors
AppointmentDate	date	No	—	—
AppointmentTime	time	No	—	—
Status	string (20)	No	Pending	—

MeetingLink	string (255)	Yes	NULL	—

Prescriptions Table

Field Name	Data Type	Null	Default	Extra
PrescriptionId	int	No	—	Primary Key, Auto Increment
AppointmentId	int	No	—	FK → Appointments
MedicineDetails	string	No	—	—
Notes	string	Yes	NULL	—
CreatedDate	string	No	—	—

Payments Table

Field Name	Data Type	Null	Default	Extra
PaymentId	int	No	—	Primary Key, Auto Increment
AppointmentId	int	No	—	FK → Appointments
Amount	decimal (10,2)	No	—	—
PaymentStatus	string (20)	No	Pending	—
PaymentDate	date	Yes	NULL	—

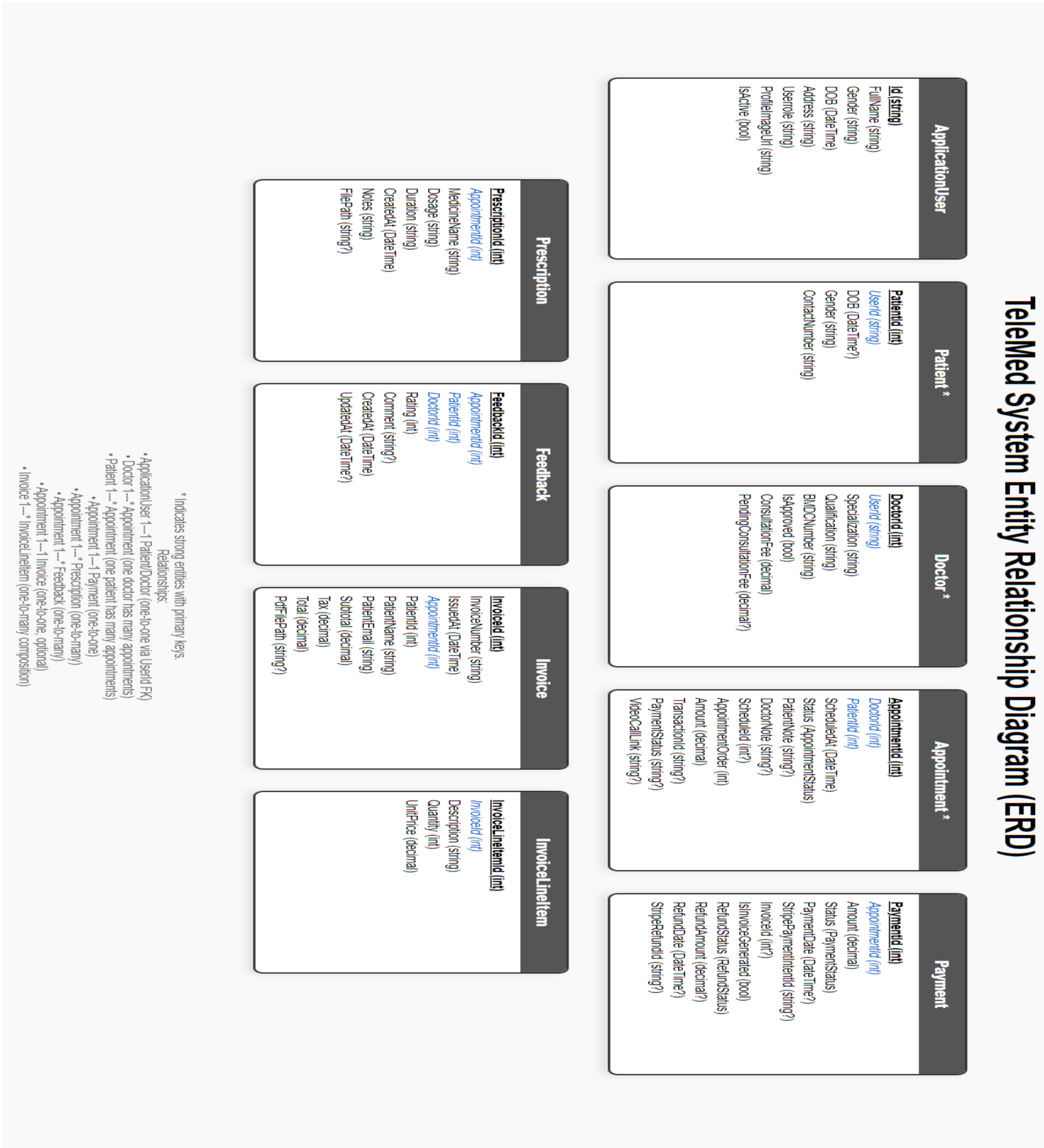
DoctorAvailability Table

Field Name	Data Type	Null	Default	Extra
AvailabilityId	int	No	—	Primary Key, Auto Increment
DoctorId	int	No	—	FK → Doctors
AvailableDate	date	No	—	—
TimeSlot	string (50)	No	—	—

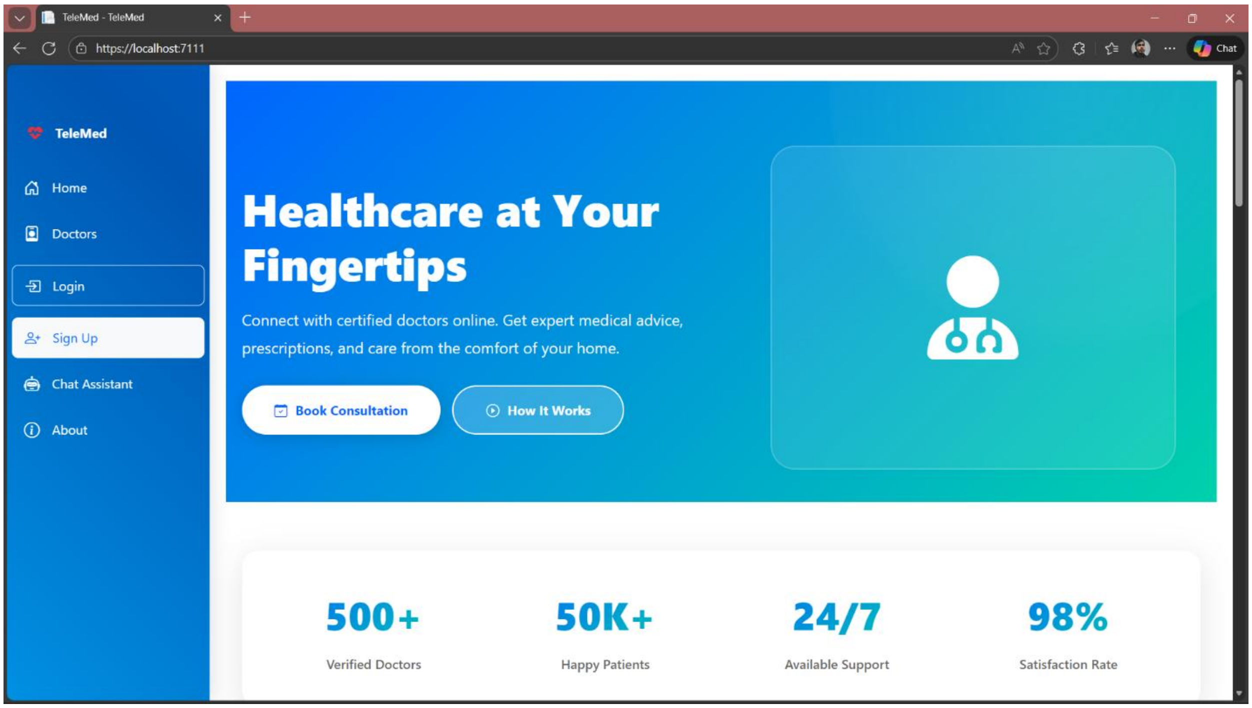
Summary of Database Design

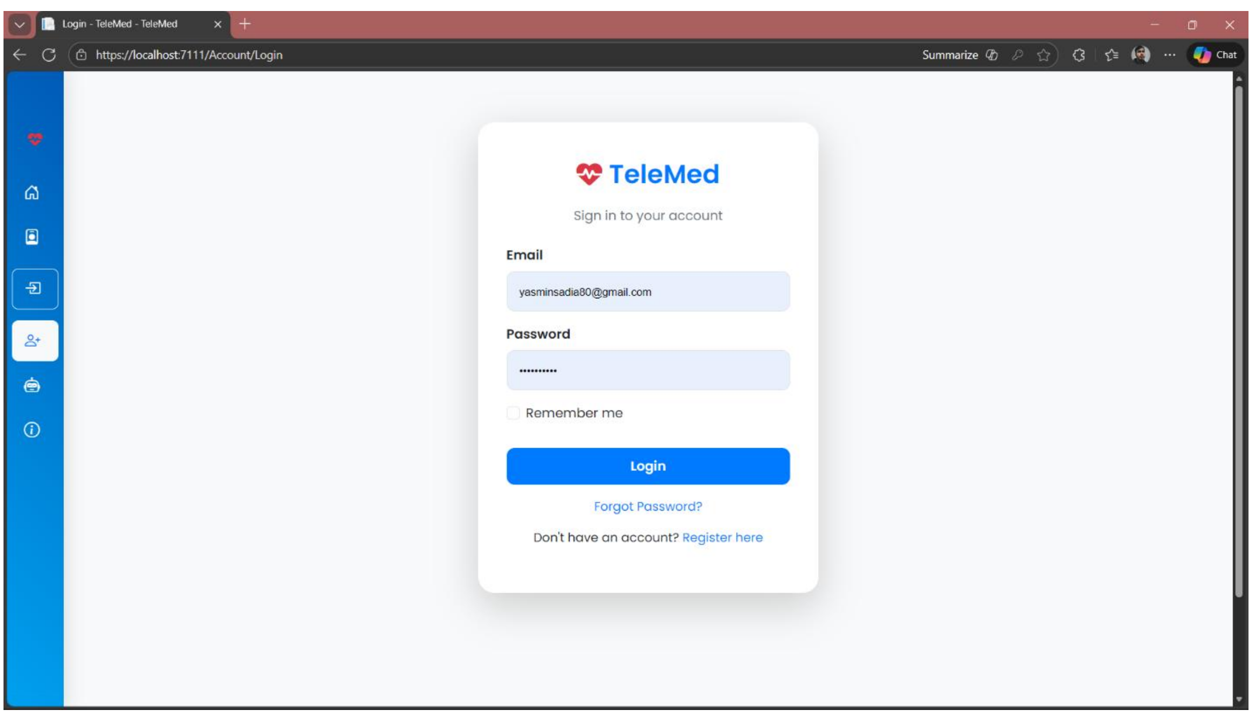
The database design supports secure role-based access, appointment-centered workflows, and proper data relationships. Using foreign keys ensures referential integrity between users, appointments, prescriptions, and payments. This structured design improves scalability, maintainability, and data consistency, making it suitable for a telemedicine system.

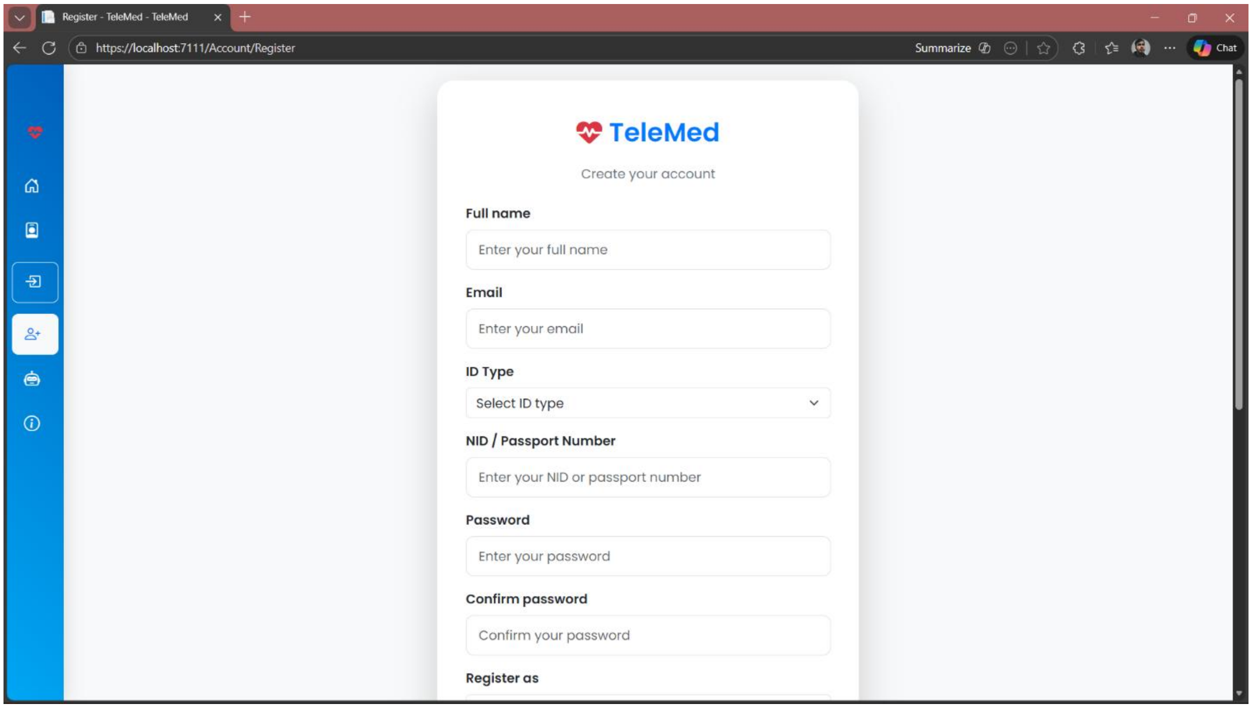
7.3 ERD

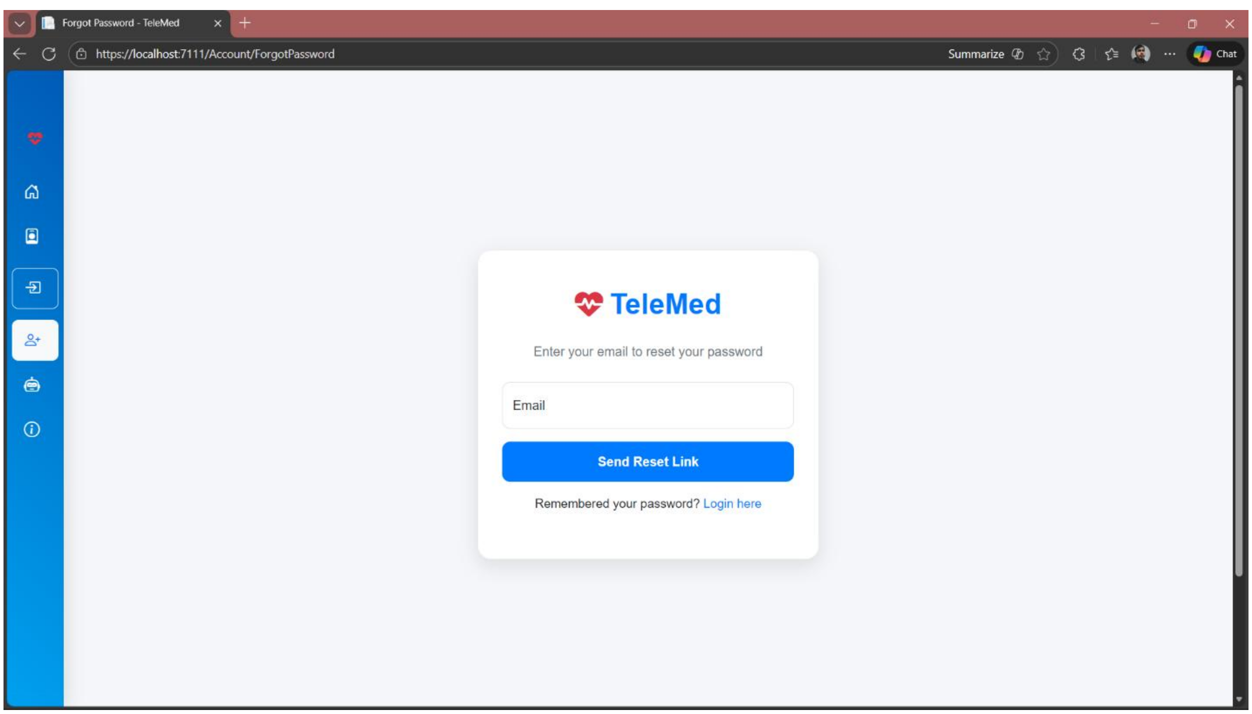

TeleMed System Entity Relationship Diagram (ERD)
ApplicationUser
Id (string)
FullName (string)
Gender (string)
DOB (DateTime)
Address (string)
Userrole (string)
ProfileImageUrl (string)
IsActive (bool)
Patient *
PatientId (int)
UserId (string)
DOB (DateTime?)
Gender (string)
ContactNumber (string)
Doctor *
DoctorId (int)
UserId (string)
Specialization (string)
Qualification (string)
BMDCNumber (string)
IsApproved (bool)
ConsultationFee (decimal)
PendingConsultationFee (decimal?)
Appointment *
AppointmentId (int)
DoctorId (int)
PatientId (int)
ScheduledAt (DateTime)
Status (AppointmentStatus)
PatientNote (string?)
DoctorNote (string?)
ScheduleId (int?)
AppointmentOrder (int)
Amount (decimal)
TransactionId (string?)
PaymentStatus (string?)
VideoCallLink (string?)
Payment
PaymentId (int)
AppointmentId (int)
Amount (decimal)
Status (PaymentStatus)
PaymentDate (DateTime?)
StripePaymentIntentId (string?)
InvoiceId (int?)
IsInvoiceGenerated (bool)
RefundStatus (RefundStatus)
RefundAmount (decimal?)
RefundDate (DateTime?)
StripeRefundId (string?)
Prescription
PrescriptionId (int)
AppointmentId (int)
MedicineName (string)
Dosage (string)
Duration (string)
CreatedAt (DateTime)
Notes (string)
FilePath (string?)
Feedback
FeedbackId (int)
AppointmentId (int)
PatientId (int)
DoctorId (int)
Rating (int)
Comment (string?)
CreatedAt (DateTime)
UpdatedAt (DateTime?)
Invoice
InvoiceId (int)
InvoiceNumber (string)
IssuedAt (DateTime)
AppointmentId (int)
PatientId (int)
PatientName (string)
PatientEmail (string)
Subtotal (decimal)
Tax (decimal)
Total (decimal)
PdfFilePath (string?)
InvoiceLineItem
InvoiceLineItemId (int)
InvoiceId (int)
Description (string)
Quantity (int)
UnitPrice (decimal)
* Indicates strong entities with primary keys.
Relationships:
• ApplicationUser 1—1 Patient/Doctor (one-to-one via UserId FK)
• Doctor 1—* Appointment (one doctor has many appointments)
• Patient 1—* Appointment (one patient has many appointments)
• Appointment 1—1 Payment (one-to-one)
• Appointment 1—* Prescription (one-to-many)
• Appointment 1—* Feedback (one-to-many)
• Appointment 1—1 Invoice (one-to-one, optional)
• Invoice 1—* InvoiceLineItem (one-to-many composition)

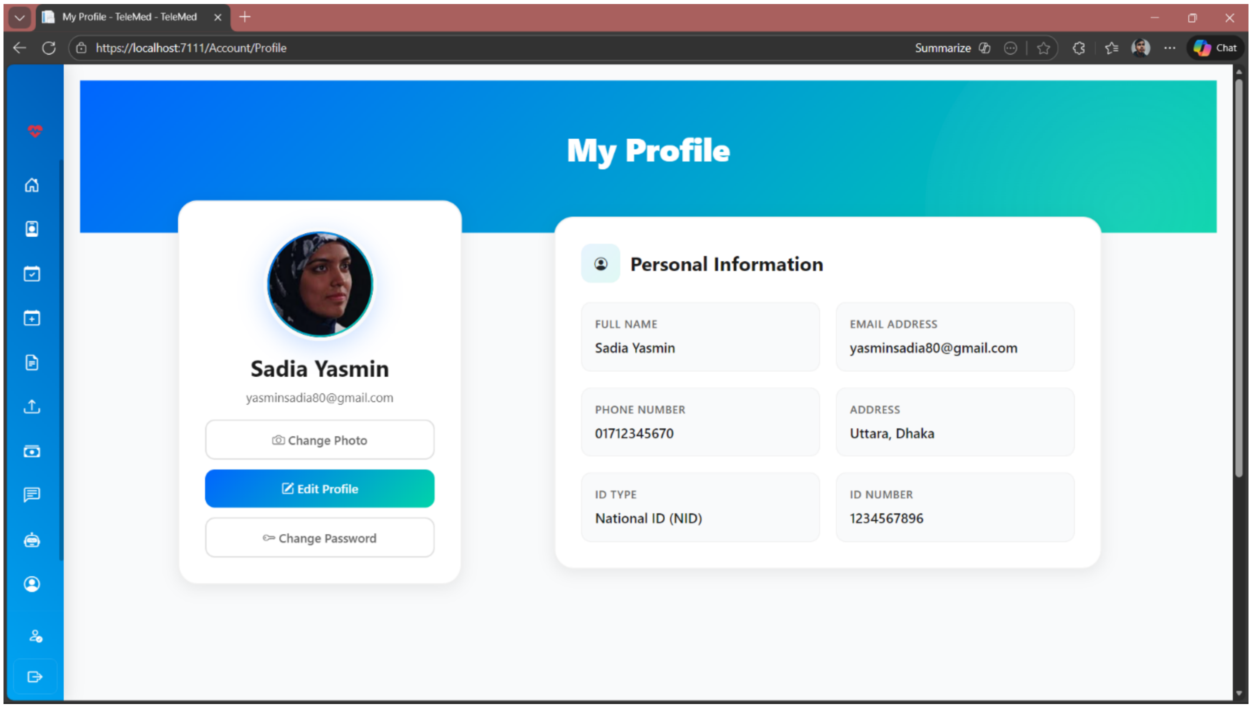
7.4 Interface Design

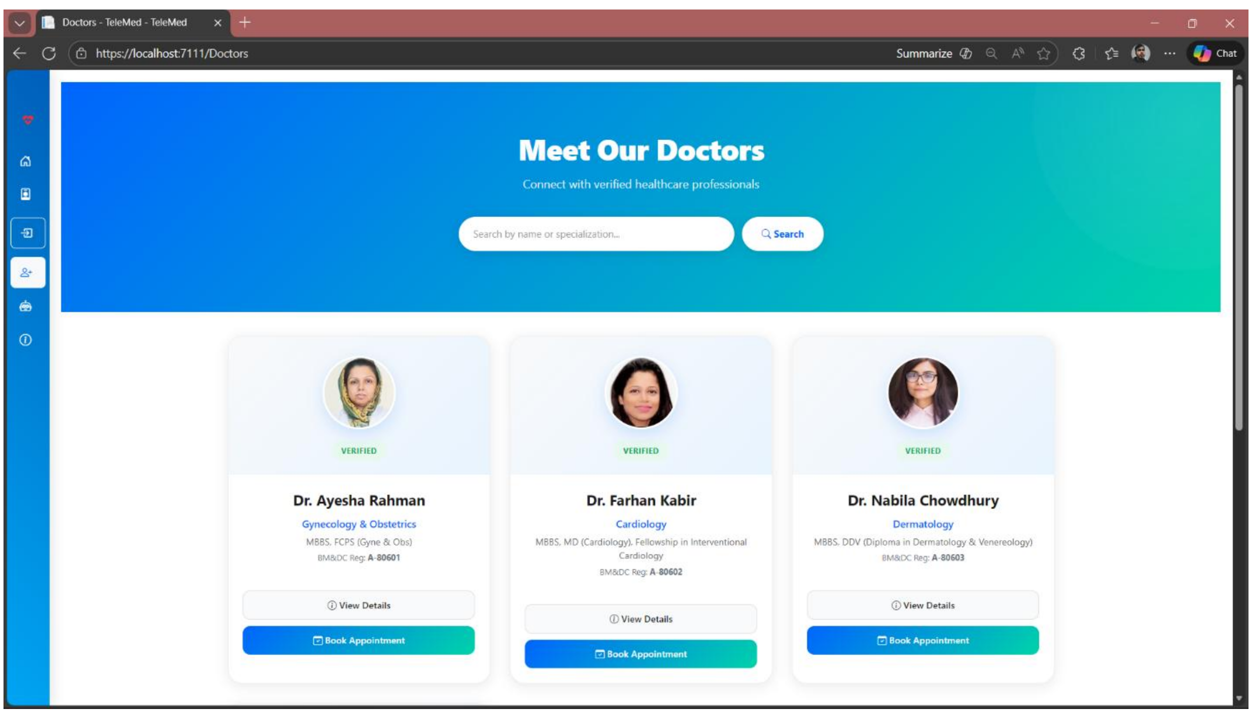


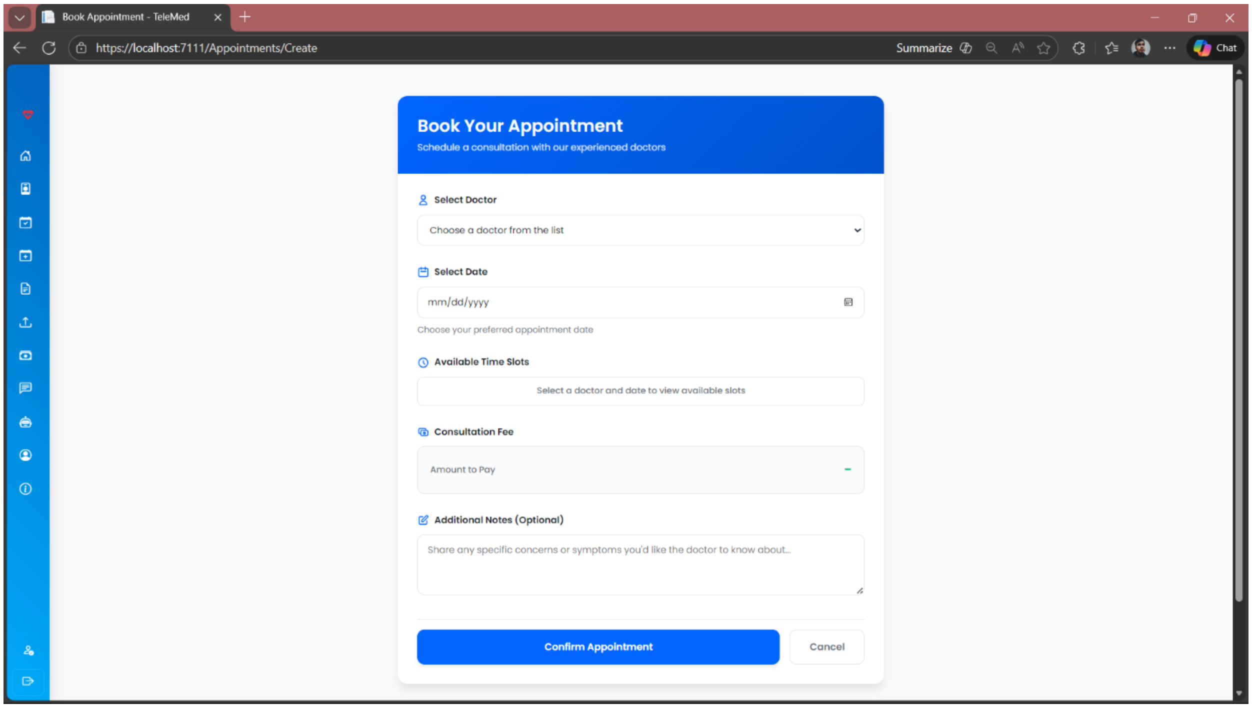


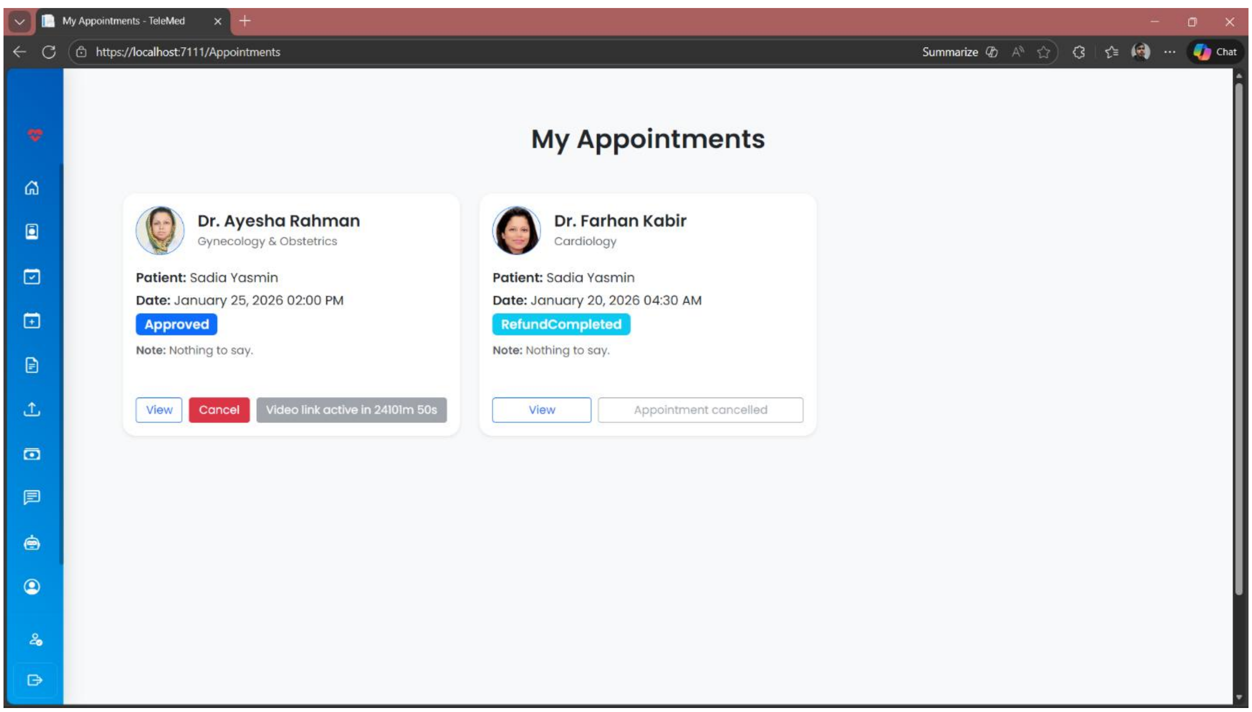

Register - TeleMed - TeleMed
https://localhost:7111/Account/Register
Summarize
Chat
TeleMed
Create your account
Full name
Enter your full name
Email
Enter your email
ID Type
Select ID type
NID / Passport Number
Enter your NID or passport number
Password
Enter your password
Confirm password
Confirm your password
Register as

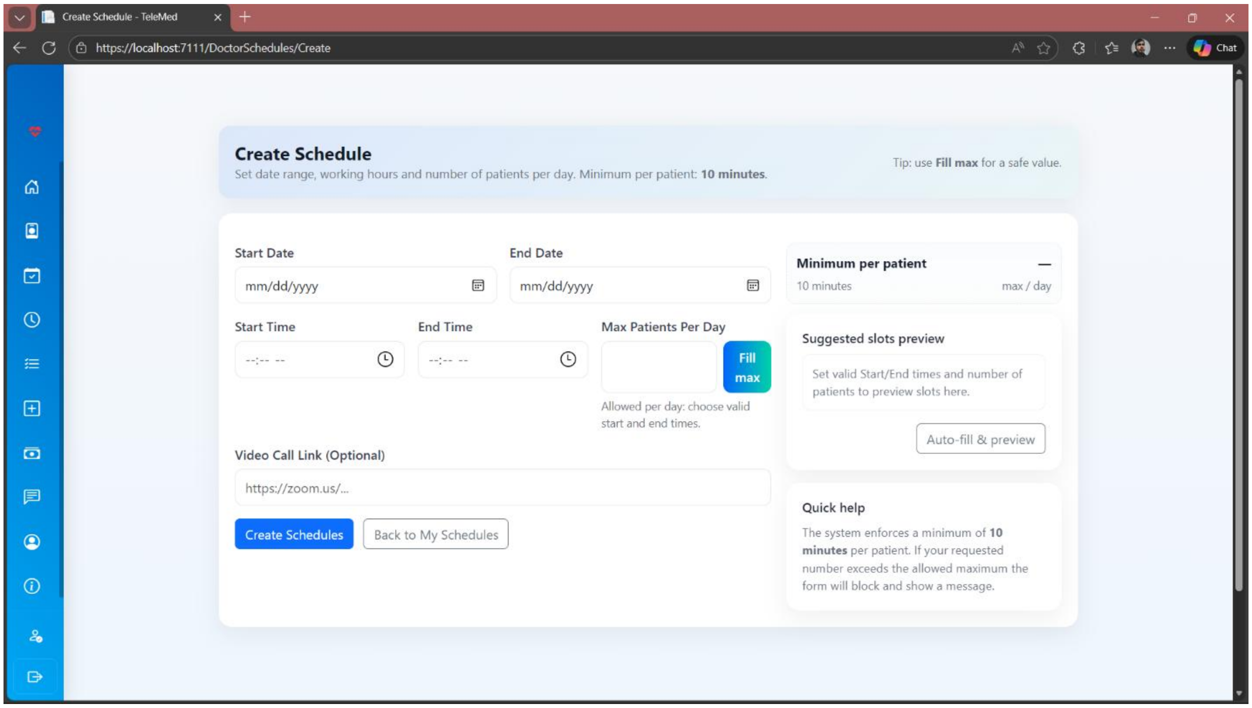

Forgot Password - TeleMed
https://localhost:7111/Account/ForgotPassword
Summarize
Chat
TeleMed
Enter your email to reset your password
Email
Send Reset Link
Remembered your password? Login here

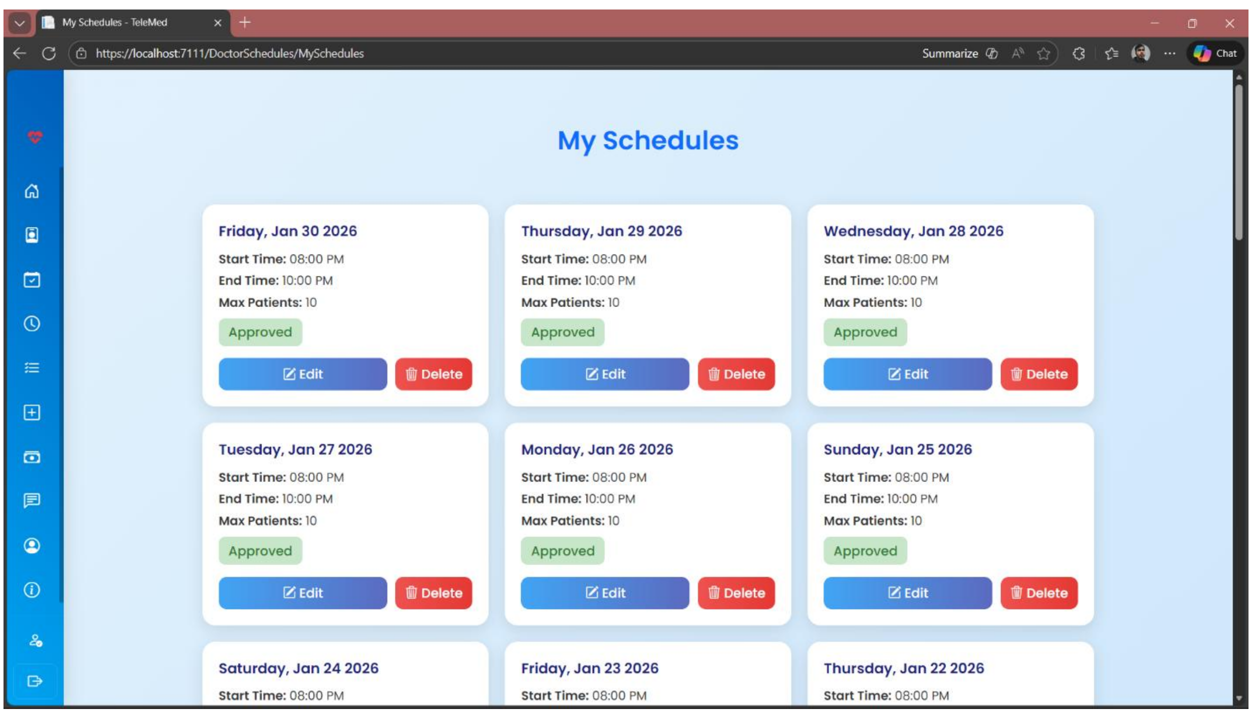

My Profile - TeleMed - TeleMed
https://localhost:7111/Account/Profile
My Profile
Sadia Yasmin
yasminsadia80@gmail.com
Change Photo
Edit Profile
Change Password
Personal Information
FULL NAME
Sadia Yasmin
EMAIL ADDRESS
yasminsadia80@gmail.com
PHONE NUMBER
01712345670
ADDRESS
Uttara, Dhaka
ID TYPE
National ID (NID)
ID NUMBER
1234567896

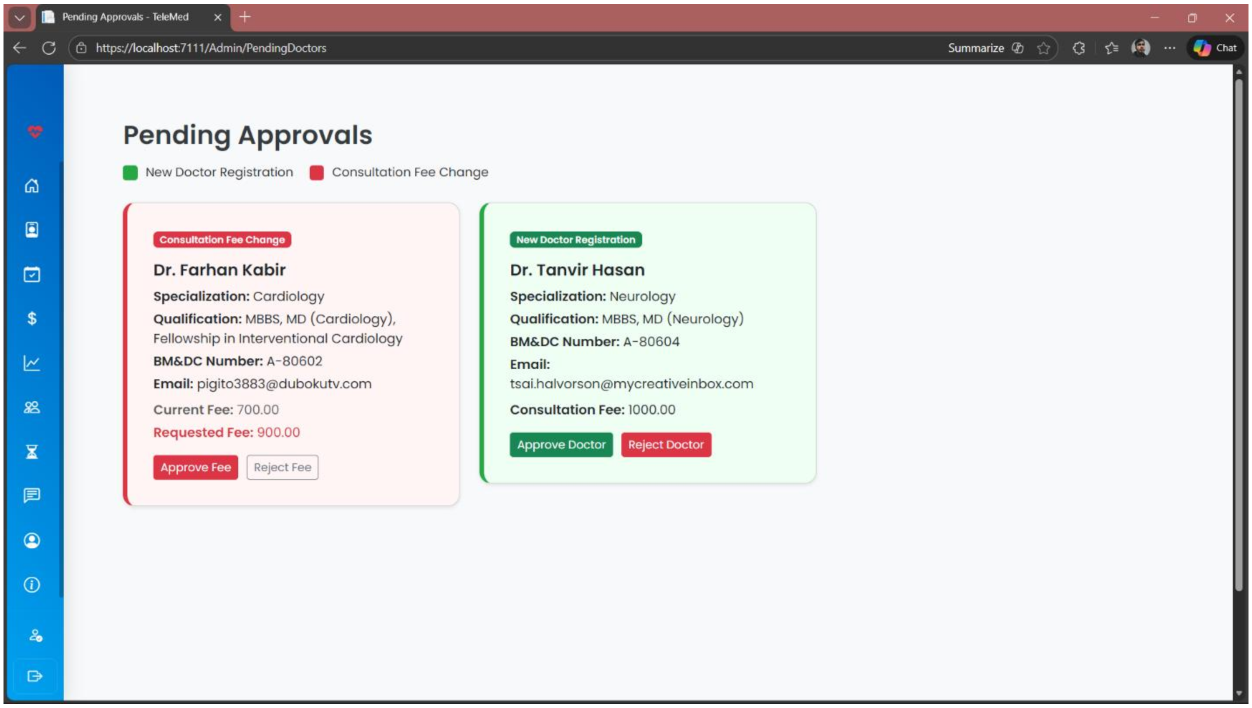

Doctors - TeleMed - TeleMed
https://localhost:7111/Doctors
Meet Our Doctors
Connect with verified healthcare professionals
Search by name or specialization...
Search
VERIFIED
Dr. Ayesha Rahman
Gynecology & Obstetrics
MBBS, FCPS (Gyne & Obs)
BM&DC Reg: A-80601
View Details
Book Appointment
VERIFIED
Dr. Farhan Kabir
Cardiology
MBBS, MD (Cardiology), Fellowship in Interventional Cardiology
BM&DC Reg: A-80602
View Details
Book Appointment
VERIFIED
Dr. Nabila Chowdhury
Dermatology
MBBS, DDV (Diploma in Dermatology & Venereology)
BM&DC Reg: A-80603
View Details
Book Appointment

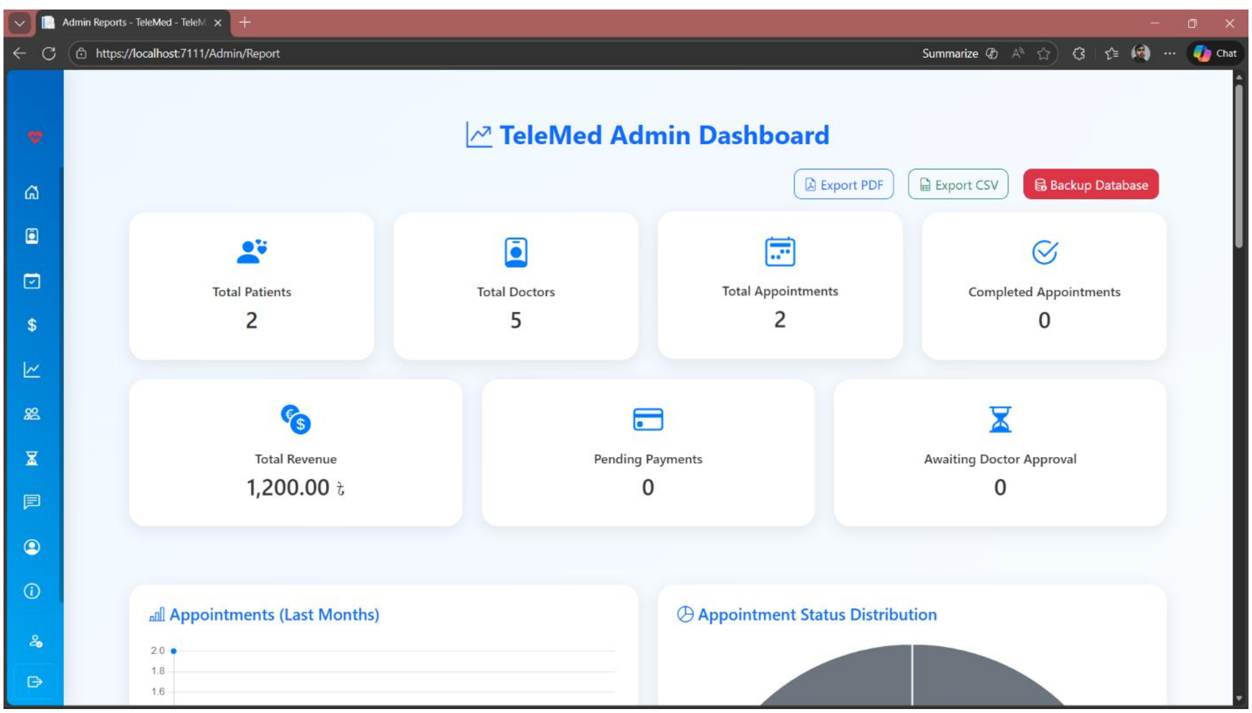

Book Appointment - TeleMed
https://localhost:7111/Appointments/Create
Book Your Appointment
Schedule a consultation with our experienced doctors
Select Doctor
Choose a doctor from the list
Select Date
mm/dd/yyyy
Choose your preferred appointment date
Available Time Slots
Select a doctor and date to view available slots
Consultation Fee
Amount to Pay
Additional Notes (Optional)
Share any specific concerns or symptoms you'd like the doctor to know about...
Confirm Appointment
Cancel

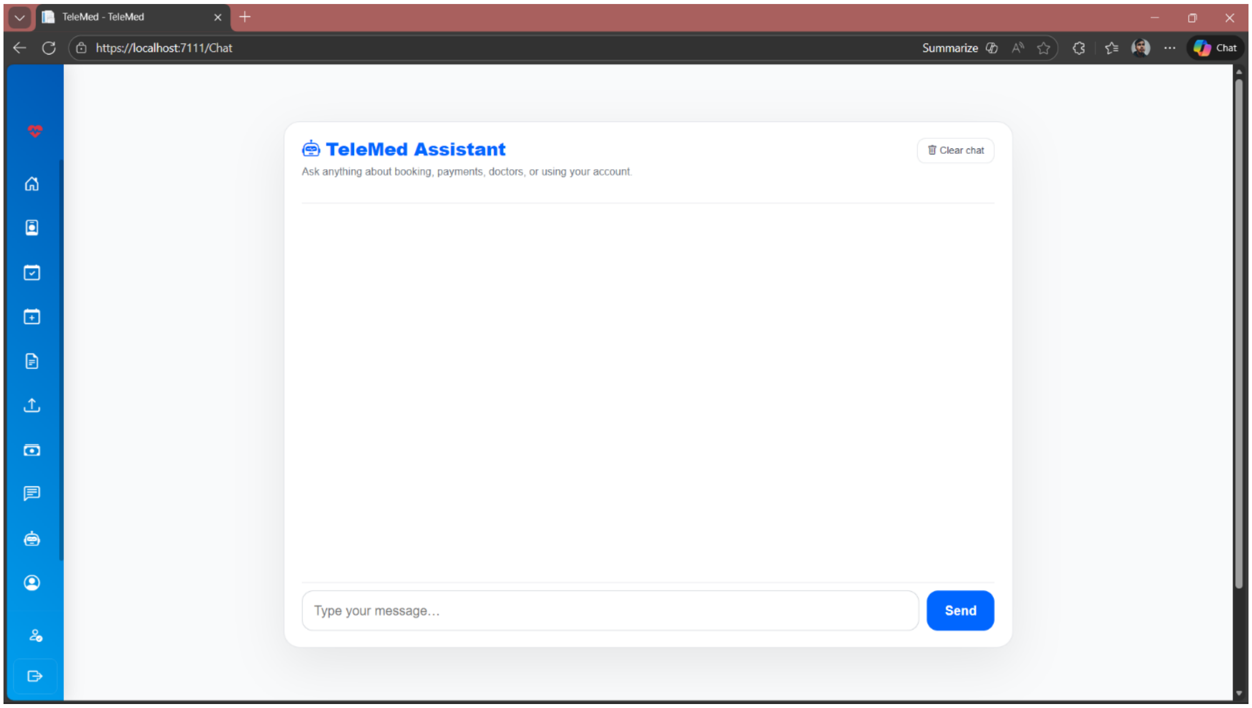

My Appointments - TeleMed
https://localhost:7111/Appointments
My Appointments
Dr. Ayesha Rahman
Gynecology & Obstetrics
Patient: Sadia Yasmin
Date: January 25, 2026 02:00 PM
Approved
Note: Nothing to say.
View
Cancel
Video link active in 2410m 50s
Dr. Farhan Kabir
Cardiology
Patient: Sadia Yasmin
Date: January 20, 2026 04:30 AM
RefundCompleted
Note: Nothing to say.
View
Appointment cancelled


Create Schedule - TeleMed
https://localhost:7111/DoctorSchedules/Create
Create Schedule
Set date range, working hours and number of patients per day. Minimum per patient: 10 minutes.
Tip: use Fill max for a safe value.
Start Date
mm/dd/yyyy
End Date
mm/dd/yyyy
Minimum per patient
10 minutes
max / day
Start Time
End Time
Max Patients Per Day
Fill max
Allowed per day: choose valid start and end times.
Suggested slots preview
Set valid Start/End times and number of patients to preview slots here.
Auto-fill & preview
Video Call Link (Optional)
https://zoom.us/...
Create Schedules
Back to My Schedules
Quick help
The system enforces a minimum of 10 minutes per patient. If your requested number exceeds the allowed maximum the form will block and show a message.


My Schedules - TeleMed
https://localhost:7111/DoctorSchedules/MySchedules
My Schedules
Friday, Jan 30 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Thursday, Jan 29 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Wednesday, Jan 28 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Tuesday, Jan 27 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Monday, Jan 26 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Sunday, Jan 25 2026
Start Time: 08:00 PM
End Time: 10:00 PM
Max Patients: 10
Approved
Edit
Delete
Saturday, Jan 24 2026
Start Time: 08:00 PM
Friday, Jan 23 2026
Start Time: 08:00 PM
Thursday, Jan 22 2026
Start Time: 08:00 PM


Pending Approvals - TeleMed
https://localhost:7111/Admin/PendingDoctors
Pending Approvals
New Doctor Registration
Consultation Fee Change
Consultation Fee Change
Dr. Farhan Kabir
Specialization: Cardiology
Qualification: MBBS, MD (Cardiology), Fellowship in Interventional Cardiology
BM&DC Number: A-80602
Email: pigito3883@dubokutv.com
Current Fee: 700.00
Requested Fee: 900.00
Approve Fee
Reject Fee
New Doctor Registration
Dr. Tanvir Hasan
Specialization: Neurology
Qualification: MBBS, MD (Neurology)
BM&DC Number: A-80604
Email: tsai.halvorson@mycreativeinbox.com
Consultation Fee: 1000.00
Approve Doctor
Reject Doctor


Admin Reports - TeleMed - TeleM
https://localhost:7111/Admin/Report
TeleMed Admin Dashboard
Export PDF
Export CSV
Backup Database
Total Patients
2
Total Doctors
5
Total Appointments
2
Completed Appointments
0
Total Revenue
1,200.00 ৳
Pending Payments
0
Awaiting Doctor Approval
0
Appointments (Last Months)
Appointment Status Distribution


TeleMed - TeleMed
https://localhost:7111/Chat
Summarize
Chat
TeleMed Assistant
Ask anything about booking, payments, doctors, or using your account.
Clear chat
Type your message...
Send

Chapter 8: Testing

Functional Testing:

Project: Telemedicine & Consultation Booking System

Created by: Sadia Yasmin

Date of creation: 09/01/26

Date of review: 09/01/26

Module 1: User Registration & Authentication (UR#1)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_AUTH_01	User Registration	Register with valid email & password	User not registered	1. Open registration page 2. Enter valid details 3. Submit form	Valid email, valid password	Account created successfully	User account exists	Account created successfully	Pass
TC_AUTH_02	User Registration	Register with existing email	Email already exists	1. Enter existing email2. Submit	Existing email	Error message shown	No new account created	Error message shown	Pass
TC_AUTH_03	Login	Login with valid credentials	User account exists	1. Enter email & password2. Click Login	Valid credentials	Login successful	Dashboard displayed	Login successful	Pass
TC_AUTH_04	Login	Login with invalid password	User account exists	1. Enter email2. Enter wrong password	Invalid password	Error message shown	User remains logged out	Error message shown	Pass
TC_AUTH_05	Password Recovery	Reset password using	User account exists	1. Click Forgot Password2.	Valid OTP	Password reset allowed	New password set	Password reset allowed	Pass

		email/ OTP		Verify OTP					

Project: Telemedicine & Consultation Booking System

Created by: Samiul Basher Shaan

Date of creation: 09/01/26

Date of review: 09/01/26

Module 2: Doctor Registration & Approval (UR#2)

Test Case ID	**Test Scenario**	**Test Case**	**Pre-Condition**	**Test Steps**	**Test Data**	**Expected Result**	**Post-Condition**	**Actual Result**	**Status**
TC_DOC_01	Doctor Registration	Submit doctor registration	Doctor not registered	1. Fill doctor form 2. Upload documents	Valid documents	Registration submitted	Pending approval	Registration submitted	Pass
TC_DOC_02	Document Validation	Upload invalid file type	Doctor registering	Upload .exe / invalid file	Invalid document	Error message	Upload rejected	Error message	Pass
TC_DOC_03	Admin Approval	Approve doctor registration	Doctor pending	1. Admin login 2. Approve doctor	Valid doctor	Doctor activated	Doctor can login	Doctor activated	Pass
TC_DOC_04	Admin Rejection	Reject doctor registration	Doctor pending	Admin rejects	Invalid credentials	Rejection recorded	Doctor inactive	Rejection recorded	Pass

Project: Telemedicine & Consultation Booking System

Created by: Md Shamiul Kabir Alvi

Date of creation: 09/01/26

Date of review: 09/01/26

Module 3: Doctor Search & Filtering (UR#3, UR#7)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_SEARCH_01	Doctor Search	Search by specialization	Doctors exist	Enter specialization	Cardiology	Relevant doctors shown	Results displayed	Relevant doctors shown	Pass
TC_SEARCH_02	Filtering	Filter by availability	Doctors exist	Apply availability filter	Available today	Filtered list shown	Results refined	Filtered list shown	Pass

Project: Telemedicine & Consultation Booking System

Created by: Abir Hasan

Date of creation: 09/01/26

Date of review: 09/01/26

Module 4: Appointment Booking & Management (UR#3, UR#4, UR#5)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_APPT_01	Booking	Book online appointment	Doctor available	Select doctor & slot	Valid slot	Appointment booked	Appointment created	Appointment booked	Pass
TC_APPT_02	Conflict Check	Book overlapping slot	Slot already booked	Select same slot	Same time	Error shown	Booking denied	Error shown	Pass

TC_APPT_04	Cancel	Cancel appointment	Appointment exists	Click cancel	Valid Appointment ID	Appointment canceled	Status updated	Appointment canceled	Pass

Project: Telemedicine & Consultation Booking System

Created by: Umana Tahrim

Date of creation: 09/01/26

Date of review: 09/01/26

Module 5: Online Consultation (UR#4, UR#6)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_CONS_01	Video Session	Join online consultation	Appointment time valid	Click join link	Valid link	Session starts	Consultation active	Session starts	Pass
TC_CONS_02	Session Control	Doctor ends session	Session active	End session	Valid link	Session closed	Session logged	Session closed	Pass
TC_CONS_03	Logging	Store consultation time	Session completed	End session	Valid link	Timestamps saved	Audit log updated	Timestamps saved	Pass

Project: Telemedicine & Consultation Booking System

Created by: Abir Hasan

Date of creation: 09/01/26

Date of review: 09/01/26

Module 6: Prescription & Medical Records (UR#5, UR#7)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test	Expected Result	Post-Condition	Actual Result	Status

					Data				
TC_RX_01	Prescription	Doctor issues prescription	Consultation done	Upload prescription	Valid file	Prescription saved	Patient can view	Prescription saved	Pass
TC_RX_02	Access Control	Patient views prescription	Prescription exists	Open record	Valid file	Prescription visible	Access granted	Prescription visible	Pass
TC_RX_03	Security	Unauthorized access attempt	Wrong role	Access record	Valid file	Access denied	No data leak	Access denied	Pass

Project: Telemedicine & Consultation Booking System

Created by: Abir Hasan

Date of creation: 09/01/26

Date of review: 09/01/26

Module 7: Payment & Receipt (UR#6, UR#8)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_PAY_01	Payment	Successful payment	Appointment booked	Pay fee	Valid payment	Payment success	Status updated		Pass
TC_PAY_02	Payment Failure	Payment fails	Gateway error	Attempt payment	Invalid card	Error shown	Payment pending		Pass
TC_PAY_03	Invoice	Generate invoice	Payment complete	Download invoice	Valid Transaction ID	Receipt downloaded	Record saved		Pass

Project: Telemedicine & Consultation Booking System

Created by: Abir Hasan

Date of creation: 09/01/26

Date of review: 09/01/26

Module 8: Admin Monitoring & Reports (UR#8, UR#9)

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Post-Condition	Actual Result	Status
TC_ADMIN_01	Dashboard	View system KPIs	Admin logged in	Open dashboard	System KPI Data (Users, Appointments, Payments)	KPIs displayed	Data shown	KPIs displayed	Pass
TC_ADMIN_02	Reports	Generate monthly report	Data exists	Select date range	Month	Report generated	File ready	Report generated	Pass
TC_ADMIN_03	Backup	Auto database backup	System running	Wait 24 hrs	Database = TeleMedDB, Backup Schedule = Daily	Backup created	Backup stored	Backup created	Pass

Non-Functional Testing

In addition to functional testing, selected non-functional test cases were designed to evaluate performance, security, usability, and reliability. These tests ensure that the system not only functions correctly but also meet quality, safety, and user experience expectations.

Project: Telemedicine & Consultation Booking System

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Performance Testing

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Status

NFR_PERF_01	Page Load Time	Load dashboard under normal load	User logged in	Open dashboard	50 concurrent users	Page loads ≤ 5 sec	Pass
NFR_PERF_02	Search Response	Doctor search speed	Doctors exist	Search by specialty	Cardiology	Results < 5 sec	Pass

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Security Testing

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Status
NFR_SEC_01	Unauthorized Access	Access admin page as patient	Patient logged in	Open /admin URL	User Role = Patient, URL = /admin	Access denied	Pass
NFR_SEC_02	Session Timeout	Auto logout after inactivity	User logged in	Stay idle 10 mins	Session Timeout = 10 minutes	User logged out	Pass

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Usability Testing

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Test Data	Expected Result	Status
NFR_USE_01	Ease of Use	New user registration clarity	New user	Register account	Valid inputs	No confusion/errors	Pass

NFR_USE_02	Error Messages	Invalid login feedback	User exists	Enter wrong password	Wrong pass	Clear error shown	Pass

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Reliability & Availability

Test Case ID	Test Scenario	Test Case	Pre-Condition	Test Steps	Expected Result	Status
NFR_REL_01	System Recovery	Restart server	System running	Restart service	System restores	Pass
NFR_REL_02	Data Integrity	Crash during booking	Appointment in progress	Simulate crash	No data loss	Pass

Chapter 9: Conclusion

9.1 Brief Overview of the Project

This lab project focused on the design, analysis, and structured development of a Telemedicine & Consultation Booking **System** using established software engineering principles. The primary goal of the project was not only to implement a functional software system, but also to demonstrate how theoretical concepts such as requirement analysis, process modeling, estimation, quality management, and software testing can be applied effectively in a practical development environment.

The system was designed as a web-based platform that enables patients to search for doctors, book appointments, participate in online consultations, and access digital prescriptions. Doctors can manage schedules, review appointment requests, communicate with patients, and issue prescriptions, while administrators oversee system operations, approve doctor registrations, and maintain security and user management. Role-based access control and modular design were used to ensure clarity, security, and proper separation of responsibilities.

Throughout development, the project followed the Iterative Process Model, allowing features to be developed incrementally and refined through continuous testing and review. This approach supported early identification of design issues, gradual improvement of system functionality, and better alignment with project requirements. Emphasis was placed on disciplined planning rather than rapid implementation, including process-based estimation, personnel and cost analysis, and quality assurance activities.

Overall, the project serves as a comprehensive academic exercise that demonstrates how structured software engineering practices contribute to the development of reliable, maintainable, and user-oriented systems. It also provides valuable hands-on experience in applying engineering concepts to a realistic problem domain within the constraints of a lab-based environment.

9.2 Limitations of the Project

Although the Telemedicine & Consultation Booking System fulfills its primary objectives and demonstrates the practical application of software engineering principles, it has several limitations that are important to acknowledge. These limitations mainly arise due to the academic nature of the project, time constraints, and the focus on learning core engineering concepts rather than building a fully commercial system.

One major limitation of the project is the absence of an integrated real-time video consultation module. Instead of implementing an in-built video conferencing system, the platform relies on externally generated meeting links for online consultations. While this approach simplifies development and avoids the complexity of managing real-time media streaming, encryption, and bandwidth optimization, it limits seamless integration and reduces control over communication features within the system.

The system also includes only a basic and simulated payment mechanism. Although payment status and records are maintained for appointments, there is no integration with real-world payment gateways or secure financial APIs. This limits the system's ability to handle actual monetary transactions and reflects the decision to prioritize security and feasibility within a lab environment.

Another limitation lies in the manual doctor verification process. Doctor accounts require administrative approval, but credential verification is not automated or linked to any official medical registry. In a real-world healthcare system, such verification would need to be more robust and automated to prevent misuse and ensure regulatory compliance.

From a security perspective, the project applies essential measures such as role-based access control and restricted permissions, but it does not implement advanced security mechanisms. Features such as data encryption at rest, multi-factor authentication, comprehensive audit trails, and secure key management are beyond the scope of this project. These features would be necessary for a production-level telemedicine platform handling sensitive medical data.

The system's testing scope is another limitation. Functional and selected non-functional tests were performed using limited and simulated data. Large-scale performance testing, stress testing, and real-world user behavior analysis were not conducted. As a result, the system's performance under high traffic or simultaneous consultations has not been fully validated.

In addition, the user interface prioritizes functionality over accessibility and visual refinement. While the interface is usable and structured, it has not been optimized for accessibility standards such as screen reader compatibility, advanced responsiveness across diverse devices, or localization for different languages and regions.

Finally, the system does not include several advanced features commonly found in commercial telemedicine platforms, such as automated notifications (email or SMS), analytics dashboards, intelligent scheduling optimization, emergency case handling, or AI-assisted medical decision support. These features were intentionally excluded to keep the project within a manageable academic scope.

Overall, these limitations reflect conscious design decisions made to balance feasibility, learning objectives, and project constraints. They also highlight clear opportunities for future enhancement if the system were to be extended beyond an academic lab project into a real-world deployment.